CLINICAL STUDY PROTOCOL

Primary study intervention(s)	Belantamab mafodotin
Other study intervention(s)	Lenalidomide, dexamethasone, daratumumab
Study identifier	214828
Abbreviated title	DREAMM-10
EU CT number	2024-516030-35-00
Approval date	12 Nov 2024
Title	A Phase 3, randomized, open-label study of belantamab mafodotin administered in combination with lenalidomide and dexamethasone (BRd) versus daratumumab, lenalidomide, and dexamethasone (DRd) in participants with newly diagnosed multiple myeloma who are ineligible for autologous stem cell transplantation (TI-NDMM)-DREAMM-10
Compound number/name	GSK2857916/belantamab mafodotin
Brief title	Phase 3 study of belantamab mafodotin, lenalidomide, and dexamethasone (BRd) versus daratumumab, lenalidomide, and dexamethasone (DRd) in participants with newly diagnosed multiple myeloma who are ineligible for autologous stem cell transplantation
Sponsor	GlaxoSmithKline Research & Development Limited 980 Great West Road, Brentford, Middlesex, TW8 9GS, UK
Sponsor signatory	Brandon Kremer Clinical Development Lead

Medical Monitor name and contact information can be found in the local study contact information document.

Based on TMF-17430267 Oncology Protocol v2.0

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Protocol Amendment 1 Investigator Agreement

I agree:

- To assume responsibility for the proper conduct of the study at this site.
- That I am aware of and will comply with GCP and all applicable regulatory requirements.
- That I will comply with the terms of the clinical study site agreement.
- To ensure that all persons assisting me with the study are adequately informed about the GSK study intervention and other study-related duties and functions as described in the protocol.
- To cooperate with representative(s) of GSK in the monitoring and data management processes of the study with respect to data entry and resolution of queries about the data.

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Investigator name	_____
Signature	_____
Date of signature (DD Month YYYY)	_____

PROTOCOL AMENDMENT SUMMARY OF CHANGES TABLE

DOCUMENT HISTORY	
Document (Type)	**Date of Issue**
Amendment 1 (Global)	12 Nov 2024
Original Protocol	11 July 2024

Amendment 1: (12 Nov 2024)

Overall rationale for the current amendment:

Protocol Amendment 01 applies to all participants in all countries and all sites i.e., globally; that addresses health authority feedback. Additional clarifications have also been provided. A description and a rationale for all changes are provided in the table that follows.

LIST OF MAIN CHANGES IN THE PROTOCOL AND THEIR RATIONALE

Section # and title	Description of change	Brief rationale
Global	Corrected grammar, formatting, or spelling	To correct minor errors
Headers, cover page, and Protocol Amendment Summary of Changes Table	Updated headers and cover page with the new TMF number, protocol Amendment number, and approval date	Administrative changes
	Created section for the Protocol Amendment Summary of Changes Table and added details and rationale for this amendment	
List of abbreviations and definitions of terms	Clarified that the LSLV date is inclusive of the PACT phase Added new abbreviations based on the new text	Administrative changes
Section 1.1 Synopsis Section 2.1 Study rationale Section 5.1 Inclusion criteria	Removed "over 70" or ≥70 years of age from the eligibility criteria	To only use age as a criterion for transplant-eligibility where locally appropriate
Section 1.1 Synopsis Section 1.3 Schedule of activities Section 4.1 Overall design Section 4.4 End-of-study definition Section 4.4.1 Final data cut-off Section 6.7 Continued access to study intervention after the end of the study Section 8.4.1 Time period and frequency for collecting AE, SAE, and other safety information	Added details regarding the conduct of the PACT phase	To allow flexibility for eligible participants to receive study treatment after study completion and clarify data collection related to the PACT phase
Section 1.1 Synopsis Section 4.1 Overall design	Added randomization stratification factor of geographic location (North America vs. Rest of the World)	To align with regulatory agency feedback
Section 1.2 Schema	Added geographical region (North America vs. Rest of the World) in stratification factors	To align with regulatory agency feedback
	Clarified that the Follow-up Period for PFS is Q4W until 24 months, then Q8W until either PD, death, withdrawal of consent, or EOS, and irrespective of new anti-myeloma therapy initiation	Clarification for study conduct
	Updated footnote for dexamethasone treatment to delete criteria regarding BMI, diabetes mellitus, or prior intolerance/AE to steroid therapy	Simplify trial design for study conduct

Section # and title	Description of change	Brief rationale
Section 1.3 Schedule of activities	Clarified that upon randomization, study intervention should be initiated as soon as possible, no later than 72 hours	Clarification for study conduct
	Clarified that at PFS follow-up, subsequent anti-myeloma therapies and further progressions should be collected	Clarification for study conduct
	Updated Home Healthcare and Telemedicine approaches to be 'de-centralized'	Clarification for study conduct
Section 1.3 Schedule of activities, **Arm A**, Table 1 Section 1.3 Schedule of activities, **Arm B** Table 2	Updated assessment window for the Treatment Period and PFS Follow-up Period window to ±5 days	Clarification for study conduct
	Clarified that the laboratory assessments should be performed prior to drug administration and 'until 70 days' after the last dose of study intervention	Clarification for study conduct
	Clarified that the disease assessments should be performed prior to drug administration	Clarification for study conduct
	Note for serum and urine immunofixation was deleted Clarified that serum and urine immunofixation will be performed at C1D1 and at the time of first achieving serum or urine M-protein	Clarification for study conduct
	Clarified that BM aspirate for risk assessment will include tp53 mutation testing and a sample must be sent to a central lab for confirmation if the test is performed at a local lab	To align with updated IMWG risk assessment testing
	Updated timepoints for EQ-5D-3L to 12, 24, and 48 weeks during the OS follow-up	Clarification for study conduct
	Clarified timing for collection of AEs and SAEs	Clarification for study conduct
	Added window for qualitative interview before C1D1	To correct minor error
Section 1.3 Schedule of activities; **Arm A**, Table 1	Added footnote to clarify HBV DNA and HCV RNA testing criteria, and will be performed at a central lab	To align with other DREAMM study (DREAMM-8) for participants with positive HBV at baseline
	Timepoints for IgG, IgA, IgM, IgD, and IgE were extended up to the PFS follow-up period	To correct minor error

Section # and title	Description of change	Brief rationale
	Added footnote for advanced skeletal imaging to clarify the window	Clarification for study conduct
	Added the following to footnote 5: For participants who have KVA Grade ≥2 events during the Treatment Period, an ocular exam will be performed Q4W (±5 days) until full resolution of ophthalmic changes to Grade 1 or baseline	Clarification for study conduct
Section 1.3 Schedule of activities; **Arm B**, Table 2	Updated timepoint for ocular exam to Q12W from C1D1 until 'C24' and 'Q24W until EOT' after C24	Reduce burden to participants
	Added assessment at EOT for urine dipstick for protein or spot urine for albumin/creatinine ratio	To correct minor error
Section 1.3 Schedule of activities; Table 3	Updated to clarify tp53 mutation testing and bone marrow biopsy testing Updated CR/sCR to CR or better	To align with updated IMWG risk assessment testing Clarification for study conduct
Section 1.3 Schedule of activities; Table 4, Table 5	Added to clarify additional procedures for participants with evidence of prior HBV or HCV exposure	To align with other DREAMM studies (DREAMM-8) for participants with positive HBV at baseline
Section 2.3.1 Risk assessment, Table 6	Mitigation strategy for hepatotoxicity was updated to include participants with Hepatitis B and C if they meet the entry criteria and agree to the monitoring/treatment criteria	To align with other DREAMM studies (DREAMM-8) for participants with positive HBV at baseline
Section 3 Objectives, endpoints, and estimands	In Table 7, updated CR or sCR to 'CR' or 'better'	Clarification for study conduct
	Reordered key secondary endpoints	To align with regulatory agency feedback
	Updated secondary endpoint of PK i.e., plasma concentrations of belantamab mafodotin '(ADC) and cys-mcMMAF'	To align with regulatory agency feedback
	Moved the secondary objective (and its associated endpoint) to assess ADAs against BRd in participants with TI-NDMM to exploratory	To align with other DREAMM studies which reported ADA assessment as an exploratory objective
	Updated PK endpoint to derive PK and exposure parameter (e.g., CL, Cmax, AUC) values of belantamab mafodotin (ADC) and cys-mcMMAF, as data permit	To align with regulatory agency feedback
	Deleted ORR from exposure-response relationship endpoint	Expected ORR in both arms negates utility of this exploratory analysis

Section # and title	Description of change	Brief rationale
	Table 8: Updated details for the estimands of the dual primary objectives, specifically for PFS S4/S5 and MRD S1/S2	Clarification and alignment with regulatory agency feedback
	Updated rationale for PFS primary estimand	To align with regulatory agency feedback
	Table 7: Updated handling strategy for the main estimand	To align with regulatory agency feedback
	Added cross-reference to the SAP for details on estimands	To align with regulatory agency feedback
Section 4.1 Overall design	Added details on cycle length for each of the 4 drugs in reference to C1D1	Clarification for study conduct
Section 4.3.2 Daratumumab dose	Revised dose justification for daratumumab and added daratumumab-related toxicity management	Clarification for study conduct
Section 5.1 Inclusion criteria	Updated inclusion criteria definition of transplant-ineligible Updated laboratory assessment criteria for hepatic organ system function	To align with regulatory agency feedback To align with regulatory agency feedback
Section 6.1 Study intervention(s) administered	Table 12: Updated cycles for Arm A and Arm B; clarified that the dose adjustment of lenalidomide is not classified as 'a dose reduction per protocol'	Clarification for study conduct
Section 6.6.1 Belantamab mafodotin dose adjustments due to body weight Section 6.6.2 Belantamab mafodotin dose modification schedule, 6.6.3 Dose modification for overlapping hematological toxicities, 6.6.4 Dose modification guidelines for belantamab mafodotin	Updated dose modification of belantamab mafodotin due to body weights and the dose modification schedule or guidelines for belantamab mafodotin in Table 14, Table 15, and Table 16 Minor updates to Table 17 footnote	Clarification for study conduct
Section 6.6.7 Management of hepatitis B positive participants	Updated the guidelines for management of hepatitis B positive participants	To align with GSK's new oncology protocol template (Oct 2024)

Section # and title	Description of change	Brief rationale
Section 6.8.1 Belantamab mafodotin overdose Section 8.5.1 Belantamab mafodotin blood sample collection for pharmacokinetics Section 8.5.2 Belantamab mafodotin pharmacokinetic sample analysis Section 9.3.1.6 Pharmacokinetic analyses	Clarified additional PK for measurement of belantamab mafodotin '(ADC) and cys-mcMMAF'	To align with regulatory agency feedback
Section 7.1.2 Liver stopping criteria	Updated Figure 2 for the liver event study intervention stopping criteria	To align with GSK's ´latest oncology protocol template (Oct 2024) for the updated liver language guidance
Section 8.2.1 Minimal residual disease	Updated CR or sCR to 'CR' or 'better'	Clarification for study conduct
Section 8.2.4 Overall survival collection	Updated data collection for OS	To align with GSK's new oncology protocol template (Oct 2024)
Section 8.4.1 Time period and frequency for collecting AE, SAE, and other safety information	Updated SAE collection for China	Clarification for study conduct
Section 8.4.5 Regulatory reporting requirements for SAEs/AESIs	Added text for reporting of SUSAR by the sponsor	Clarification for study conduct
8.7. Genetics	Updated amount of blood sample to 5-mL	To correct a minor error
Section 9.1.1 Multiplicity adjustment	Updated PFS and MRD analyses Updated OS analysis Updated adjustment for key secondary endpoints to be described in SAP	To align with regulatory agency feedback
Section 9.2 Analysis sets	Updated definitions for 'enrolled', 'safety', and 'randomized' analysis sets	Clarification for study analysis
Section 9.3.1.4 Other safety measures	Removed BCVA measurement details	Clarification for study conduct
Section 9.4 Interim analyses	Updated PFS boundaries, added OS in the planned analyses' table, and timing for interim analyses	To align with regulatory agency feedback
Section 9.5 Sample size determination	Added sample size for OS	To align with regulatory agency feedback
Section 10.2 Appendix 2: Clinical laboratory tests, Table 28	Updated PK and biomarker measurements	To correct a minor error
Section 10.6 Appendix 6: Liver safety: Requirements and guidelines	Updated liver safety stopping criteria and monitoring criteria with continued study intervention	To align with GSK's latest oncology protocol template (Oct 2024) for the updated liver language guidance

Section # and title	Description of change	Brief rationale
Section 10.11 Appendix 11 PFS event and censoring rules	Updated censoring rules	To reflect the change in primary estimand
Section 10.13 Appendix 13: Country-specific requirements	Added followings: For studies conducted in EU, sponsor will share the SAEs or other significant safety findings with the agency	To align with regulatory requirements
Section 10.14 Appendix 14 De-centralized options for healthcare	Updated definition of de-centralized visits and direct shipment of study drug to participants	To reflect that telehealth – virtual reviews by the investigator would be permitted (where possible)

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LIST OF ABBREVIATIONS AND DEFINITIONS OF TERMS

Abbreviation	Definition
Ab	Antibody
ADA	Anti-drug antibody(ies)
ADC	Antibody drug conjugate
ADCC	Antibody-dependent cellular cytotoxicity
ADCP	Antibody-dependent cellular phagocytosis
ADL	Activities of daily living
ADR	Adverse drug reaction
AE	Adverse event
AESI	Adverse event of special interest
AIDS	Acquired immunodeficiency syndrome
ALP	Alkaline phosphatase
ALT	Alanine aminotransferase
ANC	Absolute neutrophil count
ART	Antiretroviral therapy
ASCT	Autologous stem cell transplantation
AST	Aspartate aminotransferase
AUC	Area under the concentration-time curve
AxMP	Auxiliary medicinal product
BCMA	B-cell maturation antigen
BCVA	Best corrected visual acuity
BM	Bone marrow
BMA	Bone marrow aspirate
BMI	Body mass index
BOR	Best overall response
BPd	Belantamab mafodotin in combination with pomalidomide and dexamethasone
BRd	Belantamab mafodotin in combination with lenalidomide and dexamethasone
BVRd	Belantamab mafodotin in combination with VRd (bortezomib in combination with lenalidomide plus dexamethasone)
CCI	Charlson comorbidity index
CD	Cluster of differentiation
CDC	Complement dependent cytotoxicity
CI	Confidence interval
CK	Creatinine kinase
CL	Clearance
Cmax	Maximum concentration
CONSORT	Consolidated Standards of Reporting Trials
CPK	Creatinine phosphokinase
CR	Complete response
CR+	Complete response or better
CRAB	(increased blood) Calcium, renal (damage), anemia, bone (damage)
CRF	Case report form
CRR	Complete response rate
CSR	Clinical study report
CT	Computed tomography
CTCAE	Common Term Criteria for Adverse Events
CV	Cardiovascular
Cx	Cycle x
CxDx	Cycle x Day x
Cys-mcMMAF	Cysteine maleimidocaproyl monomethyl auristatin F
CYP	Cytochrome P450
DCO	Data cut-off
D/d	Day/day

Abbreviation	Definition
DDI	Drug-drug interaction
DILI	Drug-induced liver injury
DNA	Deoxyribonucleic acid
DoR	Duration of response
DRd	Daratumumab in combination with lenalidomide and dexamethasone
DRESS	Drug reaction with eosinophilia and systemic symptoms
EAE SCS	Hellenic Society of Hematology supported collaborative study
ECG	Electrocardiogram
ECOG	Eastern Cooperative Oncology Group
eCRF	Electronic case report form
eGFR	Estimated glomerular filtration rate
EMD	Extramedullary disease
EOI	End of infusion
EORTC	European Organization for Research and Treatment of Cancer
EOS	End-of-study
EOT	End-of-Treatment
EQ-5D-3L	European Quality of Life 5 Dimensions 3 Level Scale
ESMO	European Society for Medical Oncology
FACT-GP	Functional Assessment of Cancer Therapy-General Population
FAS	Full analysis set
FDA	Food and Drug Administration
FISH	Fluorescence-in-situ hybridization
FLC	Free light chain
FSFV	First Subject First Visit
FSH	Follicle-stimulating hormone
GCP	Good clinical practice
GDPR	General Data Protection Regulation
GGT	Gamma-glutamyl transferase
GI	Gastrointestinal
GLDH	Glutamate dehydrogenase
GSK	GlaxoSmithKline
HBcAb	Hepatitis B core antibody
HBeAb	Hepatitis B e antibody
HBeAg	Hepatitis B e antigen
HBsAg	Hepatitis B surface antigen
HBV	Hepatitis B virus
HCRU	Healthcare Resource Utilization
HCV	Hepatitis C virus
HCVAb	Hepatitis C virus antibody
HDV	Hepatitis D virus
Hep C	Hepatitis C
Hep C Ab	Hepatitis C antibody
HGRAC	Human Genetic Resource Administration of China
HHS	Home healthcare services
HR	Hazard ratio
HRQoL	Health-related quality of life
HRT	Hormonal replacement therapy
IA	Interim analysis
IB	Investigator's Brochure
ICF	Informed consent form
IDMC	Independent data monitoring committee
IEC	Independent Ethics Committee
IF	Information fraction
Ig	Immunoglobulin

Abbreviation	Definition
IHC	Immunohistochemistry
IMP	Investigational medicinal product
IMWG	International Myeloma Working Group
INR	International normalized ratio
IP	Investigational product
IQR	Interquartile range
IRB	Institutional Review Board
IRC	Independent Review Committee
IRR	Infusion-related reaction
ISS	International Staging System
ITT	Intent-to-treat
KVA	Keratopathy and Visual Acuity
LDH	Lactate dehydrogenase
LFT	Liver function test
LSLV	Last Subject Last Visit
MAP	Managed access program
MDRD	Modified Diet in Renal Disease
MGUS	Monoclonal gammopathy of undetermined significance
MI	Myocardial infarction
MM	Multiple myeloma
MMAF	Monomethyl auristatin-F
mo	Month
MRD	Minimal residual disease
MRI	Magnetic resonance imaging
MSDS	Material Safety Data Sheet
NA	North America
NCCN	National Comprehensive Cancer Network
NCI-CTCAE v5.0	National Cancer Institute – Common Terminology Criteria for Adverse Events, version 5
NGS	Next-generation sequencing
NR	Not reached
NYHA	New York Heart Association
OATP	Organic Anion Transporting Polypeptide
ORR	Overall response rate
OS	Overall survival
OSDI	Ocular surface disease index
PACT	Post Analysis Continued Treatment
PBMC	Peripheral blood mononuclear cell
PCR	Polymerase chain reaction
PD	Progressive disease
PET	Positron emission tomography
PFS	Progression-free survival
PFS2	Progression-free survival on subsequent line of therapy
PGIC	Patient Global Impression of Change
PGIS	Patient Global Impression of Severity
P-gp	P-glycoprotein
PII	Personally identifiable information
PK	Pharmacokinetic(s)
PKPD	Pharmacokinetic and pharmacodynamic
PO	Per oral(ly)
PP	Per protocol
PR	Partial response
PR+	Partial response or better
PRO	Patient-reported outcome

Abbreviation	Definition
PRO-CTCAE	Patient-Reported Outcomes Version of the Common Term Criteria for Adverse Events
PS	Performance status
PVd	Pomalidomide in combination with bortezomib and dexamethasone
QD	Once daily
QoL	Quality of life
QTcF	QT interval corrected for heart rate using Fridericia's formula
QTL	Quality tolerance limit
QxW	Every x week(s)
RBC	Red blood cell
Rd	Lenalidomide plus dexamethasone
RNA	Ribonucleic acid
RoW	Rest-of-world
RRMM	Relapsed/refractory multiple myeloma
RTMS	Randomization and Trial Supply Management
RWD	Real-world data
Sx	Supplementary x (x would be 1, 2, 3, 4)
SAE	Serious adverse event
SAP	Statistical analysis plan
sBCMA	Soluble BCMA
SC	Subcutaneous
sCR	Stringent complete response
SD	Stable disease
SJS	Stevens-Johnson syndrome
sMRD	Sustained minimal residual disease
SoA	Schedule of activities
SOC	Standard of care
SOI	Start of infusion
SPD	Sum of the perpendicular diameters
SPEP	Serum protein electrophoresis
SRT	Safety Review Team
SUSAR	Suspected unexpected serious adverse reaction
TEN	Toxic epidermal necrolysis
TI	Transplant-ineligible
TI-NDMM	Transplant-ineligible newly diagnosed multiple myeloma
TLS	Tumor lysis syndrome
TM	Telemedicine
TMF	Trail Master File
TTP	Time to progression
TSST	Time to second subsequent therapy
ULN	Upper limit of normal
UPEP	Urine protein electrophoresis
VA	Visual acuity
VAS	Visual analogue scale
VGPR	Very good partial response
VGPR+	Very good partial response or better
VRd	Bortezomib in combination with lenalidomide plus dexamethasone
WBC	White blood cell
WOCBP	Woman of childbearing potential
WONCBP	Woma(e)n of non-childbearing potential

Abbreviation	Definition
ADR	An AE where a causal relationship between a medicinal product and the AE is at least a reasonable possibility, i.e., the relationship cannot be ruled out. In the context of a clinical study, an ADR can be serious or nonserious. Serious ADRs may be subject to expedited reporting if they are considered unexpected (see SUSAR definition). For marketed products, ADRs are subject to expedited reporting within the country where they are authorized.
AxMP	Medicinal products used in the context of a clinical study but not as IMPs, such as medicinal products used for background treatment, challenge agents, rescue medication, or used to assess endpoints in a clinical study. AxMPs should not include concomitant medications, i.e., medications unrelated to the clinical study and not relevant for the design of the clinical study. Authorized AxMP = Medicinal product authorized in accordance with Regulation (EC) No 726/2004, or in any member state concerned in accordance with Directive 2001/83/EC, irrespective of changes to the labeling of the medicinal product. Note: Safety reporting with regard to authorized AxMPs shall be made in accordance with Chapter 3 of Title IX of Directive 2001/83/EC. Unauthorized AxMP = Medicinal product not authorized in accordance with Regulation (EC) No 726/2004 Safety reporting for unauthorized AxMPs will follow the same processes and procedures as SUSAR safety reporting.
Background treatment	Type of medicinal product administered to each of the clinical study participants, regardless of randomization group, to treat the indication that is the object of the study. Background treatment is generally considered to be the current SOC for the particular indication. In these clinical studies, the IMP is given in addition to the background treatment and safety/efficacy are assessed. The protocol may require that the IMP plus the background treatment is compared with an active comparator or with placebo plus background treatment.
Blinding:	A procedure in which 1 or more parties to the study are kept unaware of the intervention assignment in order to reduce the risk of biased study outcomes. The level of blinding is maintained throughout the conduct of the study, and only when the data are cleaned to an acceptable level of quality will appropriate personnel be unblinded or when required in case of a SAE. In an observer-blind study, the participant, the site and sponsor personnel involved in the clinical evaluation of the participants are blinded while other study personnel may be aware of the treatment assignment.
Certified copy	A copy (irrespective of the type of media used) of the original record that has been verified (e.g., by a dated signature or by generation through a validated process) to have the same information, including data that describe the context, content, and structure, as the original.
Co-administered product	A product given to clinical study participants as required by the protocol as part of their standard care for a condition that is not the indication for which the IMP is being tested and is therefore not part of the objective of the study.

Abbreviation	Definition
Combination product	Combination product comprises any combination of: • drug • device • biological product. Each drug, device, and biological product included in a combination product is a constituent part.
Comparator	Any product used as a reference (including placebo, marketed product, GSK or non-GSK) for an investigational product being tested in a clinical study. This is any product that is being used to assess safety, efficacy, or other measurable value against the test product (IMP).
eDiary	Electronically recorded participant data and automated data entries on, for example, a handheld mobile device, tablet, or computer.
Eligible	Qualified for enrollment into the study based upon strict adherence to inclusion/exclusion criteria.
Essential documents	Documents that individually and collectively permit evaluation of the conduct of a study and the quality of the data produced.
HHS	Deployment of mobile health care professional(s) (nurses or phlebotomists) to perform study activities remotely.
Intercurrent event	Event occurring after study intervention initiation that affects either the interpretation or the existence of the measurements associated with the clinical question of interest.
Intervention number	A number identifying the intervention assigned to a participant according to intervention allocation.
IMP	A pharmaceutical form of an active substance or placebo being tested or used as a reference in a clinical study, including products already with a marketing authorization but used or assembled (formulated or packaged) in a way different from the authorized form, or when used for an unauthorized indication, or when used to gain further information about the authorized form.
Investigator	A person responsible for the conduct of the clinical study at a study site. If a study is conducted by a team of individuals at a study site, the investigator is the responsible leader of the team and may be called the principal investigator. The investigator can delegate study-related duties and functions conducted at the study site to qualified individual(s) or party(ies) to perform those study-related duties and functions.
LSLV	The date on which the last participant in a clinical study was examined (inclusive of the PACT phase) or received an intervention/treatment to collect final data for the primary outcome measures, secondary outcome measures, or AEs (that is, the last participant's last visit or LSLV).
Medicinal products used to assess endpoints	A product given to the participant in a clinical study as a tool to assess a relevant clinical study endpoint; it is not being tested or used as a reference in the clinical study.

Abbreviation	Definition
Non-IMP	Any products used in a clinical study (other than the investigational product being tested) which are stipulated to be used to evaluate the efficacy and safety of the investigational drug in the protocol including comparators, co-administration drugs, rescue drugs, and premedication drugs. • Non-IMPs products can be approved in Japan or other countries or can be products that are not approved.
Participant	Term used throughout the protocol to denote an individual who has been contacted to participate or who participates in the clinical study as a recipient of the study intervention (vaccine[s]/product[s]/control). Synonym: subject
Participant identifier	A unique identification number assigned to each participant who consents to participate in the study.
PGx	The ICH E15 Guidance for Industry defines PGx as "the study of variation of DNA and RNA characteristics as related to drug or treatment response." Pharmacogenetics, a subset of PGx, is "the study of variations in DNA sequence as related to drug response." PGx biomarkers include germline (host) DNA and RNA as well as somatic changes (e.g., mutations) that occur in cells or tissues. PGx biomarkers are not limited to human samples but include samples from viruses and infectious agents as well as animal samples. The term pharmacogenomic experiment includes both the generation of new genetic or genomic (DNA and/or RNA) data with subsequent analysis as well as the analysis of existing genetic or genomic data to understand drug or treatment response (PK, pharmacodynamics, safety, efficacy, or effectiveness, mode of action). Proteomic and metabolomic biomarker research is not PGx.
Primary completion date	This is the date that the final participant in the study was examined or received an intervention for the purposes of final collection of data for the primary outcome, whether the clinical study concluded according to the pre-specified protocol or was terminated. In other words, Primary Completion Achieved is the date of the last contact with the participant when data have been collected/intervention done for the purpose of data collection for analysis of all primary endpoints. In the case of clinical studies with more than 1 primary outcome measure with different completion dates, this term refers to the date on which data collection is completed for all of the primary outcomes. This date may occur prior to the study end or be the same date as the study end milestone.
Randomization	Process of random attribution of intervention to participants to reduce selection bias.
Remote visit	A visit conducted in a place other than the study site.
Rescue medication	Medicine(s) identified in the protocol as those that may be administered to the participants when the efficacy of the IMP is not satisfactory, or the effect of the IMP is too great and is likely to cause a hazard to the participant, or to manage an emergency situation.
Source data	All information in original records and certified copies of original records of clinical findings, observations, or other activities in a clinical study necessary for the reconstruction and evaluation of the study. Source data are contained in source documents (original records or certified copies).

Abbreviation	Definition
SOC	Medicine(s) for a specific indication, or a component of the standard care for a particular medical indication, based on national and/or international consensus; there is no regulatory significance to this term. Products/regimens considered SOC may differ country to country, depending on consensus in individual countries.
Study intervention	Term used throughout the clinical study to cover all types of investigational and non-investigational products, including medical devices and vaccines, intended to be administered to the study participants during the study conduct. Procedures conducted to manage participants or to collect data are excluded from the usage of this term.
SUSAR	In a clinical study, a serious adverse reaction that is considered unexpected, i.e., the nature or severity of which is not consistent with the reference safety information (e.g., IB for an unapproved IMP). All ADRs that are both serious and unexpected are subject to expedited reporting.
TM	The use of electronic information and telecommunications technologies (both video-based and audio-only) to facilitate remote health care delivery, participant, and professional health-related education, public health, and health administration.

1. PROTOCOL SUMMARY

1.1. Synopsis

Protocol title: A Phase 3, randomized, open-label study of belantamab mafodotin administered in combination with lenalidomide and dexamethasone (BRd) versus daratumumab, lenalidomide, and dexamethasone (DRd) in participants with newly diagnosed multiple myeloma who are ineligible for autologous stem cell transplantation (TI-NDMM)-DREAMM-10

Brief title: Phase 3 study of belantamab mafodotin, lenalidomide, and dexamethasone (BRd) versus daratumumab, lenalidomide, and dexamethasone (DRd) in participants with newly diagnosed multiple myeloma who are ineligible for autologous stem cell transplantation

Rationale: The current SOC for patients with TI-NDMM includes autologous stem cell transplantation. However, the majority of patients with NDMM are ineligible for transplant (TI-NDMM) at the time of diagnosis due to age, frailty and/or comorbidities and there is an unmet need to improve the prognosis of patients with TI-NDMM.

Prior studies have demonstrated the clinical activity of single agent belantamab mafodotin and as triplet combination in participants with RRMM (Section 2.1).

The triplet regimen of belantamab mafodotin plus Rd could provide additional therapeutic benefit over the DRd triplet regimen, which is an alternative SOC for patients with TI-NDMM. In DREAMM-10, GSK will investigate the combination of belantamab mafodotin plus Rd (BRd) as a triplet regimen that could provide additional therapeutic benefit for participants with TI-NDMM (Section 2.2).

Objectives, endpoints, and estimands: The primary objectives are to evaluate if BRd prolongs PFS and/or improves MRD negative status compared with DRd in participants with TI-NDMM. The key secondary objectives are to further assess whether BRd prolongs OS and PFS on next line of therapy when compared with DRd (Section 3).

Overall design: DREAMM-10 is a randomized, open-label, active controlled, parallel group, multicenter, Phase 3 study in adults with NDMM who are ineligible for ASCT (TI-NDMM).

The study will consist of a Screening Period, Treatment Period, and Follow-up Period (Figure 1). After final analysis, participants who are still receiving benefit from study intervention(s) may transition to either PACT, an extension study, a rollover study, and/or expanded access route.

Participants may move into the PACT phase where the study remains open only to provide continued access to study intervention(s) for study participants who are continuing to derive clinical benefit. At that time, collection of new data for participants who no longer receive study intervention(s) will stop entirely and the clinical trial database will be closed. Participants in survival follow-up will be considered to have

completed the study. Participants who are still benefiting from study intervention(s), as assessed by the investigator, may continue to receive study intervention(s) and only SAEs, AEs leading to study intervention discontinuation, overdose, pre-specified ocular data, and pregnancy cases will be reported directly to the sponsor.

Eligible participants will be centrally randomized to either receive BRd or DRd. Randomization will be stratified by age (<75 vs. ≥75 years), International Staging System (I vs. II vs. III) at screening, and geographic location (North America vs. Rest-of-World) (Section 4.1). Within each stratum, participants will be randomized using an equal allocation ratio of 1:1. No cross-over will be allowed (Section 4.1).

Treatment Arm A: Belantamab mafodotin 1.9 mg/kg IV will be administered Q8W for first 24 weeks, then 1.9 mg/kg Q12W thereafter. Lenalidomide will be administered at 25 mg once a day PO on Day 1 to Day 21 of every 28-day lenalidomide treatment cycle. Dexamethasone will be administered PO at a dose of 40 mg on Days 1, 8, 15, and 22 of each 28-day cycle.

Treatment Arm B: Daratumumab 1800 mg SC will be administered weekly from Week 1 to 8, Q2W from Week 9 to 24, and Q4W from Week 25 onwards. Lenalidomide will be administered at 25 mg once a day PO on Day 1 to Day 21 of every 28-day lenalidomide treatment cycle. Dexamethasone will be administered PO at a dose of 40 mg on Days 1, 8, 15, and 22 of each 28-day cycle.

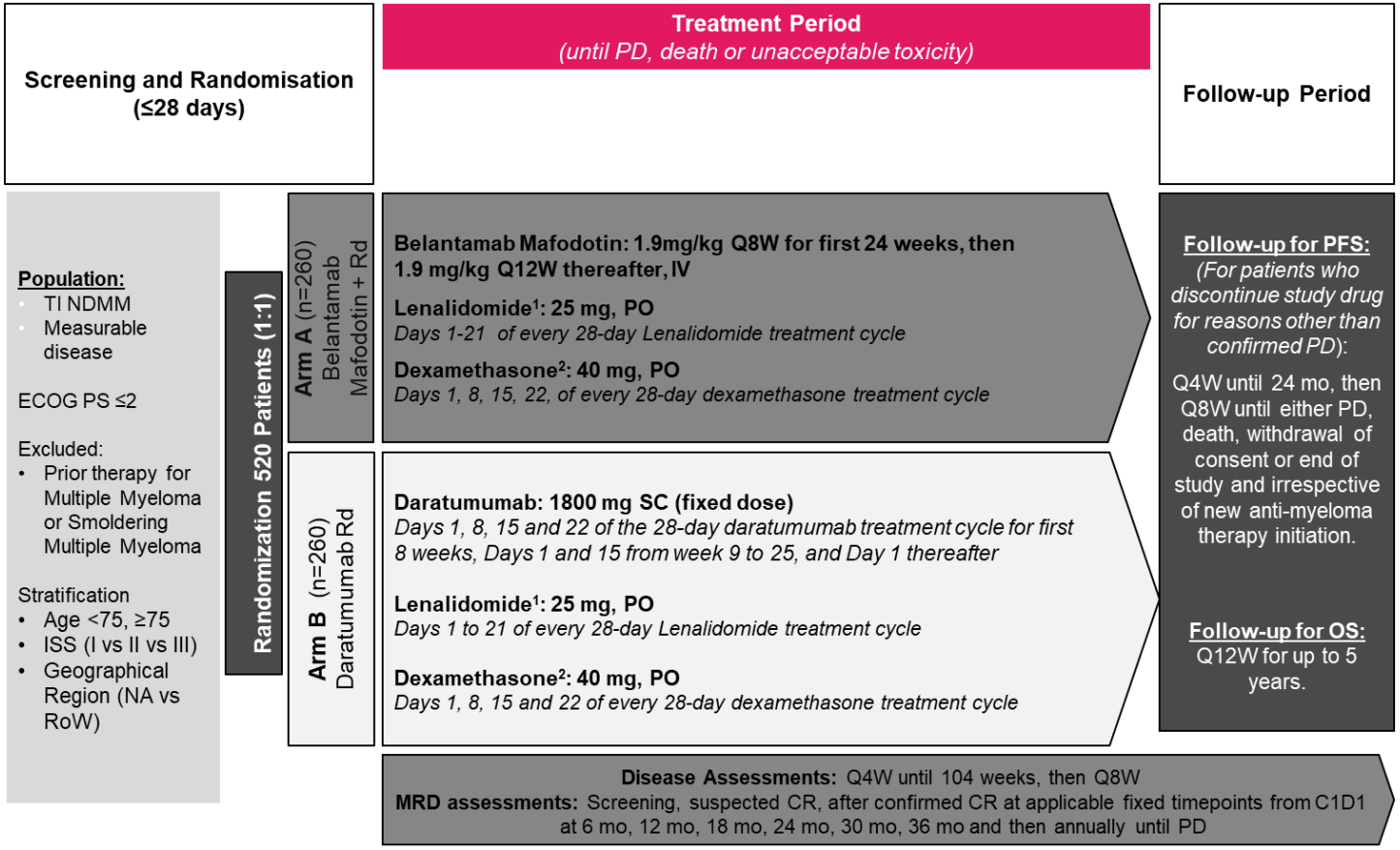
Treatment will continue in both arms until PD, death, unacceptable toxicity, withdrawal of consent, or EOS, whichever occurs first. Dose delays or reductions may be required following potential drug-associated toxicities.

Number of participants: Approximately 520 participants will be randomized into the study in a 1:1 ratio between the 2 arms.

Data monitoring/other committee: An IDMC will be established to provide independent review and assessment of the efficacy and safety data in a systematic manner and to safeguard the interest and safety of the participants in the study. The composition and responsibilities of the committee will be described in the IDMC Charter.

1.2. Schema

Figure 1 Study design overview



1.For participants with eGFR 30-60 mL/min/1.73m^2 at screening lenalidomide 10 mg PO.
2.For participants ≥75 years old, dexamethasone 20 mg PO may be used.

1.3. Schedule of activities

- **Screening** assessments do not need to be repeated on C1D1 if conducted within 72 hours of C1D1 dosing, unless otherwise specified.
- The 28-day Screening Period is initiated upon signature of the ICF (Day 1). The full 28 days can be taken to determine the eligibility of the participants. Once participants are randomized, study intervention (C1D1) should be initiated as soon as possible, no later than 72 hours.
- Assessments scheduled on days of dosing should be done prior to drug administration, unless otherwise specified. All assessments should be carried out even if a participant is not dosed **except for the following**: weight, where applicable.
- **Treatment visits:** Participants are scheduled to receive treatment cycles as described in the tables below for each arm. Precise timing of treatment cycles may vary due to treatment delays, but participants must still visit for efficacy/disease assessments as described in the table below for each arm regardless of when they receive treatment.
- **EOT visit:** to occur within 30 days from the last dose, or prior to the new anti-MM treatment (whichever occurs first).
- **PFS Follow-up:** Participants who discontinue study intervention for a reason other than confirmed PD, will continue to have disease evaluations performed at the timepoints indicated below until confirmed PD, death, withdrawal of consent, lost to follow-up, or EOS, whichever occurs first. Subsequent anti-myeloma therapies and further progressions should be collected.
- **Survival follow-up:** After PD is confirmed, participants will be followed for survival, subsequent anti-myeloma therapy, and second progression (PFS2) by chart review, telephone call, or any form of communication Q12W (14-day window) from the last Q4W assessment visit until death, withdrawal of consent, lost to follow-up, or EOS, whichever comes first. Participants do not need to come in for a visit unless they are being followed for treatment-related changes in vision that are present at the EOT.
- **PACT phase:** Participants who continue to receive study intervention(s) during the PACT phase will be monitored and will receive follow-up care in accordance with standard local clinical practice. Assessments will revert to the SOC at a participant's particular study site and only SAEs, AEs leading to discontinuation of study intervention, overdoses, pre-specified ocular data, and pregnancy cases will be reported directly to the sponsor via paper/electronic forms (see Section 8.4.5). Participants will be required to have an ocular assessment at time of dosing and/or as clinically indicated, until the EOT.

- It is advised that participants with treatment-related ocular examination findings after treatment discontinuation continue to have ocular assessments as SOC until resolution. Note that participants who continue to receive belantamab mafodotin via a commercial source or a drug supply program should follow the ocular examination guidance provided by the source/program from which they are receiving belantamab mafodotin.
- For details on de-centralized healthcare and telemedicine approaches, please refer to Section 10.14.

Table 1 Arm A: belantamab mafodotin+Rd

Procedure	Notes	Screening	Treatment Period							EOT	Follow-up Period	
Cycle / Visit			Cycle 1 (8-week cycle)		Cycle 2 and 3 (8-week cycles)		Cycle 4+[1] (12-week cycles)				PFS[2]	OS Q12W
Day			D1	D29	D1	D29	D1	D29	D57			
Window		-28 d	±5 d								±5 d	±14 d
Informed consent	Must be signed before any study-specific assessments are performed.	X										
Informed consent for tokenization[3] (optional)	Must be signed to link clinical study data to data about a participant collected outside of the clinical study (Real-World Data); as described in Section 10.15	X										
Inclusion and exclusion criteria		X										
Baseline demography		X										
Body weight		X	X		X		X					
Height		X										
Medical history, disease history and characteristics		X										
IMWG frailty score		X										
Vaccine history questionnaire[4]		X										
Informed consent for genetic research (Optional)	Must be signed before collecting a sample.	X										

Procedure	Notes	Screening	Treatment Period							EOT	Follow-up Period	
Cycle / Visit			Cycle 1 (8-week cycle)		Cycle 2 and 3 (8-week cycles)		Cycle 4+[1] (12-week cycles)				PFS[2]	OS Q12W
Day			D1	D29	D1	D29	D1	D29	D57			
Window		-28 d	±5 d								±5 d	±14 d
Safety Assessments												
ECOG PS		X	X		X		X			X	X	
Ocular exam[5]	C1D1 ocular exam does not need to be performed if C1D1 is within 28 days of screening exam. For participants who have KVA Grade ≥2 event in the Treatment Period, an ocular exam to be performed Q4W (±5 days) until full resolution of ophthalmic changes to Grade 1 or baseline.	X			X[5]		X[5]			X[5]	X[5]	
12-Lead ECG	Acceptable at screening if performed as SOC within 42 days of C1D1.	X	As clinically indicated									
Laboratory Assessments During study participation, until 70 days after the last dose of study intervention, all laboratory test values considered clinically significantly abnormal should be reported as AEs and tests should be repeated until values return to normal or baseline. Assessments scheduled on days of dosing should be done prior to drug administration unless otherwise specified.												
Hematology	May be taken up to 72 hours prior to dosing. Results must be received and evaluated prior to dosing with any drug and may be repeated as clinically necessary.	X	X	X	X	X	X	X	X	X		
Clinical chemistry		X	X	X	X	X	X	X	X	X		
eGFR		X	X	X	X	X	X	X	X	X		
Urine dipstick for protein OR spot urine for albumin/creatinine ratio[6]		X	X	X	X	X	X	X	X	X		

Procedure	Notes	Screening	Treatment Period							EOT	Follow-up Period	
Cycle / Visit			Cycle 1 (8-week cycle)		Cycle 2 and 3 (8-week cycles)		Cycle 4+[1] (12-week cycles)				PFS[2]	OS Q12W
Day			D1	D29	D1	D29	D1	D29	D57			
Window		-28 d	±5 d								±5 d	±14 d
HBsAg, HBcAb, HCVAb, Hep C RNA, HBV DNA[7]	Hepatitis C RNA testing will be done only if Hep C Ab is positive; results must be received prior to dosing. HBV DNA (if confirmed history of hepatitis B or positive HBsAg or HBcAb) performed at a central lab.	X	For additional procedures during Screening and during treatment for participants who have a history of Hep B and/or Hep C see Section 5.2 and Section 6.6.									
Pregnancy test for WOCBP[8]	Perform only on WOCBP	X	X	X	X	X	X	X	X	X		
Disease Evaluations: All laboratory disease assessments will be done as specified in Table 28. Imaging completed within 42 days as part of SOC acceptable at screening. For participants who are discontinuing IP due to PD, the confirmation based on laboratory parameters must be performed from a different blood collection within 14 days of the original date of the disease progression, preferably before institution of any new anti-myeloma therapy. Visits are anchored to C1D1 regardless of cycle delay. Assessments scheduled on days of dosing should be done prior to drug administration unless otherwise specified.												
B2 Microglobulin		X										
IgG, IgA, IgM		X	At C1D1 and Q4W from C1D1 for 104 weeks then Q8W until PD and at the EOT visit									
IgD, IgE	Only for participants with IgD/IgE myeloma.	X	At C1D1 and Q4W from C1D1 for 104 weeks then Q8W until PD and at the EOT visit									
SPEP		X	At C1D1 and Q4W from C1D1 for 104 weeks then Q8W until PD and at the EOT visit									
UPEP (on 24-h collected urine)		X	At C1D1 and Q4W from C1D1 for 104 weeks then Q8W until PD and at the EOT visit									
Serum M-protein calculation		X	At C1D1 and Q4W from C1D1 for 104 weeks o then Q8W until PD and at the EOT visit									
Serum immunofixation		X	At C1D1 and at time of first achieving serum or urine M-protein ~0 g/dL (suspected CR) up to confirmation of CR and then to continue for as long as serum or urine M-proteins remain ~0 g/dL									
Urine immunofixation		X										
Serum kappa lambda FLC, FLC ratio		X	At C1D1 and Q4W from C1D1 for 104 weeks then Q8W until PD and at the EOT visit									

Procedure	Notes	Screening	Treatment Period							EOT	Follow-up Period	
Cycle / Visit			Cycle 1 (8-week cycle)		Cycle 2 and 3 (8-week cycles)		Cycle 4+[1] (12-week cycles)				PFS[2]	OS Q12W
Day			D1	D29	D1	D29	D1	D29	D57			
Window		-28 d	±5 d								±5 d	±14 d
Calcium corrected for albumin (serum)		X	At C1D1 and Q4W from C1D1 for 104 weeks then Q8W until PD and at the EOT visit									
Imaging for EMD[9]		X	If Extramedullary Disease detected at screening, Q12W from C1D1 until PD using the same modality as used at screening									
Advanced skeletal imaging[10]		X[11]	As clinically indicated using the same modality as used at screening									
Response assessment by IMWG	Assessed, based on central laboratory tests, using IMWG criteria [Kumar, 2016].		Q4W from C1D1 for 104 weeks then Q8W until PD									
BM Assessments: See Table 3 for details on collection and timepoints												
BM aspirate for MRD testing	Sample for MRD Testing must be the first pull of the BM aspirate. Testing will be performed at a central laboratory.	X	Frequency of collection detailed in Table 3									
BM aspirate for risk assessment (FISH + tp53 mutation) testing	Screening only. Testing for screening can be performed at a local lab but a sample must be sent to central lab for confirmation. If no local lab is available, central testing only is acceptable.	X										
BM biopsy and aspirate for disease assessments	Fresh sample required at screening to confirm diagnosis of MM.	X	Frequency of collection detailed in Table 3									
BM biopsy to confirm sCR by IHC[12]			Frequency of collection detailed in Table 3									

Procedure	Notes	Screening	Treatment Period							EOT	Follow-up Period	
Cycle / Visit			Cycle 1 (8-week cycle)		Cycle 2 and 3 (8-week cycles)		Cycle 4+[1] (12-week cycles)				PFS[2]	OS Q12W
Day			D1	D29	D1	D29	D1	D29	D57			
Window		-28 d	±5 d								±5 d	±14 d
Treatments Administered												
Belantamab mafodotin **Dose schedule Q8W/Q12W**	Administered as an IV infusion (see Section 6.1 for details), 1- to 2-hour observation period required on Day 1 of every cycle when belantamab mafodotin is administered.		X		X		X					
Lenalidomide	If dose(s) are held within a cycle they will be skipped and will not be made up later in the cycle. The 7-day break Q4W should be observed.		Days 1-21	Days 29-49	Days 1-21	Days 29-49	Days 1-21	Days 29-49	Days 57-77			
Dexamethasone	Administered weekly		Days 1, 8, 15, 22	Days 29, 36, 43, 50	Days 1, 8, 15, 22	Days 29, 36, 43, 50	Days 1, 8, 15, 22	Days 29, 36, 43, 50	Days 57, 64, 71, 78			
Pregnancy prevention counseling	Counseling on pregnancy precautions and the potential risks of fetal exposure from study drugs must be conducted at D1 of each cycle.	X	X	X	X	X	X	X	X	X		
Supportive medication			Supportive medication to be provided for each agent as per institutional guidelines									
Dispense/Return medication diary	Participant diary will be used to keep a record of self-administered oral lenalidomide and dexamethasone.		X	X	X	X	X	X	X	X		

Procedure	Notes	Screening	Treatment Period							EOT	Follow-up Period	
Cycle / Visit			Cycle 1 (8-week cycle)		Cycle 2 and 3 (8-week cycles)		Cycle 4+[1] (12-week cycles)				PFS[2]	OS Q12W
Day			D1	D29	D1	D29	D1	D29	D57			
Window		-28 d	±5 d								±5 d	±14 d
Ocular Supportive Care for Belantamab Mafodotin												
Preservative-free artificial tears			4 to 8 times daily in each eye. Daily beginning on C1D1 until EOT. In the event of ocular symptoms (i.e., dry eyes), the use of artificial tears may be increased up to every 2 hours as needed									
Cooling eye masks			At the start of each belantamab mafodotin infusion, apply for approximately 1 hour or as long as tolerated to both eyes. Participants may continue to apply for up to 4 hours in the clinic									
PROs PROs will be completed at the start of every applicable visit and must be prior to any other assessments or provision of results (Section 8.10).												
EORTC QLQ-C30, EORTC QLQ-MY20, PGIS, PGIC	PGIC is not collected at C1D1.		At C1D1 and Q4W from C1D1 for 24 weeks, after Q8W until 52 weeks, Q12W after 52 weeks until PD and at the EOT visit									X[13]
OSDI	Participants with no treatment-related ocular AEs at the EOT ocular assessment will have no further OSDI collection after EOT. Participants with treatment-related ocular AEs present at the EOT will have OSDI collected in PFS and OS until ocular signs resolve (to Grade 1 or baseline).		At C1D1 and Q4W from C1D1 for 24 weeks, Q8W until 52 weeks, Q12W after 52 weeks until PD and at the EOT visit								X (see notes)	X (see notes)
PRO-CTCAE, FACT-GP5			At C1D1 and Q4W from C1D1 for 24 weeks, Q8W until 52 weeks, Q12W after 52 weeks until PD, and at the EOT visit									X[13]
EQ-5D-3L			At C1D1 and Q4W from C1D1 for 24 weeks, Q8W until 52 weeks, Q12W after 52 weeks until PD, and at the EOT visit. EQ 5D should also be collected at 12, 24, and 48 weeks during OS follow-up (can be collected by telephone using interviewer administration).									

Procedure	Notes	Screening	Treatment Period							EOT	Follow-up Period	
Cycle / Visit			Cycle 1 (8-week cycle)		Cycle 2 and 3 (8-week cycles)		Cycle 4+[1] (12-week cycles)				PFS[2]	OS Q12W
Day			D1	D29	D1	D29	D1	D29	D57			
Window		-28 d	±5 d								±5 d	±14 d
Medical Resource Utilization and Health Economics												
Healthcare Resource Utilization			At C1D1 and Q4W from C1D1 for 24 weeks, Q8W until 52 weeks, Q12W after 52 weeks until PD, and at the EOT visit									
Biomarkers, Genetics, PK, and ADA												
PK	PK, sBCMA and ADA sample collection is relative to dose of belantamab mafodotin which is independent of cycle number to account for the possibility that belantamab mafodotin may be held while other agents continue.		Frequency of collection detailed in Table 23									
ADA for belantamab mafodotin			Frequency of collection detailed in Table 23									
Soluble BCMA			Frequency of collection detailed in Table 23									
Immunophenotyping[14]			Sampling prior to the belantamab mafodotin dose in first year and at EOT or PD, whichever is first.									
Optional genetic sample	Informed consent for optional genetic research must be obtained before collecting a sample.		Collect at the first opportunity once a participant has met all eligibility requirements, preferably on C1D1. May be taken at another visit if not taken at C1D1									
Ongoing Participant Review												
AEs/SAEs			AEs will be collected from the start of a study intervention until 70 days after the last dose. SAEs will be collected from the start of a study intervention for at least 70 days after the last dose. All AEs and SAEs will be followed until resolution or EOS. SAEs related to study participation are collected from time of consent until EOS. Ocular AEs/SAEs are followed until resolution.									
Concomitant medications	Concomitant medications will be collected, as reported by the participant or through participant medical record review (Section 6.9)	All collected from screening until at least 70 days after the last dose. Anti-MM therapies will be collected out to the safety follow-up visit										

Procedure	Notes	Screening	Treatment Period							EOT	Follow-up Period	
Cycle / Visit			Cycle 1 (8-week cycle)		Cycle 2 and 3 (8-week cycles)		Cycle 4+[1] (12-week cycles)				PFS[2]	OS Q12W
Day			D1	D29	D1	D29	D1	D29	D57			
Window		-28 d	±5 d								±5 d	±14 d
Others												
Survival status telephone call												X
Subsequent anti-MM treatment												X
Response to subsequent anti-MM treatment												X
Qualitative interview	Participants may opt-in to discussing their disease and treatment experience during telephone interviews with trained interviewers. Interviews can be performed within 28 days before C1D1 and within 60 days of C6D1.		X				C6D1					

1. 12-week cycles will begin after 24 weeks on study or upon change from 8-week to 12-week dosing schedule based on dose modification guidelines, which may occur prior to the first 6 months on study.
2. PFS visits will be performed Q4W from EOT for 24 months then Q8W until PD. Timing will take into account treatment duration.
3. Where applicable country and local regulations and infrastructure allow.
4. Questionnaire is described in Section 10.16. Source data verification is not required.
5. Ocular exams will be performed at screening, within 5 days prior to belantamab mafodotin treatment on C2 onward until EOT. For participants who have KVA Grade ≥2 event during the Treatment Period, an ocular exam will be performed Q4W (±5 days) until full resolution of ophthalmic changes to Grade 1 or baseline. Participants in PFS follow-up or OS follow-up will only have ocular exams if they had ocular signs or symptoms at EOT; these ocular exams will be Q12W until findings return to baseline. Must be performed by a qualified eye care specialist (Section 10.10).
6. Urine dipstick for protein may be used to assess for presence of urine protein. Albumin/creatinine ratio needs to be done in any participant with urine dipstick result of ≥1+ (at Screening Visit), or ≥2+ (during study intervention), or with positive protein if urine dipstick protein quantification is not available. Albumin/creatinine will be performed at a local lab. If local testing is not available, then central testing will be performed (first void). If testing to be performed at a central lab, may be conducted 7 days prior to dosing.

Procedure	Notes	Screening	Treatment Period							EOT	Follow-up Period	
Cycle / Visit			Cycle 1 (8-week cycle)		Cycle 2 and 3 (8-week cycles)		Cycle 4+[1] (12-week cycles)				PFS[2]	OS Q12W
Day			D1	D29	D1	D29	D1	D29	D57			
Window		-28 d	±5 d								±5 d	±14 d

7. Hepatitis B virus DNA testing will be done to determine a participant's eligibility if HBcAb and/or HBsAg is positive. Hepatitis C virus RNA testing will be done to determine participant eligibility if Hepatitis C Ab-positive. This will be performed at a central laboratory. For additional procedures during screening and upon enrollment of participants who have a history of Hepatitis B and/or Hepatitis C, please refer to Table 4 and Table 5, respectively.
8. For questionable cases, follicle-stimulating hormone and estradiol (as needed in WONCBP) should be performed at local lab to determine childbearing potential. The 2 highly sensitive serum pregnancy tests must be performed during screening, 1 within 10-14 days prior to C1D1 and 1 within 24 hours prior to starting lenalidomide at C1D1. Subsequent pregnancy tests on dosing days may be either serum or urine. After the first dose, pregnancy tests will be done weekly during the first month, then every cycle thereafter (or Q2W in females with irregular menstrual cycles). Each pregnancy test must be performed within 24 hours prior to any study intervention. Final pregnancy test (serum or urine) must be performed at the EOT visit. Follow-up pregnancy assessment by telephone (for WOCBP only) should be performed 4 months after the last dose of belantamab mafodotin or 4 weeks after the last dose of lenalidomide, whichever comes last.
9. Imaging is only required for participants with EMD by either CT, MRI, or PET-CT per local guidance. Screening assessment may be performed within 30 days prior to first dose. The same method should be used throughout the study (i.e., if a PET-CT scan was used as baseline scan than participants need to be followed by PET-CT scans). Selected target lesion needs to be measured and followed over time. Plasmacytoma measurements should be taken from the CT portion of the PET/CT, or MRI scans, or dedicated CT scans where applicable. For participants with only skin involvement, skin lesions should be measured with a ruler. Measurement of tumor size will be determined by the sum of the products of the perpendicular diameters of measured lesions (SPD).
10. Advanced skeletal imaging includes whole-body CT, whole-body MRI, or PET/CT (whole-body as defined by local imaging protocol).
11. Window for advanced skeletal imaging is -42 days from C1D1.
12. BM biopsy/aspirate elective if BM to confirm CR was within 12 weeks prior to suspected sCR.
13. EORTC QLQ-C30, EORTC QLQ-MY20, PRO-CTCAE, FACT-GP5, and EQ-5D-3L should be administered at 12-, 24-, and 48-week OS follow-up visits. For participants in OS follow-up not attending the site, they can be administered via a phone call in an interview format.
14. Immunophenotyping assessments will be waived in China and samples will not be collected.

Table 2 Arm B: daratumumab+Rd

Procedure	Notes	Screening	Treatment Period (28-day cycles)				EOT	Follow-up Period	
Cycle / Visit			C1	C2	C3-6	C7+		PFS[1]	OS Q12W
Day			D1	D1	D1	D1			
Window		-28 d	±5 d					±5 d	±14 d
Informed consent	Must be signed before any study -specific assessments are performed.	X							
Informed consent for tokenization[2] (optional)	Must be signed to link clinical study data to data about a participant collected outside of the study (Real-World Data); As described in Section 10.15	X							
Inclusion and exclusion criteria		X							
Baseline demography		X							
Body weight		X	X	X	X	X			
Height		X							
Medical history, disease history and characteristics		X							
IMWG frailty score	Section 10.7	X							
Vaccine history questionnaire[3]		X							
Informed consent for genetic research (optional)	Must be signed before collecting a sample.	X							

Procedure	Notes	Screening	Treatment Period (28-day cycles)				EOT	Follow-up Period	
Cycle / Visit			C1	C2	C3-6	C7+		PFS[1]	OS Q12W
Day			D1	D1	D1	D1			
Window		-28 d	±5 d					±5 d	±14 d
Safety Assessments									
ECOG PS		X	X	X	X	X	X	X	
Ocular exam[4]	C1D1 ocular exam does not need to be performed if C1D1 is within 28 days of screening exam. May be performed within 5 days of daratumumab dosing	X	Q12W from C1D1 until C24 visit. Frequency after C24 is Q24W and EOT.				X	X[4]	X[4]
12-Lead ECG	Acceptable at screening if performed as SOC within 42 days of C1D1.	X	As clinically indicated						
Laboratory Assessments During study participation until 70 days after the last dose of study intervention, all laboratory test values considered clinically significantly abnormal should be reported as AEs and tests should be repeated until values return to normal or baseline. Assessments scheduled on days of dosing should be done prior to drug administration unless otherwise specified.									
Hematology	May be taken up to 72 hours prior to dosing. Results must be received and evaluated prior to dosing with any drug and may be repeated as clinically necessary.	X	X	X	X	X	X		
Clinical chemistry		X	X	X	X	X	X		
eGFR		X	X	X	X	X	X		
Urine dipstick for protein OR spot urine for albumin/creatinine ratio[5]		X	X	X	X	X	X		

Procedure	Notes	Screening	Treatment Period (28-day cycles)				EOT	Follow-up Period	
Cycle / Visit			C1	C2	C3-6	C7+		PFS[1]	OS Q12W
Day			D1	D1	D1	D1			
Window		-28 d	±5 d					±5 d	±14 d
HbsAg, HbcAb, HCVAb, Hep C RNA, HBV DNA	Hepatitis C RNA testing will be done only if Hep C Ab is positive; results must be received prior to dosing. HBV DNA (if confirmed history of hepatitis B or positive HBsAg or HbcAb) performed at a central lab.	X	For additional procedures during Screening and during treatment for participants who have a history of Hep B and/or Hep C see Section 5.2 and Section 6.6.						
Pregnancy test for WOCBP[6]	Perform only on WOCBP	X	X	X	X	X	X		
Indirect antiglobulin test, red cell phenotype, and antibody screen		X							
Daratumumab-specific interference assay[7]			X	X	X	X		X	
Disease Evaluations: All laboratory disease assessments will be done as specified in Table 28. Screening laboratory assessments completed within 14 days of C1D1 do not need to be repeated. For participants who are discontinuing IP due to PD, the confirmation based on laboratory parameters must be performed from a different blood collection within 14 days of the original date of the disease progression, preferably before institution of any new anti-myeloma therapy. Imaging completed within 42 days as part of SOC acceptable at screening. Visits are anchored to C1D1 regardless of cycle delay. Assessments scheduled on days of dosing should be done prior to drug administration unless otherwise specified.									
B2 microglobulin		X							
IgG, IgA, IgM		X	At C1D1 and Q4W from C1D1 for 104 weeks then Q8W until PD and at the EOT visit.						
IgD, IgE	Only for participants with IgD/IgE myeloma.	X	At C1D1 and Q4W from C1D1 for 104 weeks then Q8W until PD and at the EOT visit.						
SPEP[7]		X	At C1D1 and Q4W from C1D1 for 104 weeks then Q8W until PD and at the EOT visit.						

Procedure	Notes	Screening	Treatment Period (28-day cycles)				EOT	Follow-up Period	
Cycle / Visit			C1	C2	C3-6	C7+		PFS[1]	OS Q12W
Day			D1	D1	D1	D1			
Window		-28 d	±5 d					±5 d	±14 d
UPEP (on 24-h collected urine)		X	At C1D1 and Q4W from C1D1 for 104 weeks then Q8W until PD and at the EOT visit.						
Serum M-protein calculation		X	At C1D1 and Q4W from C1D1 for 104 weeks o then Q8W until PD and at the EOT visit.						
Serum immunofixation	.	X	At C1D1 and at time of first achieving serum or urine M-protein ~0 g/dL (suspected CR) up to confirmation of CR and then to continue for as long as serum or urine M-proteins remain ~0 g/dL						
Urine immunofixation		X							
Serum kappa lambda FLC, FLC ratio		X	At C1D1 and Q4W from C1D1 for 104 weeks then Q8W until PD and at the EOT visit.						
Calcium corrected for albumin (serum)		X	At C1D1 and Q4W from C1D1 for 104 weeks then Q8W until PD and at the EOT visit.						
Imaging for EMD[8]	Imaging is only required for participants with known EMD	X	If EMD detected at screening, Q12W from C1D1 until PD using the same modality as used at screening.						
Advanced skeletal imaging[9]		X[10]	As clinically indicated using the same modality as used at screening.						
Response assessment by IMWG	Assessed, based on central laboratory tests, using IMWG criteria [Kumar, 2016]		Q4W from C1D1 for 104 weeks then Q8W until PD.						

Procedure	Notes	Screening	Treatment Period (28-day cycles)				EOT	Follow-up Period	
Cycle / Visit			C1	C2	C3-6	C7+		PFS[1]	OS Q12W
Day			D1	D1	D1	D1			
Window		-28 d	±5 d					±5 d	±14 d
BM Assessments: Table 3 for details on collection and timepoints									
BM aspirate for MRD testing	Sample for MRD Testing must be the first pull of the BM aspirate. Testing will be performed at a central laboratory.	X	Frequency of collection detailed in Table 3						
BM aspirate for Risk Assessment (FISH + tp53 mutation) testing	Screening only. Testing at screening may be performed at a local lab but a sample should be sent to central lab for confirmation. If no local lab is available, central testing only is acceptable.	X							
BM biopsy and/or aspirate for disease assessments	Fresh sample required at screening to confirm diagnosis of MM.	X	Frequency of collection detailed in Table 3						
BM biopsy to confirm sCR by IHC[11]			Frequency of collection detailed in Table 3						
Treatments Administered									
Daratumumab	Administered as a subcutaneous injection		D1, D8, D15 and D22	D1, D8, D15 and D22	D1 and D15	D1			
Lenalidomide	If dose(s) are held within a cycle they will be skipped and will not be made up later in the cycle. The 7-day break prior to the next cycle should be observed.		Days 1-21	Days 1-21	Days 1-21	Days 1-21			
Dexamethasone			Days 1, 8, 15, and 22	Days 1, 8, 15, and 22	Days 1, 8, 15, and 22	Days 1, 8, 15, and 22			

Procedure	Notes	Screening	Treatment Period (28-day cycles)				EOT	Follow-up Period	
Cycle / Visit			C1	C2	C3-6	C7+		PFS[1]	OS Q12W
Day			D1	D1	D1	D1			
Window		-28 d	±5 d					±5 d	±14 d
Pregnancy prevention counseling	Counseling on pregnancy precautions and the potential risks of fetal exposure from study drugs must be conducted at D1 of each cycle.	X	X	X	X	X	X		
Supportive medication			Supportive medication to be provided for each agent as per institutional guidelines						
Dispense/return medication diary	Participant diary will be used to keep a record of self-administered oral lenalidomide and dexamethasone.		X	X	X	X	X		
PROs PROs will be completed at the start of every applicable visit and must be prior to any other assessments or provision of results (Section 8.10)									
EORTC QLQ-C30, EORTC QLQ-MY20, PGIS, PGIC	PGIC is not collected at C1D1.		At C1D1 and Q4W from C1D1 for 24 weeks, Q8W until 52 weeks, Q12W after 52 weeks until PD, and at the EOT visit.						X[12]
OSDI	Participants with no treatment-related ocular AEs at the EOT ocular assessment will have no further OSDI collection. Participants with treatment-related ocular AEs present at the EOT will have OSDI collected in PFS and OS until ocular signs resolve (to Grade 1 or baseline).		At C1D1 and Q4W from C1D1 for 24 weeks, Q8W until 52 weeks, Q12W after 52 weeks until PD, and at the EOT visit.					X (see notes)	X (see notes)
PRO-CTCAE, FACT-GP5			At C1D1 and Q4W from C1D1 for 24 weeks, Q8W until 52 weeks, Q12W after 52 weeks until PD, and at the EOT visit.						X[12]

Procedure	Notes	Screening	Treatment Period (28-day cycles)				EOT	Follow-up Period	
Cycle / Visit			C1	C2	C3-6	C7+		PFS[1]	OS Q12W
Day			D1	D1	D1	D1			
Window		-28 d	±5 d					±5 d	±14 d
EQ-5D-3L			At C1D1 and Q4W from C1D1 for 24 weeks, Q8W until 52 weeks, Q12W after 52 weeks until PD, and at the EOT visit. EQ-5D-3L should also be collected at 12, 24, and 48 weeks during OS follow-up (can be collected by telephone using interviewer administration).						
Medical Resource Utilization and Health Economics									
Healthcare Resource Utilization			At C1D1 and Q4W from C1D1 for 24 weeks, Q8W until 52 weeks, Q12W after 52 weeks until PD, and at the EOT visit.						
Biomarkers									
Soluble BCMA			Frequency of collection detailed in Table 23						
Immunophenotyping[13]			To be collected before the daratumumab dose on C1D1 and Q8W for 24 weeks then Q12W until 52 weeks, and at EOT or PD, whichever is first.						
Optional genetic sample	Informed consent for optional genetic research must be obtained before collecting a sample.		Collect at the first opportunity once a participant has met all eligibility requirements, preferably on C1D1. May be taken at another visit if not taken at C1D1.						
Ongoing Participant Review									
AEs/SAEs			AEs will be collected from the start of a study intervention until 70 days after the last dose. SAEs will be collected from the start of a study intervention for at least 70 days after the last dose. All AEs and SAEs will be followed until resolution or EOS. SAEs related to study participation are collected from time of consent until EOS. Ocular AEs/SAEs are followed until resolution.						
Concomitant medications	Concomitant medications will be collected, as reported by the participant or through participant medical record review	All collected from screening until at least 70 days after the last dose. Anti-MM therapies will be collected out to the safety follow-up visit.							

Procedure	Notes	Screening	Treatment Period (28-day cycles)				EOT	Follow-up Period	
Cycle / Visit			C1	C2	C3-6	C7+		PFS[1]	OS Q12W
Day			D1	D1	D1	D1			
Window		-28 d	±5 d					±5 d	±14 d
Others									
Survival status telephone call									X
Subsequent anti-MM treatment									X
Response to subsequent anti-MM treatment									X
Qualitative interview	Participants may opt-in to discussing their disease and treatment experience during telephone interviews with trained interviewers. Interviews can be performed within 28 days before C1D1 and within 60 days of C6D1.		X		C6D1				

Procedure	Notes	Screening	Treatment Period (28-day cycles)				EOT	Follow-up Period	
Cycle / Visit			C1	C2	C3-6	C7+		PFS[1]	OS Q12W
Day			D1	D1	D1	D1			
Window		-28 d	±5 d					±5 d	±14 d

1. PFS visits will be performed Q4W from EOT for 24 months then Q8W until PD. Timing will take into account treatment duration.
2. Where applicable country and local regulations and infrastructure allow.
3. Questionnaire is described in Section 10.16. Source data verification is not required.
4. Ocular exams will be performed at screening, Q12W up to Cycle 24, then Q24W onwards, and EOT. Participants in PFS follow-up or OS follow-up will only have ocular exams if they had ocular signs or symptoms at EOT; these ocular exams will be Q12W until findings return to baseline. Must be performed by a qualified eye care specialist (Section 10.10).
5. Urine dipstick for protein may be used to assess for presence of urine protein. Albumin/creatinine ratio needs to be done in any participant with urine dipstick result of ≥1+ (at Screening Visit), or ≥2+ (during study intervention), or with positive protein if urine dipstick protein quantification is not available. Albumin/creatinine will be performed at a local lab. If local testing is not available, then central testing will be performed (first void). If testing to be performed at a central lab, may be conducted 7 days prior to dosing.
6. For questionable cases, follicle-stimulating hormone and estradiol (as needed in women of non-childbearing potential) should be performed at local lab to determine childbearing potential. The 2 highly sensitive serum pregnancy tests must be performed during screening, 1 within 10-14 days prior to C1D1 and 1 within 24 hours prior to starting lenalidomide at C1D1. Subsequent pregnancy tests on dosing days may be either serum or urine. After the first dose, pregnancy tests will be done weekly during the first month, then every cycle thereafter (or Q2W in females with irregular menstrual cycles). Each pregnancy test must be performed within 24 hours prior to any study intervention. Final pregnancy test (serum or urine) must be performed at the EOT visit. Follow-up pregnancy assessment by telephone (for WOCBP only) should be performed 4 months after the last dose of daratumumab or 4 weeks after the last dose of lenalidomide, whichever comes last.
7. SPEP samples from participants with IgG kappa myeloma will undergo central lab testing with a daratumumab-specific interference assay (HYDRASHIFT 2/4, FDA-approved).
8. Imaging is only required for participants with EMD by either CT, MRI, or PET-CT per local guidance. Screening assessment may be performed within 30 days prior to first dose. The same method should be used throughout the study (i.e., if a PET-CT scan was used as baseline scan than participants need to be followed by PET-CT scans). Selected target lesion needs to be measured and followed over time. Plasmacytoma measurements should be taken from the CT portion of the PET/CT, or MRI scans, or dedicated CT scans where applicable. For participants with only skin involvement, skin lesions should be measured with a ruler. Measurement of tumor size will be determined by the sum of the products of the perpendicular diameters of measured lesions (SPD).
9. Advanced skeletal imaging includes whole-body CT, whole-body MRI, or PET/CT (whole-body as defined by local imaging protocol).
10. Window for advanced skeletal imaging is -42 days from C1D1.
11. BM biopsy/aspirate elective if BM to confirm CR was within 12 weeks prior to suspected sCR.
12. EORTC QLQ-C30, EORTC QLQ-MY20, PRO-CTCAE, FACT-GP5, and EQ-5D-3L should be administered at 12-, 24-, and 48-week OS follow-up visits. For participants in OS follow-up not attending the site, they can be administered via a phone call in an interview format.
13. Immunophenotyping assessments will be waived in China and samples will not be collected.

Table 3 Bone marrow aspirate/biopsy collection (treatment Arm A and treatment Arm B)

Time point	BM aspirate and/or biopsy for disease assessment[1,2]	BM aspirate and biopsy for disease assessment[2]	BM aspirate for risk assessment (FISH + tp53 mutation)[3,5]	BM aspirate for MRD testing[4,6]
Screening Period	X		X	X
Suspected CR or better		X[7]		X
Suspected PD (only if PD is not otherwise evident)		X[8]		

1. **For Disease assessment at screening: BM aspirate and/or biopsy per institutional practice**.
2. These assessments will be performed at a local laboratory.
3. Testing at screening may be performed at a local lab but a sample should be sent to central lab for confirmation. If no local lab is available, central lab testing only is acceptable.
4. These assessments will be performed at a central laboratory.
5. FISH and tp53 mutation testing results are not required prior to randomization or dosing. Central confirmation of FISH and tp53 mutation testing results will be performed when local risk assessment testing is performed for screening.
6. BM aspirate samples are required for MRD assessment at screening, at the time of achieving suspected CR or better, then repeated at each of the following applicable timepoints after CR: post-C1D1 6, 12, 18, 24, 30, 36 month (±1 month), thereafter annually (±1 month) until PD. When a timepoint is less than 3 months from the suspected CR or better MRD assessment, a repeat BMA sample for MRD is not required.
7. For participants achieving a CR, a BM core **biopsy is required** to confirm sCR by IHC for absence of clonal cells. Only 1 BM procedure required for confirming CR and sCR assessment. If IHC testing cannot be performed at a local lab, the samples can be sent to the central lab.
8. Central assessment of BM is available if there is concern for PD that is not evident on serum/urine labs or radiographically. This is not a required test if PD is evident via serum, urine, or radiographically.

Table 4 Additional procedures for participants with evidence of prior HBV at baseline

HBV study assessments	During Screening/Prior to starting treatment	During treatment	EoT	Notes
HBV related Liver Imaging	X[1]	X[1]	X[1]	1. Liver imaging (specific test per SOC at local institution) in HBsAg+ participants to rule-out/identify cirrhosis, focal hepatic lesions, and/or biliary abnormalities at baseline. Repeat imaging at 1 year after starting treatment if still on treatment and twice yearly thereafter as long as participant remains on study-treatment or as clinically indicated.
HBV DNA testing	X[2]	X[2]	X[2]	2. HBV DNA testing prior to the start of belantamab mafodotin and subsequently every 3 months, or if LFT elevations requiring increased monitoring or stopping criteria occurs, or for any clinical suspicion of hepatitis reactivation.
Prevention of HBV reactivation	X[3]	X[3]	X[3]	3. For HBsAg+ participants, appropriate antiviral treatment per local guidance (e.g., tenofovir or entecavir) is started before starting belantamab mafodotin, continues through to completion of belantamab mafodotin therapy and should not be stopped unless advised by local hepatology or virology services.
HBeAg and HBeAb testing	X[4]			4. Only for Japan: HBsAg+ participants only.

Please refer to Section 6.6.7 for management of participants with positive serology for HBV.

Table 5 Additional schedule of procedures for participants with prior HCV exposure

HCV study assessments	During Screening /Prior to starting treatment	During treatment	Post treatment	Notes
HCV RNA testing	X[1]	X[1]		1. HCV-RNA testing prior to the start of belantamab mafodotin and subsequently every 3 months, or if LFT elevations requiring increased monitoring or stopping criteria occurs, or for any clinical suspicion of hepatitis reactivation.
Treatment of active HCV	X[2]			2. Antiviral treatment should be given to participants with HCV before enrollment using an 8 (to 12) week antiviral treatment course with curative intent per local guidance. Hep C RNA should be negative at 4 weeks washout period post anti-HCV therapy prior to enrollment.

2. INTRODUCTION

MM is a clonal proliferation of abnormal plasma cells in the BM and is the second most common hematological malignancy, accounting for approximately 10% of all cases. Belantamab mafodotin is an ADC that binds to BCMA and kills MM cells via a multimodal mechanism including MMAF-induced apoptosis, ADCC and ADCP.

2.1. Study rationale

The treatment approach to NDMM is guided mainly by risk factors based on certain chromosomal abnormalities and eligibility for ASCT, which is determined by factors such as age, general fitness, and presence of comorbidities. The current SOC for newly diagnosed younger, 'fit' patients (aged typically up to 70 years without significant frailty or comorbidities) is the sequential treatment of combination induction therapy, consolidation with high-dose melphalan and autologous stem cell transplantation, followed by maintenance therapy. However, the majority of patients are ineligible for transplant (TI-NDMM) at the time of diagnosis due to age, frailty and/or comorbidities [Dimopoulos, 2021].

Despite the advancements in treatment options, TI patients have worse clinical outcomes in terms of PFS, OS, and depth of response, including lower achievement of MRD negative status, when compared to patients who are eligible for ASCT [Attal, 1996; Child, 2003; Barlogie, 2006; Attal, 2017; Dhakal, 2018; Perrot, 2018]. As a result, there is an unmet need to improve the prognosis of patients with TI-NDMM through the addition of novel and effective therapies to the first-line SOC agents used to treat NDMM.

Prior studies have demonstrated the clinical activity of single agent belantamab mafodotin and in combination with bortezomib and dexamethasone in participants with RRMM, as well as the potential for lenalidomide to combine with the multimodal mechanisms of action of belantamab mafodotin.

We hypothesize that for patients with TI-NDMM, the triplet regimen of belantamab mafodotin plus Rd could provide additional therapeutic benefit over the DRd triplet regimen by improving outcomes of depth and durability of response, without compromising tolerability. In DREAMM-10, GSK will investigate the combination of belantamab mafodotin plus Rd as a triplet regimen that could provide additional therapeutic benefit for participants with TI-NDMM.

2.2. Background

2.2.1. Clinical management of TI-NDMM

In recent years, there have been significant advancements in the treatment of patients with TI-NDMM. In particular, proteasome inhibitors (such as bortezomib, carfilzomib [Bringhen, 2018] or ixazomib [Moreau, 2016]), immunomodulatory drugs (such as lenalidomide), and monoclonal antibodies (such as daratumumab [Facon, 2019]) have transformed the treatment of NDMM and are considered SOC for TI patients, in addition to the previously established use of alkylating agents (melphalan or cyclophosphamide), and corticosteroids (such as prednisone and dexamethasone).

The ESMO-European Hematology Association [Dimopoulos, 2021] and National Comprehensive Cancer Network guidelines [National Comprehensive Cancer Network, 2024], have well-defined recommendations for primary regimens for patients with TI-NDMM:

- **Rd**: Lenalidomide (Revlimid, R) plus dexamethasone (d) [Benboubker, 2014]
- **VRd**: Bortezomib (Velcade, V) plus Rd, followed by Rd [Durie, 2017; Durie, 2020]
- **DRd:** Daratumumab (D) plus Rd [Facon, 2019; Facon, 2021]
- **DVMP:** D plus V, melphalan (M) and prednisone (P) [Mateos, 2018; Mateos, 2020; Dimopoulos, 2021].

These regimens can also be used as a backbone for novel therapies in patients with TI-NDMM.

The recommended treatment for patients who are not eligible for ASCT involves triplet induction regimens aiming to achieve remission, followed by maintenance therapy until PD. Triplets have been shown to be more effective than doublet regimens in producing deeper and more durable responses with prolonged survival time. The VRd regimen has been a widely accepted clinical SOC regimen in TI-NDMM [Mikhael, 2020; Dimopoulos, 2021]. VRd has been found to be highly active compared to Rd doublet treatment, resulting in a significant increase in median PFS from 29 to 41 months [Durie, 2017; Durie, 2020]. Furthermore, the SWOG S0777 Phase 3 clinical study demonstrated that VRd was superior to Rd, with a median OS reported as 69 months for Rd and not reached for VRd [Durie, 2020].

Recently, the combination DRd has become an alternative SOC for patients with TI-NDMM. This is based on the findings of the MAIA Phase 3 clinical study which evaluated DRd against Rd doublet treatment [Facon, 2019]. After a median follow-up of 56.2 months (IQR: 52.7 to 59.9), median PFS was not reached (95% CI: 54.8, NR) in the DRd group vs. 34.4 months (29.6 to 39.2) in the control group (HR 0.53 [95% CI: 0.43, 0.66]; p<0.0001) [Facon, 2021]. At a median follow-up of 47.9 months (IQR: 44.2 to 51.3), MRD negativity rate was 31% in the DRd group vs. 10% in the control group (overall response: 3.91 [95% CI: 2.62, 5.84]; p<0.0001). Furthermore, median OS was not reached in either group (DRd group, [95% CI: NR, NR]; control group, [95% CI: 55.7, NR]; HR: 0.68 [95% CI: 0.53, 0.86]; p=0.0013). Due to the MAIA data, NCCN

Guidelines now includes DRd as a Category 1 recommended treatment regimen for patients with TI-NDMM and EHA/ESMO guidelines recommend DRd with Level 1A evidence.

2.2.2. Antibody drug conjugate belantamab mafodotin

Belantamab mafodotin (GSK2857916, BLENREP) is an ADC currently in development for the treatment of MM. Belantamab mafodotin is a BCMA-directed antibody conjugated to a microtubule disrupting agent, MMAF with antitumor activity in MM cells. Belantamab mafodotin mediates the killing of tumor cells through MMAF-induced apoptosis as well as by tumor cell lysis through ADCC and ADCP.

2.2.3. Clinical experience with belantamab mafodotin in combination with SOC regimens

Results from the DREAMM-7 Phase 3 head-to-head clinical study evaluating belantamab mafodotin combined with bortezomib plus dexamethasone (BVd) vs. daratumumab plus Vd (DVd) in second line and later treatment of RRMM showed that the BVd combination nearly tripled median PFS vs. DVd. A statistically significant and clinically meaningful improvement was observed with BVd, showing a 59% reduction in the risk of disease progression or death (HR: 0.41, p-value <0.00001) compared to DVd. With a median follow-up of 28.2 months, the median PFS was 36.6 months (95% CI: 28.4, NR) with BVd compared to 13.4 months (11.1 to 17.5) in the DVd arm. The MRD negative rate (sensitivity of 10^{-5}) in participants treated with BVd was more than double that in participants treated with DVd (24.7% vs. 9.6%). The positive results of the DREAMM-7 clinical study displayed the synergistic efficacy and manageable toxicities of combining belantamab mafodotin with SOC therapies. Given these results, a head-to-head investigation of belantamab mafodotin in combination with SOC compared with daratumumab in combination with SOC in the front-line setting is of clinical importance [Hungria, 2024].

Positive results from the DREAMM-8 Phase 3 head-to-head study evaluating belantamab mafodotin, in combination with the immunomodulatory agent pomalidomide plus dexamethasone (Pd), vs. bortezomib plus Pd, as a second line and later treatment for RRMM has shown that 1.9 mg/kg belantamab mafodotin combination significantly extended the time to disease progression or death vs. the SOC combination [Dimopoulos, 2024]. At a median follow-up of 21.8 months (range, <0.1 to 39.2), the 12-month estimated PFS with BPd was 71% (95% CI, 63, 78), as compared with 51% (95% CI, 42, 60) with PVd (HR for disease progression or death, 0.52; [95% CI, 0.37, 0.73; p<0.001]). The percentage of participants with a response to treatment (PR+) was 77% (95% CI, 70, 84) in the BPd group and 72% (95% CI, 64, 79) in the PVd group; 40% (95% CI, 32, 48) and 16% (95% CI, 11, 23), respectively, had CR+. The study also shows a positive OS trend favoring the belantamab mafodotin combination was observed at the time of this analysis and continues to be followed up for OS.

Preliminary data recently published from the DREAMM-9 (NCT04091126) ongoing Phase 1, open-label, randomized study of belantamab mafodotin in combination with VRd (BVRd) in participants with TI-NDMM, has shown that this combination, being evaluated in 7 belantamab mafodotin dose cohorts [Cohort 1 (1.9 mg/kg Q3/4W), Cohort 2 (1.4 mg/kg Q6/8W), Cohort 3 (1.9 mg/kg Q6/8W), Cohort 4 (1.0 mg/kg Q3/4W), Cohort 5 (1.4 mg/kg Q3/4W), Cohort 6 (1.4 mg/kg QW9 to 1.0 mg/kg Q9/12W), and Cohort 7 (1.9 mg/kg QW9 to 1.4 mg/kg[-] Q9/12W)] has the potential to improve clinical outcomes compared to current SOC in TI-NDMM by dosing belantamab mafodotin at longer intervals (Q9W and Q12W), which would make the treatment as effective and more tolerable. Participants received BVRd for the first 8 cycles and continued on BRd for the remainder of time on study. As of 04 March 2024, results from dose level Cohorts 6 and 7 showed a significant decrease in participants presenting with Grade 3 and 4 ocular findings per the KVA scale, (7% of participants in each cohort) while maintaining a good ORR, (Cohort 6=87% ORR, of which 73% were VGPR+ and 27% CR+/MRD[-]; Cohort 7=71% ORR, of which 59% were VGPR+, and 29% CR+/MRD[-]). Cohort 8 (1.0 mg/kg Q12W) is also currently being evaluated in this study.

Similarly, the dose, schedule, safety, and efficacy of BRd is also being evaluated in Phase 1/2, open-label, single center collaborative study in participants with treatment-naive, TI-NDMM. Preliminary efficacy data at a median follow-up of 20.3 months demonstrated an ORR of 100% across the 2.5 mg/kg Q8W, 1.9 mg/kg Q8W and 1.4 mg/kg Q8W cohorts [Terpos, 2024]. The overall population had a median time to first response of 1 month (range: 0.9 to 3.8), median time to CR+ of 13.4 months (range: 2.8 to 24.8), and median time to VGPR+ of 11.9 months (range: 2.8 to 24.8). Thirty participants (83.0%) achieved VGPR+ and 19 participants (52.8%) achieved CR+. No disease progression occurred at median follow-up of 20.3 months.

2.3. Benefit-risk assessment

Information regarding the known and expected benefits and adverse reactions associated with belantamab mafodotin are provided in the current version of the IB [GSK2857916 Investigator's Brochure]. A summary of risk assessment for potential overlapping toxicities with the combination of BRd are described in Table 6.

Details on risks for lenalidomide, dexamethasone, and daratumumab can be found in respective prescribing information; refer to the local/regional prescribing information (where available) for daratumumab, lenalidomide, and dexamethasone, for further information.

2.3.1. Risk assessment

Table 6 Summary of risk assessment

Potential Risk of Clinical Significance	Summary of Data/Rationale for Risk	Mitigation Strategy
Potential overlapping toxicities of belantamab mafodotin/lenalidomide/dexamethasone (treatment Arm A)		
Thrombocytopenia	Belantamab mafodotin may cause transient thrombocytopenia in some participants which, for most cases, recovered between doses. In DREAMM-2, which included participants treated with belantamab mafodotin 2.5 mg/kg, thrombocytopenia was noted in 38% of participants and ranged between Grade 1-4 in severity. Lenalidomide can cause significant thrombocytopenia.	Routine monitoring of hematologic panels as outlined in SoA. (Section 1.3). Supportive therapy per local medical practice (e.g., platelet transfusion). Dose modification guidelines are outlined in Section 6.6.
Increased infections due to immunosuppression or neutropenia	**Belantamab mafodotin:** In nonclinical studies, belantamab mafodotin has been associated with decrease in immunoglobulins in monkeys, at all doses. An increase in immunoglobulins was seen in rats (rats are not an antigen-specific species for belantamab mafodotin). Immunosuppression is frequently associated with an increased risk of infection. Serious and nonserious infections have been reported in belantamab mafodotin studies, including respiratory infections, pneumonia, and sepsis. Neutropenic events, including febrile neutropenia have been observed with belantamab mafodotin. **Dexamethasone**: Patients who are on corticosteroids are more susceptible to infections or exacerbation of latent infections than healthy individuals. Chickenpox and measles can have a more serious or even fatal course in pediatric and adult patients on corticosteroids. In a study of belantamab mafodotin in combination with lenalidomide/dexamethasone, 2 fatal cases of severe infections associated with neutropenia were observed.	Participants with an active infection are excluded. Monitoring for infections and immediate treatment of immunosuppression according to standard practice. Routine monitoring of hematologic panels as outlined in SoA. (Section 1.3). Supportive therapy per local medical practice (e.g., growth factors). Prophylactic antibiotics, per local institutional guidance, in participants with Grade 3-4 neutropenia. Immediate hospitalization of participants with febrile neutropenia. Dose modification guidelines are outlined in Section 6.6.

Potential Risk of Clinical Significance	Summary of Data/Rationale for Risk	Mitigation Strategy
Keratopathy (changes to the corneal epithelium, potentially resulting in vision changes)	**Belantamab mafodotin**: Changes in corneal epithelium on ocular examination have been frequently observed with belantamab mafodotin and was most commonly associated with keratopathy (changes in the corneal epithelium upon examination), blurred vision, dry eyes, photophobia, and changes in VA. Participants with a history of dry eye were more prone to develop changes in the corneal epithelium. Based on available follow-up data, vision returned to, or near baseline in most cases. **Dexamethasone:** Use of corticosteroids may produce posterior subcapsular cataracts, glaucoma with possible damage to the optic nerves, and may enhance the establishment of secondary ocular infections due to bacteria, fungi, or viruses.	Active monitoring of the corneal epithelium and VA as outlined in the SoA (Section 1.3). Evaluation and management by an eye care professional. Dose modification guidelines are outlined in Section 6.6.
Risks related to belantamab mafodotin (not listed under potential overlapping toxicity)		
Pneumonitis	Nonclinical safety experiments have demonstrated the presence of progressive microscopic changes in the lungs (prominent alveolar macrophages associated with eosinophilic material; mixed perivascular/neutrophilic inflammation) in rats at all doses tested. Cases of pneumonitis, including fatal events, have been observed with belantamab mafodotin although a causal association has not been established.	Monitoring for clinical signs and symptoms related to pulmonary toxicity. If a participant experiences new or worsening pulmonary symptoms, (e.g., cough, dyspnea) without obvious etiology, further diagnostic tests and management should be performed as described in Section 6.6.and further treatment with belantamab mafodotin delayed (refer to Section 6.6). An overall benefit/risk assessment should be considered for the participant prior to continuing belantamab mafodotin treatment. Further diagnostic tests and management will be implemented immediately in cases of suspected pneumonitis as described in Section 6.6.4.

Potential Risk of Clinical Significance	Summary of Data/Rationale for Risk	Mitigation Strategy
Nephrotoxicity	Nonclinical safety experiments have demonstrated primary glomerular injury and tubular degeneration/regeneration (in rat and monkey). These morphologic changes were accompanied by large molecular weight proteinuria (albuminuria) and enzymuria. Single cell necrosis of the kidney and bladder urothelium was also noted in the chronic study. The renal changes were dose-dependent and reversible. Severe tubular degeneration/regeneration and marked glomerulonephritis as a result of immune complex disease associated with ADA led to the early euthanasia of 1 monkey following 5 weekly doses of 10 mg/kg. Increase albumin/creatinine ratio (albuminuria) has been reported in participants receiving belantamab mafodotin not indicative of disease progression, and in such cases, appropriate monitoring and dose modification should be considered.	Kidney function monitoring including albumin/creatinine ratio. Education of participants on the need to maintain adequate urinary output. Dose modification criteria for increased serum creatinine and urinary albumin/creatinine ratio are provided in Section 6.6.4.
Risks related to daratumumab		
Interference with serological testing	Daratumumab binds to CD38 on RBCs and results in a positive indirect antiglobulin test, which may persist for up to 6 months after the last daratumumab infusion. Daratumumab bound to RBCs masks detection of antibodies to minor antigens in the patient's serum. Determination of ABO and Rh blood type not impacted.	Participants receiving daratumumab will be typed and screened prior to daratumumab administration.
Thrombocytopenia/ neutropenia	Daratumumab may increase neutropenia and thrombocytopenia induced by background therapy.	Hematologic panels will be routinely assessed as outlined in the SoA (Section 1.3).
Viral infections, especially herpes zoster	Cases of herpes zoster infection have been reported in patients receiving daratumumab.	Participants assigned to treatment Arm B will receive antiviral prophylaxis (for example acyclovir or other according to institutional guidelines) from C1D1 until 3 months after the completion of daratumumab treatment.
Risks related to lenalidomide (not listed under overlapping toxicity with belantamab mafodotin)		
Venous and arterial thromboembolism	Increased risk of venous thromboembolic events, such as deep vein thrombosis, pulmonary embolism, and arterial thromboembolic events such as MI and stroke has been reported in participants with MM receiving lenalidomide with dexamethasone.	Antithrombotic prophylaxis will be implemented in participants at risk, according to investigator judgment. Use of drugs that may increase thromboembolic risk (e.g., erythropoietic agents, estrogens, etc.) should be avoided, if possible.
Second primary malignancies	Higher incidences of secondary primary malignancies were observed in controlled studies with participants with MM receiving lenalidomide.	Monitor for the occurrence of new malignancies.

Potential Risk of Clinical Significance	Summary of Data/Rationale for Risk	Mitigation Strategy
Severe cutaneous reactions	Severe cutaneous reactions including SJS, TEN, and DRESS have been reported.	Consider dose interruption or discontinuation for Grade 2 or 3 skin rash. Permanently discontinue lenalidomide for Grade 4 rash, exfoliative or bullous rash, or for other severe cutaneous reactions such as SJS, TEN, or DRESS.
Hypersensitivity	Hypersensitivity, angioedema, anaphylaxis, and anaphylactic reactions have been reported.	Discontinue lenalidomide for angioedema and anaphylaxis.
TLS including fatalities	Fatal instances of TLS have been reported during treatment with lenalidomide.	Participants at risk for TLS (i.e., high tumor burden) should be monitored and appropriate precautions instituted per local practice.
Altered thyroid function	Both hypothyroidism and hyperthyroidism have been reported with lenalidomide treatment.	Monitor thyroid function, and implement treatment as clinically indicated.
Dose adjustment for renal impairment	The starting dose of lenalidomide should be adjusted for participants with renal impairment.	For participants with eGFR 30-60 mL/min/1.73 m^2, lenalidomide10 mg daily, Days 1-21 of 28-day cycle will be used. If renal impairment develops during the clinical study, dose adjustment based on the creatinine CL will be implemented and for participants on dialysis [Refer to Pharmacy Manual for lenalidomide prescribing information for guidance on Dosage and Administration].
Hepatotoxicity	Hepatic failure, including fatal cases, have occurred in patients treated with lenalidomide in combination with dexamethasone.	Only participants with well-preserved liver function per the inclusion/exclusion criteria will be enrolled. Participants with Hepatitis B and C may be enrolled if they meet the entry criteria and if they agree to the monitoring/treatment criteria. See Table 4 and Table 5. Exclusion criteria are provided in Section 5.2. Liver function tests will be monitored per SoA (Section 1.3). In case of liver abnormalities, refer to Section 7.1.1 and Section 10.6 for further details. Liver stopping criteria are provided in Section 7.1.1.
Risks related to dexamethasone (not listed under overlapping toxicity with belantamab mafodotin)		
Vaccination	Administration of live or live, attenuated vaccines is contraindicated in patients receiving immunosuppressive doses of corticosteroids.	Killed or inactivated vaccines may be administered. However, the response to such vaccines cannot be predicted. Immunizations with live/live-attenuated vaccines are not allowed in the study.
Allergic or hypersensitivity reactions	Contraindicated in patients who are hypersensitive to any components of this product. Rare instances of anaphylactoid reactions have occurred in patients receiving corticosteroid therapy.	Enrollment of participants with known hypersensitivity reactions to any component of the formulation is prohibited.

Potential Risk of Clinical Significance	Summary of Data/Rationale for Risk	Mitigation Strategy
Endocrine	Hypothalamic-pituitary adrenal axis suppression with the potential for corticosteroid insufficiency after withdrawal of treatment.	Avoid rapid withdrawal of corticosteroids if used at high doses for prolonged period of time.
Gastrointestinal	Use with caution in active or latent peptic ulcers, diverticulitis, fresh intestinal anastomoses, and nonspecific ulcerative colitis, since they may increase the risk of a perforation, or GI bleeding.	Hematologic parameters will be monitored as outlined in the SoA (Section 1.3). New cases of anemia should be fully investigated, including the possibility of GI bleeding
Musculoskeletal	Decreased bone formation and increase bone resorption. Acute myopathy has been observed with the use of high doses of corticosteroids, most often occurring in patients with neuromuscular disorders, or in patients receiving neuromuscular blocking drugs.	Ongoing monitoring of AEs.
Neuropsychiatric	Psychic derangements may appear when corticosteroids from mild symptoms to severe depression, and frank psychotic manifestations. Also, existing emotional ranging from mild symptoms to severe depression, to frank psychotic manifestations. Existing emotional instability or psychotic tendencies may be aggravated by corticosteroids.	Ongoing monitoring of AEs.
Potential for cardiotoxicity	Average and large doses of corticosteroids can cause elevation of blood pressure, sodium and water retention, and increased excretion of potassium.	Participants with significant cardiovascular risk factors will be excluded from study participation.
Study procedures		
BM aspiration/biopsy	Pain, infection, bleeding may occur after the procedure.	Participants will be treated according to institution's practice.
Incidental findings during imaging data acquisition	During the acquisition of imaging data (e.g., MRI, PET-CT), non-MM disease or drug-related clinical abnormalities could be found by the radiographer performing the exams.	All imaging scans will be reported at the site by a radiographer (non-anonymized) for non-MM disease or drug-related clinical abnormalities.

Refer to the belantamab mafodotin Investigator's Brochure and local/regional prescribing information (where available) for daratumumab, lenalidomide, and dexamethasone, for further information.

2.3.2. Benefit assessment

In DREAMM-7 and DREAMM-8 Phase 3 studies, belantamab mafodotin in combination with SOC therapies for MM (Vd in DREAMM-7 and Pd in DREAMM-8) demonstrated a statistically significant and clinically meaningful improvement in median PFS when compared to SOC triplet regimens (DVd in DREAMM-7 and PVd in DREAMM-8) for treatment of RRMM in participants who have received at least 1 prior line of therapy [Hungria, 2024; Dimopoulos, 2024]. In DREAMM-7, mPFS with BVd treatment was 36.6 months (95% CI: 28.4, NR) vs. 13.4 months (95% CI: 11.1, 17.5) with DVd treatment. In DREAMM-8, mPFS with BPd was NR vs. 12.7 months (95% CI: 9.1, 18.5) with PVd treatment. The HR for disease progression or death for BPd vs. PVd treatment was 0.52 (95% CI: 0.37, 0.73). Overall, there were no new safety signals identified in the DREAMM-7 or DREAMM-8 studies, and the profile of AEs was similar to the experience in prior DREAMM studies. Data show that both combination treatments conferred significant benefit compared to the comparator SOC. Ocular events were common but controllable by belantamab mafodotin dose modifications.

Overall, given the strong clinical activity data of belantamab mafodotin in combination with other SOC myeloma therapies in patients with RRMM, it is reasonable to hypothesize that a belantamab mafodotin-containing triplet combination with a Rd backbone could potentially improve therapeutic outcomes for the TI-NDMM population compared to the current SOC.

2.3.3. Overall benefit-risk conclusion

Considering the observed clinical activity and safety profile of belantamab mafodotin in participants with RRMM, the combination of belantamab mafodotin with Rd has the potential to provide increased clinical benefit for patients than can currently be attained with DRd in TI-NDMM.

Taking into account the measures taken to minimize risk to participants from this study, the potential risks identified in association with belantamab mafodotin in combination with Rd are justified by the anticipated benefits that may be afforded to participants with TI-NDMM.

3. OBJECTIVES, ENDPOINTS, AND ESTIMANDS

Table 7 Objectives and endpoints

Objectives	Endpoints
Primary	
To evaluate the clinical activity of BRd compared with DRd in participants with TI-NDMM	• PFS, defined as the time from the date of randomization to the date of first documented disease progression per IMWG criteria by IRC or death from any cause in the absence of progression, whichever occurs first • MRD negative status, defined as achieving MRD negativity at 10^{5} sensitivity threshold assessed by NGS at least once during the time of confirmed CR or better response per IMWG criteria by IRC
Key secondary	
To further evaluate the clinical efficacy of BRd compared with DRd in participants with TI-NDMM	• OS, defined as the time from the date of randomization until the date of death due to any cause • PFS2, defined as the time from the date of randomization to the date of documented disease progression following the first subsequent anti-myeloma therapy or death from any cause, whichever is earlier
Secondary	
To evaluate the clinical efficacy of BRd compared with DRd in participants with TI-NDMM	• CR+, defined as confirmed CR or better per IMWG criteria by IRC • VGPR+, defined as confirmed VGPR, CR or better per IMWG criteria by IRC • sMRD, defined as achieving MRD negative status at 10^{-5} sensitivity threshold assessed by NGS at least twice, a minimum of 1 year apart and with no MRD positive result in between, during the time of confirmed CR or better response per IMWG criteria by IRC • DoR, defined as the time from first documented evidence of PR or better until PD or death due to PD (among participants who achieve confirmed PR+ by IRC) • TSST, defined as time from randomization until the date of start of second subsequent line of anti-myeloma therapy (irrespective of PD) or death due to any cause, whichever is earlier
To evaluate the safety and tolerability of BRd vs. DRd in participants with TI-NDMM	• Incidence and severity of AEs and SAEs • Incidence of AEs leading to dose modifications or study intervention discontinuation • Incidence and severity of ocular findings on ophthalmic exam (changes in VA and corneal findings)
To evaluate the safety and tolerability of the study intervention based on self-reported symptomatic adverse effects in participants with TI-NDMM	• Maximum post-baseline PRO-CTCAE score for each item attribute
To evaluate and compare changes in symptoms and impact on function and HRQoL in participants with TI-NDMM	• Change from baseline in HRQoL as measured by EORTC QLQ-C30 • Change from baseline in HRQoL as measured by EORTC QLQ-MY20 (Disease Symptoms domain)

Objectives	Endpoints
To characterize the exposure to belantamab mafodotin when administered in combination with lenalidomide/dexamethasone in participants with TI-NDMM	• Plasma concentrations of belantamab mafodotin (ADC) and cys-mcMMAF
Exploratory	
To characterize the clinical efficacy of BRd vs. DRd in participants with TI-NDMM	• TTP, defined as the time from the date of randomization until the earliest date of documented PD per IMWG criteria by IRC or death due to PD • TTR, defined as the time between the date of randomization and the first documented evidence of response (PR+ per IMWG criteria by IRC) • MRD negativity rate at sensitivity threshold of 10^{-6}, defined as the proportion of participants who are MRD negative at 10^{-6} as assessed by NGS
To evaluate self-reported ocular symptomatic adverse effects of BRd in participants with TI-NDMM	• Change from baseline in self-reported ocular symptoms and related impacts as measured by OSDI
To evaluate and compare changes in HRQoL and symptoms in participants with TI-NDMM	• Change from baseline in EQ-5D-3L • Change from baseline in PGIS and change in PGIC over time
To evaluate the impact of side effects on QoL in participants with TI-NDMM	• Change from baseline in FACT-GP5
To evaluate and compare HCRU in participants with TI-NDMM	• Number of office/outpatient/hospital clinic visits by specialty • Number of emergency room/urgent care facility visits • Number and duration of inpatient hospitalizations (total nights, including duration by wards [intensive care unit vs. general ward]) • Use of supportive care medication
To assess ADAs against BRd in participants with TI-NDMM	• Overall incidence and titers of ADAs against belantamab mafodotin
To explore the mechanistic and biological relationship with belantamab mafodotin and SOC treatments	• Assess biomarkers at baseline and on-treatment, by blood-based analysis including, but not limited to, evaluating immune cell populations[1], and sBCMA
To further describe the PK of belantamab mafodotin when administered in combination with lenalidomide/dexamethasone	• Derived PK and exposure parameter (e.g., CL, Cmax, AUC) values of belantamab mafodotin (ADC) and cys-mcMMAF, as data permit
To explore the exposure-response relationship between belantamab mafodotin exposure and clinical endpoints (efficacy and safety) in participants treated with BRd	• Belantamab mafodotin exposure (e.g., concentration, Cmax, or AUC) vs. efficacy and/or safety endpoints (e.g., PFS, MRD negative status, CRR, ocular events)
To qualitatively assess Disease Symptoms and HRQoL and treatment burden from direct patient feedback	• Qualitative telephone interview(s)

1. Immunophenotyping assessments will be waived in China and samples will not be collected.

DUAL PRIMARY ESTIMANDS

The study includes dual primary endpoints PFS and MRD negative rate. The primary clinical questions of interest are:

1. Does BRd improve PFS, as assessed per IMWG criteria by an IRC, compared with DRd in patients with TI-NDMM, regardless of treatment discontinuation due to clinical PD before IRC PD or death, or due to other reasons, and regardless of subsequent anti-myeloma therapy initiated before IRC PD or death?
2. Does BRd improve MRD negative rate, as assessed per IMWG criteria, compared with DRd in patients with TI-NDMM, regardless of treatment discontinuation?

The treatment condition is BRd or DRd. The population is participants with TI-NDMM. All other estimand attributes are defined in Table 8.

Table 8 Estimands for the dual primary objectives

Estimand	Variable/ Endpoint	Intercurrent Event(s)		Population-level Summary Measure
		Description*	Handling Strategy	
Main estimand	PFS (IRC)	1. Subsequent anti-myeloma therapy started[1] 2. Study intervention discontinuation due to clinical progression[2] 3. Study intervention discontinuation due to reasons other than clinical progression[3] 4. Death or progression after ≥2 missed assessments[4]	1. Treatment policy 2. Treatment policy 3. Treatment policy 4. Hypothetical	HR of BRd relative to DRd
Main estimand	MRD (CR or better by IRC, 10^{-5} sensitivity)	1. Subsequent anti-myeloma therapy started 2. Study intervention discontinuation due to any reason	1. While on-treatment 2. Treatment policy	Difference of MRD negativity rate (ΔMRD) between BRd and DRd
PFS Supplementary 1 (S1)	PFS (IRC)	1. Subsequent anti-myeloma therapy started[1] 2. Study intervention discontinuation due to clinical progression[2] 3. Study intervention discontinuation due to reasons other than clinical progression[3] 4. Death or progression after ≥2 missed assessments[4]	1. Hypothetical 2. Treatment policy 3. Treatment policy 4. Hypothetical	HR of BRd relative to DRd

Estimand	Variable/ Endpoint	Intercurrent Event(s)		Population-level Summary Measure
		Description*	**Handling Strategy**	
PFS Supplementary 2 (S2)	PFS (IRC)	1. Subsequent anti-myeloma therapy started[1] 2. Study intervention discontinuation due to clinical progression[2] 3. Study intervention discontinuation due to reasons other than clinical progression[3] 4. Death or progression after ≥2 missed assessments[4]	**1. Composite** 2. Treatment policy 3. Treatment policy 4. Hypothetical	HR of BRd relative to DRd
PFS Supplementary 3 (S3)	PFS (IRC)	1. Subsequent anti-myeloma therapy started[1] 2. Study intervention discontinuation due to clinical progression[2] 3. Study intervention discontinuation due to reasons other than clinical progression[3] 4. Death or progression after ≥2 missed assessments[4]	1. Treatment policy **2. Composite** 3. Treatment policy 4. Hypothetical	HR of BRd relative to DRd
PFS Supplementary 4 (S4)	PFS (IRC)	1. Subsequent anti-myeloma therapy started[1] 2. Study intervention discontinuation due to clinical progression[2] 3. Study intervention discontinuation due to reasons other than clinical progression[3] 4. Death or progression after ≥2 missed assessments[4]	1. Treatment policy 2. Treatment policy 3. Treatment policy **4. Composite**	HR of BRd relative to DRd
PFS Supplementary 5 (S5)	PFS (**Investigator**)	5. Subsequent anti-myeloma therapy started[1] 6. Study intervention discontinuation due to clinical progression[2] 7. Study intervention discontinuation due to reasons other than clinical progression[3] 8. Death or progression after ≥2 missed assessments[4]	As per main estimand	HR of BRd relative to DRd
MRD Supplementary 1 (S1)	MRD (**CR or better by investigator**, 10^{-5} sensitivity)	1. Subsequent anti-myeloma therapy started 2. Study intervention discontinuation due to any reason	As per main estimand	Difference of MRD negativity rate (ΔMRD) between BRd and DRd

Estimand	Variable/ Endpoint	Intercurrent Event(s)		Population-level Summary Measure
		Description*	Handling Strategy	
MRD Supplementary 2 (S2)	MRD (CR or better by IRC, **10^{-6} sensitivity**)	1. Subsequent anti-myeloma therapy started[1] 2. Study intervention discontinuation due to any reason	As per main estimand	Difference of MRD negativity rate (ΔMRD) between BRd and DRd

* Complete intercurrent event descriptions are as follows:
1. Subsequent anti-myeloma therapy started before confirmed PD or death;
2. Study intervention discontinuation due to clinical progression before confirmed PD or death;
3. Study intervention discontinuation due to reasons other than clinical progression before confirmed PD or death;
4. Two or more missed consecutive scheduled response assessments directly before confirmed PD or death.

The rationale for the PFS primary estimand is that, interest lies in evaluating superiority regardless of study intervention discontinuation and regardless of initiation of subsequent anti-myeloma therapies, and in the absence of extended loss to follow-up, to reflect clinical practice. The supplementary estimands enable an assessment of whether the rate and nature of censoring and the method of data collection have resulted in bias.

The rationale for the MRD primary estimand is that, interest lies in evaluating superiority regardless of study intervention discontinuation to reflect clinical practice, and in the absence of confounding due to other anti-myeloma therapies.

Additional estimands may be described in the SAP.

SECONDARY ESTIMANDS

Estimands for the key secondary objectives are defined in Table 9. For these estimands, the treatment condition and population are the same as for the primary PFS estimand.

Additional estimands for OS are described in the SAP.

Table 9 Estimands for the key secondary objectives

Estimand	Variable/ Endpoint	Intercurrent Event(s)		Population-level Summary Measure
		Description*	Handling Strategy	
Main estimand	OS	1. Subsequent anti-myeloma therapy started prior to PFS event[1] 2. Study intervention discontinuation due to any reason	1. Treatment policy 2. Treatment policy	HR of BRd relative to DRd
Main estimand	PFS2	1. Second subsequent line of anti-myeloma therapy started before progression (investigator) on the first subsequent line of anti-myeloma therapy[1] or death 2. Discontinuation of first subsequent anti-myeloma therapy before progression on the first subsequent anti-myeloma therapy[1] or death 3. Death prior to progression on first subsequent anti-myeloma therapy	1. Treatment policy 2. Treatment policy 3. Composite	HR of BRd relative to DRd

1. Subsequent anti-myeloma therapy before confirmed PD or death.

4. STUDY DESIGN

4.1. Overall design

DREAMM-10 is a randomized, open-label, active controlled, parallel group, multicenter, Phase 3 study in adults with NDMM who are ineligible for ASCT (TI-NDMM). The primary objective is to evaluate the efficacy and safety of BRd compared with DRd (see Table 7).

- Approximately 520 participants will be centrally randomized to either receive BRd or DRd. Randomization will be stratified by age (<75 vs. ≥75 years), International Staging System (I vs. II vs. III) and region (North America vs. Rest-of-World). ISS stage should be determined as per central lab results available *at the randomization visit.* Within each stratum, participants will be randomized using an equal allocation ratio of 1:1. No cross-over will be allowed.
- The study will consist of a Screening Period, Treatment Period, and Follow-up Period (Figure 1). The 28-day Screening Period is initiated upon signature of the ICF. During the 28-day Screening Period, participants will be evaluated for study eligibility per protocol as defined in the inclusion/exclusion criteria. Eligible participants must have a confirmed diagnosis of TI-NDMM with measurable disease and ECOG PS ≤2.
- During the Treatment Period, safety and disease assessments will be performed regularly according to the SoA for each arm. Treatment will continue in both arms until PD, death, unacceptable toxicity, withdrawal of consent, or EOS, whichever occurs first. Dose interruptions or reductions may be required to address potential drug-associated toxicities.
- Response will be assessed according to the IMWG criteria [Kumar, 2016] by the investigator as well as by an IRC. For participants who discontinue study intervention for reasons other than confirmed PD, disease evaluations will continue to be performed Q4W (±5 days) until confirmed PD, death, withdrawal of consent, loss to follow-up, or end of the study whichever occurs first. Upon confirmed PD participants will be followed to ascertain subsequent anti-myeloma therapy and progression on new therapy (PFS2), and survival status Q12W (±14 days) until withdrawal of consent, loss to follow-up, death, or the end of the study.
- GSK may request that updated survival data be collected on all randomized participants outside the protocol window noted in the SoA (Section 1.3). At the time of the request, the site will determine the survival status for each participant by the method agreed with the participant, unless the participant has withdrawn consent for survival follow-up. Where permitted, data from publicly available death registers will be used for participants withdrawn from study or lost to follow-up prior to death.
- An IDMC will be established to provide independent review and assessment of the efficacy and safety data in a systematic manner and to safeguard the interest and safety of the participants in the study. The composition and responsibilities of the committee will be described in the IDMC Charter.

- After final analysis, participants who are still receiving benefit from study intervention may transition to either, PACT, an extension study, a rollover study, and/or expanded access route.
- Participants may move into the PACT phase where the study remains open only to provide continued access to study intervention(s) for study participants who are continuing to derive clinical benefit. At that time, collection of new data for participants who no longer receive study intervention(s) will stop entirely and the clinical trial database will be closed. Participants in survival follow-up will be considered to have completed the study. Those participants still benefiting from study intervention(s), in the opinion of their treating physician, may continue to receive study intervention(s) and only SAEs, AEs leading to study intervention discontinuation, overdose, pre-specified ocular data and pregnancy cases will be reported directly to the sponsor.

For definition of EOS see Section 4.4.

Note: The start of a cycle is defined by when a participant takes the first dose of a drug for the given cycle. This study is assessing 3 drugs given in combination; thus, within the context of this protocol "cycle" refers to the planned cycle of doses for the 3 drugs per their individual planned dosing schedules.

Cycle length for each drug in reference to C1D1:

- Each belantamab mafodotin cycle is 56 days for the first 24 weeks and 84 days thereafter
- Each lenalidomide cycle is 28 days (i.e., 4 weeks)
- Each dexamethasone cycle is 28 days (i.e., 4 weeks)
- Each daratumumab cycle is 28 days (i.e., 4 weeks)

If administration of 1 or 2 of the drugs are delayed for any reason, the dosing of the other drug(s) should continue within the cycle as per the planned dosing schedule. If drug(s) that were delayed can be administered again during an ongoing planned cycle, dosing would resume according to what day it is relative to Day 1 of the planned cycle, and only the remaining planned doses for that cycle would be given. See Section 6.6 for dose modifications guidelines for management of common toxicities associated with the study interventions.

Type and number of participants

Approximately 520 participants will be randomized into the study in a 1:1 ratio between the 2 arms.

4.2. Scientific rationale for study design

Based on clinical data from the DREAMM-7 and DREAMM-8 studies, which demonstrated high ORR, together with rapid time to first response and durability of response for belantamab mafodotin in combination with SOC therapies in participants with RRMM, as well as nonclinical in vivo data which showed that both lenalidomide and bortezomib separately synergizes the anti-myeloma activity of belantamab mafodotin [GSK2857916 Investigator's Brochure], the combination belantamab mafodotin plus Rd has been selected for investigation in the TI-NDMM population.

DRd is an alternative SOC for patients with TI-NDMM based on the findings of the MAIA Phase 3 study which showed that DRd treatment had superior efficacy compared to doublet Rd [Facon, 2019; Facon, 2021]. Based on the MAIA study, the NCCN Guidelines now includes DRd as a Category 1 recommended treatment regimen for patients with TI-NDMM and EHA/ESMO guidelines recommend DRd with Level 1A evidence.

The study's dual primary objectives are to demonstrate superiority of BRd over DRd in terms of MRD negative rate and/or PFS.

Achieving MRD negative status has been identified as a strong prognostic factor in MM studies, independent of treatment [Perrot, 2018]. The IMWG has defined criteria for MRD negativity as a response category [Kumar, 2016]. In a meta-analysis conducted to examine the effect of MRD on the prognosis of MM [Munshi, 2020], it was found that in over 8 000 cases from 44 studies, the MRD negative group (3 111 cases) had significantly extended PFS compared to the MRD positive group (4 987 cases) (HR 0.33 [95% CI: 0.29, 0.37]; p<0.001).

Furthermore, Munshi et al. performed an analysis involving over 4000 cases from 23 studies, the MRD negative group (1605 cases) had significantly extended OS compared to the MRD positive group (2692 cases) (HR 0.45 [95% CI: 0.39, 0.51]; p<0.001). When the analysis was focused to TI-NDMM, the MRD negative group had significantly extended PFS (HR 0.32 [95% CI: 0.27, 0.39]; p<0.001) and OS (HR 0.40 [95% CI: 0.31, 0.51]; p<0.001) compared to the MRD positive group. A similar trend was observed in transplant eligible NDMM (PFS; HR 0.33 [95% CI: 0.28, 0.40]; p <0.001, OS; HR 0.50 [95% CI: 0.42, 0.59]; p<0.001). In RRMM, those who achieved MRD negativity had significantly extended PFS (HR 0.34 [95% CI: 0.24,0.47]; p<0.001) and OS (HR 0.28 [95% CI: 0.18, 0.45]; p<0.001).

These results indicate that for both transplant eligible and not eligible, and regardless of initial or salvage treatment, achieving MRD negativity leads to superior PFS and OS. In this study, we aim to detect a 20% difference in the MRD negative rate between the BRd and DRd groups. Upon investigating clinical studies that measured the MRD negative rate with a threshold of 10^{-5}, similar to this study, it was found that regardless of the mechanism of action of the drug, a statistically significant difference in PFS was shown when the difference in MRD negative rate was 13% to 19%, and consistent results were shown in subsequent OS evaluations. In addition, consistent results were obtained in all studies with the MRD negative rate and the response rate of CR or higher, and VGPR or

higher. In the study where the difference in the MRD negative rate was 3%, no extension of PFS and OS was observed.

While MRD negativity has been shown to predict improved PFS [Avet-Loiseau, 2020; Ficek, 2023] and OS [Munshi, 2020], PFS remains the core clinical outcome for patients being treated for NDMM. Anderson et al published a summary of the recommendations of the American Society and Hematology/US FDA Workshop on Clinical Endpoints in MM in 2008, describing OS, TTP, and PFS as the desirable clinical endpoints in studies for NDMM [Anderson, 2008]. In that summary, it was noted that improvements in therapy have lengthened OS by years, and the continued development of salvage therapies likely acts as a confounder to the OS endpoint. PFS is a well described endpoint in clinical studies for patients with MM [Mohyuddin, 2021]. The continued improvement in OS for patients with MM has made PFS an integral endpoint for studies for newly diagnosed patients in particular. Cartier et al performed a meta-analysis of 21 studies in patients with MM (14 of which were studies in NDMM, 4 studies of maintenance, and 3 studies in RRMM) which showed a positive association of treatment effect on PFS and OS [Cartier, 2015]. For this reason, PFS is retained as a dual primary endpoint, and the clinical study will continue regardless of statistically significant improvement in MRD negativity.

4.2.1. Participant input into design

Participant input was obtained to minimize participant burden associated with study participation and to provide input on study outcomes that are considered most important to participants. The study design was shared with the GSK Multiple Myeloma Patient Council to obtain feedback on the design.

It is GSK's intent to continually engage participants whose feedback may influence future study designs for MM and may improve the participant experience within clinical research.

4.3. Justification for dose

4.3.1. Belantamab mafodotin dose

The dosing regimen for this study will be belantamab mafodotin 1.9 mg/kg Q8W for first 24 weeks (Cycles 1 to 3 Q8W), then 1.9 mg/kg (Cycles 4+ Q12W) thereafter.

The dose, schedule, and management guidelines for TI-NDMM participants in the DREAMM-10 study are justified from the totality of data across the DREAMM program and multiple belantamab mafodotin collaborative studies, as well as population PK and E-R analyses.

DREAMM-9 is a GSK-sponsored dose exploration and optimization study in participants with TI-NDMM, evaluating belantamab mafodotin in addition to the SOC VRd backbone (BVRd) over a wide range of doses (1.0 to 1.9 mg/kg) and schedules (Q3/4W, Q6/8W, Q9/12W, Q12W) with fixed or step-down dosing. As of 04 March 2024, 105 participants

have been enrolled in the study and were evaluated on an intention-to-treat basis, with 10 to 17 participants per dosing cohort.

Overall, DREAMM-9 demonstrated high efficacy across all cohorts, with a trend towards deeper responses (CRR and MRD[-]) in the 1.9 mg/kg cohorts compared to 1.4 mg/kg and 1.0 mg/kg cohorts. While extending the dosing interval from Q3/4W to Q6/8W did not seem to impact efficacy (as assessed by CRR) at doses of 1.9 mg/kg or 1.4 mg/kg, a trend was observed for reduced efficacy with step-down regimen to 1.4 or 1.0 Q9/12W. Clinically meaningful changes in VA (defined as Grade ≥3 BCVA changes) were mitigated in cohorts with less frequent dosing of Q9/12W and dose reduction at a KVA Grade ≥2 corneal event, with especially low rates in cohorts that initiated dosing Q9W and stretched to Q12W after 6 months of therapy (Cohorts 6 and 7). Data also indicate that while most participants (at highest doses) experience a clinically meaningful VA change at some point in their treatment, the percentage of KVA Grade ≥2 corneal events is significantly lower. Measured as a percentage of total assessments, it suggests that participants, once they experience a treatment-emergent change to their VA, can be effectively managed via dose holds/reductions and continue receiving therapy for an extended period of time.

The EAE SCS study is evaluating belantamab mafodotin in addition to the SOC Rd backbone (BRd) in TI-NDMM participants [Terpos, 2024]. As of 05 June 2023, 36 participants were enrolled in 3 cohorts: 2.5 mg/kg, 1.9 mg/kg, and 1.4 mg/kg, all dosed Q8W and reduced to Q12W at a KVA Grade ≥2 corneal event.

Overall, the EAE SCS study demonstrated high efficacy across all cohorts, with a trend towards deeper responses (MRD[-]) in the 2.5 mg/kg and 1.9 mg/kg cohorts compared to 1.4 mg/kg cohort. Symptomatic changes in VA as reflected by Grade ≥3 BCVA changes were comparable across all 3 cohorts and were similar to the rates observed for the same doses/schedules in the DREAMM-9 study.

DREAMM-6 is a GSK-sponsored dose exploration and optimization study in participants with RRMM that evaluated belantamab mafodotin in addition to lenalidomide and dexamethasone in Arm A (n=45). Overall, the efficacy of cohorts evaluating 1.9 mg/kg (Q4W or Q8W) was comparable to the efficacy of cohorts evaluating 2.5 mg/kg (Q4W or split dosing Day 1/Day 8 Q4W). Extensive PK collection for belantamab mafodotin and lenalidomide showed that belantamab mafodotin and cys-mcMMAF PK were not impacted by co-administration of lenalidomide and dexamethasone and that lenalidomide PK was not impacted by co-administration of belantamab mafodotin.

The PK of belantamab mafodotin is associated with participant baseline characteristics (i.e., IgG, sBCMA, albumin, body weight) and was found to be similar in participants with TI-NDMM (DREAMM-9) and participants with RRMM. Belantamab mafodotin CL was reduced in relation to changes in disease burden on average by 44% over time. The average elimination half-life increased from 13 days to 19 days. The time to 50% change in CL was 66 days with time to 95% change of approximately 25 weeks.

The E-R analyses based on all participants across all cohorts in DREAMM-9 showed a positive trend between Cycle 1 belantamab mafodotin exposure and probability of achieving a response of CR+. In addition, step-down Q9/12W (Cohorts 6 and 7) and Q12W (Cohort 8c) schedules were significantly associated with reduced probability of achieving a response of CR+ (p=0.002) and longer time to response (p<0.001) independent of Cycle 1 belantamab mafodotin exposure. The E-R analyses for ocular safety showed that the probability of Grade ≥2 or Grade ≥3 corneal event (KVA scale) was significantly associated with schedules but not with Cycle 1 belantamab mafodotin exposure. Step-down Q9/12W and Q12W schedules were significantly associated with lower probability and occurrence of Grade ≥2 corneal events. In addition, the probability of Grade ≥2 ocular AESI (CTCAE) was not significantly associated with schedule or Cycle 1 belantamab mafodotin exposure. Overall, the E-R analyses suggest that higher early belantamab mafodotin exposure may provide higher probability of deeper response with no associated increase in probability of ocular event.

Based on the totality of the above information, belantamab mafodotin will be administered at 1.9 mg/kg Q8W until Week 24, then 1.9 mg/kg Q12W thereafter. If dose modifications are required, dosing schedule will be followed as per Section 6.6.1.

4.3.2. Daratumumab dose

Daratumumab will be administered at the approved dose regimen, according to the most recently approved country-specific label.

Daratumumab will be administered at 1800 mg SC (fixed dose) in 28-day cycles as per the schedule in Table 10. If a daratumumab administration does not commence within the pre-specified window of the scheduled administration date, then the dose will be considered a missed dose. Administration may resume at the next planned dosing date.

Table 10 Daratumumab-related toxicity management

Cycles (Weeks)	Schedule	Dose held	Dosing restart
1 and 2 (Weeks 1-8)	Q1W (total 8 doses)	>3 days	Next planned dose Q1W dosing date
3-6 (Weeks 9-24)	Q2W (total 8 doses)	>1 week	Next planned dose Q2W dosing date
7+ (Week 25 onwards until disease progression)	Q4W	>2 weeks	Next planned dose Q4W dosing date

For further information, refer to the local/regional prescribing information (where available) for daratumumab.

4.3.3. Lenalidomide dose

The dosing regimen of lenalidomide in TI-NDMM will follow approved schedules. Lenalidomide treatment will be administered QD for 3 weeks followed by 1 week without lenalidomide treatment. Lenalidomide will be administered according to the most recently approved country-specific label.

In Arm A and Arm B, lenalidomide will be administered at 25 mg once a day PO on Day 1 to Day 21 of every 28-day lenalidomide treatment cycle (Table 12).

For participants with eGFR 30 to 60 mL/min/1.73m^2, lenalidomide should be administered at 10 mg on the days indicated above.

For further information, refer to the local/regional prescribing information (where available) for lenalidomide.

4.3.4. Dexamethasone dose

The dosing regimen of dexamethasone in TI-NDMM will follow approved schedules. Dexamethasone treatment will be administered QD on the first day of every week. Dexamethasone will be administered at the approved dose regimen, according to the most recently approved country-specific label. Dexamethasone will be administered according to the regimen approved for use in combination with pomalidomide or daratumumab (for further information, refer to the local/regional prescribing information [where available] for dexamethasone, pomalidomide, and daratumumab.

In Arm A and Arm B dexamethasone will be administered PO at a dose of 40 mg on Days 1, 8, 15, and 22 of each 28-day cycle (Table 12). For participants who are ≥75 years old, are underweight (BMI <18.5), have poorly controlled diabetes mellitus, or have prior intolerance/AE to steroid therapy, dexamethasone may be administered at the lower dose of 20 mg at the discretion of the investigator. Refer to Section 6.6.6 for instructions on dose modification guidelines.

4.4. End-of-study definition

The EOS is defined as the date of the last scheduled procedure in the SoA (Section 1.3) for the last participant enrolled in the study globally.

EOS will be when the last participant on study has completed their last study visit, including any necessary follow-up for all primary and secondary endpoints. For the evaluation of OS, the last participant should be in follow-up for a minimum of 5 years.

A participant will be considered to have completed the study if the participant dies during the Treatment Period or Follow-up Period or has been in follow-up for a minimum of 5 years, whichever is sooner. In addition, participants who are in the Follow-up Period at the time of the sponsor's decision to close the study will be considered to have completed the study. Participants who are receiving study interventions may continue to receive study interventions through PACT, an extension study, a rollover study, and/or expanded

access route if they are gaining clinical benefit as assessed by the investigator until they meet any protocol-defined study intervention discontinuation criteria.

4.4.1. Final data cut-off

A final DCO representing the end of data collection, prior to the EOS, is defined as the DCO date for the final analysis. Following the final DCO date, the study may move into PACT, an extension study, a rollover study, and/or expanded access route and the clinical study database will be closed to new data. Participants who are receiving study intervention may continue to receive study intervention if they are gaining clinical benefit as assessed by the investigator until they meet any protocol-defined treatment discontinuation criteria. Participants in survival follow-up at the time of the final DCO date will be considered to have completed the study.

Although the clinical study database will be closed at the time of the final DCO date, for participants who receive continued study intervention during the PACT phase, the study will remain open until all participants have discontinued study intervention and completed the 70-day safety follow-up and the EOS definition has been reached. For participants who are benefiting from study intervention and who are transferred to a rollover study, and/or expanded access route, the study will not move into the PACT phase and EOS will be reached after final DCO.

5. STUDY POPULATION

Prospective approval of protocol deviations to recruitment and enrollment criteria, also known as protocol waivers or exemptions, is not permitted.

The study will enroll adult participants with NDMM. Full criteria are included below.

5.1. Inclusion criteria

Participants are eligible to be included in the study only if all of the following criteria apply:

Age

1. Is at least 18 years or the legal age of consent in the jurisdiction in which the study is taking place, at the time of signing the informed consent.

Informed consent

2. Capable of giving signed informed consent as described in Section 10.1.3, which includes compliance with the requirements and restrictions listed in the ICF and in this protocol.

   NOTE: In France a participant will be eligible for inclusion in this study only if either affiliated to or a beneficiary of a social security category.

Diagnosis and disease status

3. Newly diagnosed MM with a requirement for treatment as documented per IMWG criteria [Rajkumar, 2014].

   a) Monoclonal plasma cells in the BM ≥10% or presence of a biopsy proven plasmacytoma and documented MM satisfying at least 1 of the CRAB criteria. CRAB criteria: Hypercalcemia: serum calcium >0.25 mmol/L (>1 mg/dL) higher than ULN or >2.75 mmol/L (>11 mg/dL); Renal insufficiency: creatinine CL <40 mL/min or serum creatinine >177 µmol/L (>2 mg/dL); Anemia: hemoglobin >2 g/dL below the lower limit of normal or hemoglobin <10 g/dL; Bone lesions: 1 or more osteolytic lesions on skeletal radiography, CT), or PET-CT.

   OR

   b) Biomarkers of malignancy criteria: Clonal BM plasma cell percentage ≥60%; Involved: uninvolved serum FLC ratio ≥100; >1 focal lesion on MRI studies.

4. Must have at least **1** aspect of measurable disease, as assessed by the central laboratory, defined as 1 of the following:

   a) Urine M-protein excretion ≥200 mg/24 hours (≥0.2 g/24 hours).

   And/or

b) Serum M-protein concentration ≥0.5 g/dL (≥5.0 g/L)

And/or

c) Serum FLC assay: involved FLC level ≥10 mg/dL (≥100 mg/L) and an abnormal serum FLC ratio (<0.26 or >1.65).

5. Newly diagnosed and not considered candidate for high-dose chemotherapy with ASCT due to any of the following:

   a) Exclusion from treatment with ASCT due to country- or site-specific age restriction.

   b) Presence of comorbid condition(s) likely to have a negative impact on tolerability of high-dose chemotherapy with ASCT.

6. ECOG PS of 0 to 2.

Organ system function

7. Adequate organ system function as defined by the laboratory assessments listed below.

Organ System and Laboratory Tests	Laboratory Values
Hematologic[1]	
ANC[2]	≥1.0×10^9/L
Platelets	≥75×10^9/L
Hepatic	
Total bilirubin	≤1.5×ULN; (Isolated bilirubin >1.5×ULN is acceptable if bilirubin is fractionated and direct bilirubin is <35%)
ALT/AST	≤2.5×ULN
Renal	
eGFR[3]	≥30 mL/min/1.73 m^2
Urine dipstick for protein OR Albumin/creatinine ratio (from spot urine)	Negative/trace (if 1+ only eligible if confirmed 500 mg/g (56 mg/mmol) by albumin/creatinine ratio (spot urine from first void) ≤500 mg/g (56 mg/mmol)

Note: Laboratory results obtained during Screening should be used to determine eligibility criteria. In situations where laboratory results are outside the permitted range, the investigator may retest the participant and the subsequent within range screening result may be used to confirm eligibility.

1. Patients with hematologic laboratory values below these thresholds may enroll if the investigator attests that these abnormalities are a consequence of MM. In this scenario, transfusion, growth factor support, or platelet stimulating agents are allowed.
2. Unless health care professional confirms stable baseline neutropenia <1.0×10^9/L due to Duffy-null phenotype.
3. As calculated by MDRD formula.

Contraception

8. Male participants:

   - Contraceptive use by men should be consistent with local regulations regarding the methods of contraception for those participating in clinical studies.
   - Male participants are eligible to participate if they agree to the following during the Treatment Period and for at least 6 months after the last dose of study intervention to allow for clearance of any altered sperm:
     - Refrain from donating fresh unwashed semen

PLUS either:

- Be abstinent from heterosexual intercourse as their preferred and usual lifestyle (abstinent on a long-term and persistent basis) and agree to remain abstinent.

OR

- Must agree to use contraception/barrier as detailed below
- Agree to use a male condom, even if they have undergone a successful vasectomy, and female partner to use an additional highly effective contraceptive method with a failure rate of <1% per year when having sexual intercourse with a WOCBP who is not currently pregnant. Male participants should also use a condom when having sexual intercourse with pregnant females.

9. Female participants

- Contraceptive use by women should be consistent with local regulations regarding the methods of contraception for those participating in clinical studies.
- A female participant is eligible to participate if she is not pregnant or breastfeeding, and at least 1 of the following conditions applies:
  - Is not a WOCBP

  OR

  - Is a WOCBP and using a contraceptive method that is highly effective (with a failure rate of <1% per year), preferably with low user dependency during the Treatment Period and for 4 months after the last dose of study intervention and agrees not to donate eggs (ova, oocytes) for the purpose of reproduction during this period. The investigator should evaluate the effectiveness of the contraceptive method in relationship to the first dose of study intervention.
  - A WOCBP must have 2 negative highly sensitive serum pregnancy tests before starting treatment, the first may be performed within 14 days from C1D1, the second within 24 hours before the first dose of study intervention.

Should pregnancy occur in a female on-treatment or the female partner of a male on-treatment, treatment must be stopped, and it is advised to seek advice from a physician specialized or experienced in teratology.

Additional requirements for pregnancy testing during and after study intervention are described in Section 10.4 and SoA (Section 1.3). The investigator is responsible for review of medical history, menstrual history, and recent sexual activity to decrease the risk for inclusion of a woman with an early undetected pregnancy.

Lenalidomide is known or expected to be teratogenic and use is contraindicated in pregnancy. Conditions of the Pregnancy Prevention Program must be fulfilled unless there is reliable evidence that the participant does not have childbearing potential (refer to the local/regional prescribing information (where available) for lenalidomide for further information).

In addition to the pregnancy testing and contraception requirements for participants treated with belantamab mafodotin, WOCBP receiving lenalidomide, must commit either to abstain continuously from heterosexual sexual intercourse or to use 2 methods of reliable birth control simultaneously: 1 highly effective form of contraception - tubal ligation, intrauterine device, hormonal (birth control pills, injections, hormonal patches, vaginal rings, or implants), or partner's vasectomy, and 1 additional effective contraceptive method - male latex or synthetic condom, diaphragm, or cervical cap. Contraception must begin 4 weeks prior to initiating treatment with lenalidomide, during therapy, during dose interruptions, and continuing for 4 weeks following discontinuation of lenalidomide therapy. Reliable contraception is indicated even where there has been a history of infertility, unless due to hysterectomy. WOCBP should be referred to a qualified provider of contraceptive methods, if needed.

Lenalidomide is present in the semen of participants receiving the drug. Therefore, males must always use a latex or synthetic condom during any sexual contact with females of reproductive potential while taking lenalidomide and for up to 4 weeks after discontinuing lenalidomide, even if they have undergone a successful vasectomy. Male participants taking lenalidomide must not donate sperm during treatment and for up to 4 weeks after discontinuing lenalidomide.

5.2. Exclusion criteria

Participants are excluded from the study if any of the following criteria are met:

Participants with prior or concomitant diseases or conditions

1. Diagnosis of systemic amyloid light chain amyloidosis, Waldenstrom's disease, POEMS (polyneuropathy, organomegaly, endocrinopathy, monoclonal plasma proliferative disorder, skin changes) or Primary Plasma Cell Leukemia (defined as circulating plasma cells >5%) [De Larrea, 2021].
2. Prior systemic therapy for MM, or smoldering MM.

   NOTE: An emergency course of steroids (defined as no greater than 40 mg of dexamethasone, or equivalent per day for a maximum of 4 days (that is, a total of 160 mg) is permitted.

   NOTE: Focal palliative radiation is permitted prior to enrollment, provided that it occurred at least 2 weeks prior to the first dose of study drug, that the participant has recovered from radiation-related toxicities, and that the participant did not require greater than 40 mg of dexamethasone (or equivalent) per day over 4 days (that is, a total of 160 mg) for radiation-induced AEs.
3. Signs of meningeal or central nervous system involvement with MM.

4. Major surgery within 2 weeks prior to the first dose of study drugs or has not recovered fully from surgery. Kyphoplasty is not considered major surgery.
5. Any serious and/or unstable pre-existing medical, psychiatric disorder or other conditions (including lab abnormalities) that could interfere with participant's safety, obtaining informed consent, or compliance with study procedures.
6. Current active liver or biliary disease (except for Gilbert's syndrome or asymptomatic gallstones, or otherwise stable chronic liver disease as per the investigator's assessment).
7. Participants with previous or concurrent malignancies other than MM are excluded. Exceptions are any other malignancy that has been considered medically stable for at least 2 years, after discussion with the GSK Medical Monitor. The participant must not be receiving active therapy, other than hormonal therapy for this disease.

   NOTE: Participants with curatively treated non-melanoma skin cancer or surgically treated cervical carcinoma in situ are allowed without a 2-year restriction.
8. Evidence of cardiovascular risk including any of the following:
   a) Evidence of current clinically significant untreated arrhythmias, including clinically significant ECG abnormalities including second-degree (Mobitz Type II) or third-degree atrioventricular block.
   b) Recent history (within 3 months of screening) of MI, acute coronary syndromes (including unstable angina), coronary angioplasty or stenting, or bypass grafting.
   c) Class III or IV heart failure as defined by the NYHA functional classification system.
9. Known HIV infection, unless the participant can meet all of the following criteria:
   a) Established ART for at least 4 weeks and HIV viral load <400 copies/mL within Screening Period.
   b) CD4+ T-cell (CD4+) counts ≥350 cells/µL.
   c) No history of AIDS-defining opportunistic infections within the last 12 months.

   NOTE: consideration must be given to ART and prophylactic antimicrobials that may have a drug-drug interaction and/or overlapping toxicities with belantamab mafodotin or other combination products as relevant.
10. Positive hepatitis C antibody test result or positive hepatitis C RNA test result at screening or within 3 months prior to first dose of study intervention unless the participant can meet the following criteria:
    a) RNA test negative.
    b) Successful antiviral treatment (usually 8 weeks duration) is required, followed by a negative HCV RNA test after a washout period of at least 4 weeks.

11. Participants with hepatitis B will be excluded unless the following criteria can be met:

Serology	Screening	During Study Intervention
HBcAb+, HBsAg-	HBV DNA undetectable	Monitoring per protocol (Section 6.6.7) Antiviral treatment instituted if HBV DNA becomes detectable
HBsAg+ at screen or within 3 months prior to first dose	HBV DNA undetectable Highly effective antiviral treatment started at least 4 weeks prior to first dose of study intervention Baseline imaging per protocol Participants with cirrhosis are excluded	Antiviral treatment maintained throughout study intervention Monitoring and management per protocol (Section 6.6.7)

Note: Presence of Hep B surface antibody (HBsAb) indicating previous vaccination will not exclude a participant.

12. Current corneal epithelial disease except for mild punctate keratopathy.

    NOTE: Participants with mild punctate keratopathy are allowed.
13. Intolerance or contraindications to antiviral prophylaxis.
14. Unable to tolerate antithrombotic prophylaxis.
15. Known immediate or delayed hypersensitivity reaction or idiosyncratic reaction to drugs chemically related to belantamab mafodotin, or any of the components of the study intervention.

Prior or Concomitant Therapies

16. Plasmapheresis within 7 days prior to the first dose of study intervention.
17. Participants must not have received a live or live-attenuated vaccine within 30 days prior to first dose of belantamab mafodotin.

5.3. Lifestyle considerations

The following lifestyle restrictions apply while participants are in the study:

- Contact lenses are prohibited while on study intervention (from first dosing to the EOS intervention). Contact lenses may be restarted after the last dose of study intervention once the eye care specialist confirms there are no other contradictions.
- Use of bandage contact lenses is permitted during study intervention as directed by the treating eye care specialist.
- Participants must not donate blood when receiving lenalidomide, during dose interruptions, and for 28 days following discontinuation of lenalidomide, as transfused blood might be given to a pregnant female whose fetus must not be exposed to lenalidomide.

5.4. Screen failures

Screen failures are defined as participants who consent to participate in the clinical study but are not subsequently enrolled. Laboratory results obtained during screening should be used to determine eligibility criteria. In situations where laboratory results are outside the permitted range, the investigator may opt to retest the participant and the subsequent screening result, if within range, may be used to confirm eligibility.

A minimal set of screen failure information is required to ensure transparent reporting of screen failure participants to meet the CONSORT publishing requirements and to respond to queries from regulatory authorities. Minimal information includes demography, screen failure details, eligibility criteria, and any SAEs.

Individuals identified as Screen Failures may be rescreened if the failure was based on elements of eligibility that may change, e.g., laboratory test results. Rescreened participants should be assigned a different participant number from the 1 assigned for the initial screening. Rescreening of a participant more than once requires discussion with the Medical Monitor.

5.5. Criteria for temporarily delaying enrollment/randomization/ administration of study intervention

Not applicable.

6. STUDY INTERVENTION(S) AND CONCOMITANT THERAPY

The definition of study intervention is provided in the table of definitions.

6.1. Study intervention(s) administered

Table 11 Study intervention(s) administered

Intervention Label				
Intervention Name	**Belantamab Mafodotin)**	**Daratumumab**	**Lenalidomide**	**Dexamethasone**
Intervention description	Belantamab mafodotin is a powder for solution for infusion	Daratumumab is a solution for SC administration	Lenalidomide is capsules for oral administration	Dexamethasone is tablets for oral administration
Type	Drug	Drug	Drug	Drug
Dose formulation	Powder for solution	Colorless to yellow and clear to opalescent solution in a single-dose vial	Capsule	Tablet
Unit dose strength(s)	100 mg lyophilized powder	1800 mg and 30 000 units hyaluronidase per 15 mL (120 mg and 2000 units/mL)	5 mg, 10 mg, and 25 mg	2 mg and 8 mg
Dosage level(s)	1.9 mg/kg Q8W from Week 1-24 then 1.9 mg/kg Q12W onwards until PD	1800 mg SC (fixed dose) weekly from Week 1-8, Q2W from Week 9-24, and Q4W from Week 25 onwards until PD	25 mg PO[1], QD on Days 1-21 of every 28-day cycle	40 mg PO[2], QD on Days 1, 8, 15, 22 of every 28-day cycle
Route of administration	IV	SC	oral	oral
Use	IMP	Comparator	Co-administered product	Co-administered product
Non-IMP	Not applicable	Not applicable	Not applicable	Not applicable
Authorized AxMP/ unauthorized AxMP	Not applicable	Not applicable	Authorized AxMP	Authorized AxMP
Sourcing	Provided centrally by the sponsor	Provided centrally by the sponsor or locally by the study site	Provided centrally by the sponsor or locally by the study site	Provided centrally by the sponsor or locally by the study site
Packaging and labeling	Study intervention will be provided in vials. Each vial will be labeled as required per country requirement.	Study intervention will be provided in vials. Each vial will be labeled as required per country requirement.	Study intervention will be provided in blisters. Each blister will be labeled as required per country requirement.	Study intervention will be provided in blisters and bottles. Each blister and bottle will be labeled as required per country requirement.
Current/ former name(s) or alias(es)	Not applicable	Darzalex	Revlimid	Not applicable

Note: For participants with eGFR 30-60 mL/min/1.73m^2 lenalidomide should be administered at 10 mg on the days indicated.

Note: For participants who are ≤75 years old, are underweight (BMI <18.5), have poorly controlled diabetes mellitus, or have prior intolerance/AE to steroid therapy, dexamethasone may be administered at the lower dose of 20 mg at the discretion of the investigator.

Table 12 Study arm(s)

Arm Title	**Arm A**	**Arm B**
Arm type1	Experimental	Active comparator
Arm description	Belantamab mafodotin 1.9 mg/kg will be administered Q8W for first 24 weeks (Cycles 1-3 Q8W), then 1.9 mg/kg (Cycles 4+ Q12W) thereafter. Lenalidomide[1] will be administered at 25 mg once a day PO on Days 1-21 of every 28-day lenalidomide treatment cycle. Dexamethasone[2] will be administered PO at a dose of 40 mg on Days 1, 8, 15, and 22 of each 28-day cycle.	Daratumumab 1800 mg SC will be administered weekly from Week 1-8 (Cycles 1 and 2), Q2W from Week 9-24 (Cycles 3-6), and Q4W from Week 25 (Cycles 7+) onwards. Lenalidomide[1] will be administered at 25 mg once a day PO on Days 1-21 of every 28-day lenalidomide treatment cycle. Dexamethasone[2] will be administered PO at a dose of 40 mg on Days 1, 8, 15, and 22 of each 28-day cycle.
Associated intervention labels	Refer to the local/regional prescribing information.	Refer to the local/regional prescribing information.

1. For participants with eGFR 30-60 mL/min/1.73 m^2 lenalidomide should be administered at 10 mg on the days indicated. Of note, this is NOT classified as a dose reduction per protocol.
2. For participants who are ≥75 years old, are underweight (BMI <18.5), have poorly controlled diabetes mellitus, or have prior intolerance/AE to steroid therapy, dexamethasone may be administered at the lower dose of 20 mg at the discretion of the investigator. Of note, this is NOT classified as a dose reduction per protocol.

6.1.1. Belantamab mafodotin

6.1.1.1. Belantamab mafodotin pre/post-medication

Premedication is not required prior to infusion unless deemed medically appropriate by the investigator. Premedication should be considered in any participant who experienced an IRR at first or any subsequent infusion with belantamab mafodotin.

Corneal events, which commonly manifest as superficial microcystic keratopathy, have been observed with ADCs conjugated to MMAF.

To minimize corneal events, preservative-free artificial tears should be administered Table 13.

6.1.1.2. Corneal supportive care guidelines for belantamab mafodotin

Prophylactic interventions for mitigation/management of ocular AEs associated with belantamab mafodotin are provided below.

Table 13 Prophylactic measures for ocular events associated with belantamab mafodotin

Prophylactic Measure	Dose and Administration	Timing
Preservative-free artificial tears	Administer in each eye at least 4-8 times daily	Administer daily beginning on C1D1 until EOT
Cooling eye mask	May apply cooling eye mask to both eyes for approximately 1 hour or as much as tolerated by the participant	During belantamab mafodotin infusion administration in the first hour for up to 4 hours, as tolerated

6.1.2. Daratumumab

6.1.2.1. Antiviral prophylaxis

Antiviral prophylaxis is recommended in accordance with local prescribing information in participants being treated with daratumumab.

6.1.2.2. Interference with serological testing

Daratumumab binds to CD38 on RBCs and results in a positive indirect antiglobulin test. Daratumumab-mediated positive indirect antiglobulin test may persist for up to 6 months after the last daratumumab infusion. Daratumumab bound to RBCs masks detection of antibodies to minor antigens in the participant's serum. The determination of a participant's ABO and Rh blood type are not impacted (refer to the local/regional prescribing information [where available] for daratumumab for further information.

Notify blood transfusion centers of this interference with serological testing and inform blood banks that a participant has received daratumumab.

As specified in the SoA (Section 1.3) and Table 28, participants on-treatment in Arm B should undergo blood typing and screening prior to starting daratumumab, and should carry a card with their blood antigen profile with them at all times.

6.1.2.3. Interference with serum protein electrophoresis and immunofixation tests

Daratumumab is a human IgG kappa monoclonal antibody that can be detected on both the serum SPEP and immunofixation assays used for the clinical monitoring of endogenous M-protein. This interference can impact the determination of complete response and of disease progression in some participants with IgG kappa myeloma protein (refer to the local/regional prescribing information [where available] for daratumumab for further information.

Participants in Arm B with IgG kappa MM are required to undergo further testing with a daratumumab-specific interference assay (SoA, Section 1.3 and Section 8.2.3).

6.2. Preparation, handling, storage, and accountability

- The investigator or designee must confirm appropriate conditions (e.g., temperature) have been maintained during transit for all study intervention received, and that any discrepancies are reported and resolved before use of the study intervention.
- Only participants enrolled in the study may receive study intervention, and only authorized site staff may supply, prepare, or administer study intervention.
- All study interventions must be stored in a secure, environmentally controlled, and monitored (manual or automated) area in accordance with the labeled storage conditions, with access limited to the investigator and authorized site staff.
- The investigator, institution, or the head of the medical institution (where applicable) is responsible for study intervention accountability, reconciliation, and record maintenance (i.e., receipt, reconciliation, and final disposition records).
- Further guidance and information for the final disposition of unused study interventions are provided in the Pharmacy Manual.
- Under normal conditions of handling and administration, study intervention is not expected to pose significant safety risks to site staff. Take adequate precautions to avoid direct eye or skin contact and the generation of aerosols or mists. In the case of unintentional occupational exposure, site personnel are to notify the monitor, GSK Medical Monitor and/or GSK study contact.
- An MSDS/equivalent document describing occupational hazards and recommended handling precautions will either be provided to the investigator, where this is required by local laws, or is available upon request from GSK.
- Precautions will be taken to avoid direct contact with the study intervention. An MSDS describing occupational hazards and recommended handling precautions will be provided to the investigator. In the case of unintentional occupational exposure, site personnel are to notify the monitor, GSK Medical Monitor and/or GSK study contact.

6.3. Assignment to study intervention

All participants will be centrally randomized to study intervention using RTMS.

Study intervention will be dispensed at the study visits as summarized in the SoA (Section 1.3).

Returned study intervention should not be re-dispensed to the participants.

Participants will be assigned a unique number (randomization number) that will not be reassigned to another participant in ascending numerical order at each study site. If the participant is a screen failure, the number will remain for the duration of the study. The randomization number encodes the participant's assignment to 1 of the 2 arms of the study, according to the randomization schedule generated prior to the study by the

statistics department at GSK. Each participant will be dispensed study intervention, labeled with the participant's unique randomization number, throughout the study.

6.4. Blinding

This is an open-label study; therefore, no blinding of treatment identity is needed for either BRd or DRd arm.

However, to ensure clinical study integrity, steps will be taken to restrict access to key information while the study is ongoing and prevent data aggregation except for where specified in the protocol.

The specific intervention to be received by a participant will be assigned using RTMS. The site will contact the RTMS prior to the start of study intervention administration for each participant. The site will record the intervention assignment on the applicable eCRF.

Potential bias will be reduced via central randomization, as well as by IRC disease assessments (which will take place with no knowledge of participants treatment arm assignment).

Designated independent representative(s) may be unblinded for preparing population PK and PKPD dataset(s) and for performing PK and PKPD model development/refinement using PK and PKPD unblinded datasets. For details see Section 9.4.3.

GSK's Global Safety staff may unblind the intervention assignment for any participant with an SAE. If the SAE requires that an expedited safety report be sent to 1 or more regulatory agencies, a copy of the report, identifying the participant's intervention assignment, may be sent to investigators in accordance with local regulations and/or GSK policy.

6.5. Study intervention compliance

When participants are dosed at the site, they will receive study intervention directly from the investigator or designee, under medical supervision. The date and time of each dose administered in the clinic will be recorded in participant's CRF and the source documents. The dose of study intervention and study participant identification will be confirmed at the time of dosing by a member of the study site staff other than the person administering the study intervention.

When participants self-administer oral study intervention(s) at home, such dosing will be recorded in the Participant's Study Diary. The entries in the Diary will be assessed through querying the participant during the site visits and documented in the source documents and CRF. A record of the number of doses dispensed to and taken by each participant must be maintained and reconciled with study intervention and compliance records. Treatment start and stop dates, including dates for treatment delays and/or dose reductions will also be recorded in the CRF.

6.6. Dose modification

The severity of AEs will be graded utilizing the NCI-CTCAE v5.0 [US Department of Health and Human Services, 2017]. Guidelines for dose modifications and interruptions for management of common toxicities associated with the study interventions are provided in this section.

6.6.1. Belantamab mafodotin dose adjustments due to body weight

The actual body weight in kg at baseline (assessed on C1D1 prior to dosing) will be used for dose calculation of belantamab mafodotin in all participants during the Treatment Period. If the change of body weight (increase or decrease) is greater than 10%, the dose should be re-calculated based on the actual body weight at the time of dosing and thereafter will serve as the new baseline for weight comparison at subsequent visits. Participants must be weighed prior to dosing for each cycle.

6.6.2. Belantamab mafodotin dose modification schedule

An overview of the recommended dose modification of belantamab mafodotin is presented in Table 14. Dose modification guidelines are provided in Section 6.6.4.

If the first dose modification is required during the first 24 weeks, the dosing interval will change from Q8W to Q12W; and belantamab mafodotin dosing will not restart until at least 12 weeks have passed since the most recent dose. Should a dose modification be required after Week 24, the dose may be reduced to 1.4 mg/kg Q12W.

Table 14 Recommendations for dose modification of belantamab mafodotin

	Q8W Dosing	Q12W Dosing
Assigned dose	1.9 mg/kg	1.9 mg/kg
Modification	Delay belantamab mafodotin and restart at 1.9 mg/kg but at Q12W interval[1]	Reduce dose to 1.4 mg/kg

1. Dose should be restarted at the same dose with Q12W interval which is from the last dose of belantamab mafodotin NOT from the ocular examination.

6.6.3. Dose modification for overlapping hematological toxicities

There may be situations where the investigator will not be able to attribute hematological toxicity to a single agent. In this event full guidance is given in Table 15.

Where the relatedness of the hematological toxicity is clear e.g., due to timing of dosing, the investigator may apply judgment to reduce any of the agents as appropriate.

Table 15 Dose modification guidelines for overlapping hematological toxicities

Hematological Toxicity	Recommended Actions
Thrombocytopenia	
Interrupt all treatments until platelet count returns to 50×10^9/L. • Conduct weekly hematology assessments • Consider additional supportive treatment (e.g., transfusion), as clinically indicated and per local practice.	
Platelets count decrease to $<50\times10^9$/L at the start of new cycle or $<30\times10^9$/L at any point during cycle	Following recovery, resume lenalidomide and belantamab mafodotin at the most recent dose.
Further drop in platelets to $<50\times10^9$/L at the start of new cycle or $<30\times10^9$/L at any point during cycle	Following recovery, resume lenalidomide at the next lower dose[1]. Do not dose below 5 mg daily. Resume belantamab mafodotin at the most recent dose.
Subsequent drop in platelets to $<50\times10^9$/L at the start of new cycle or $<30\times10^9$/L at any point during cycle	Following recovery, resume lenalidomide at the most recent dose. Follow guidance for belantamab mafodotin: Day 1 Dosing: • No bleeding: if planned for 1.9 mg/kg, continue treatment at 1.4 mg/kgdose[2]. If planned for treatment at 1.4 mg/kg, withhold belantamab mafodotin until thrombocytopenia has recovered to Grade 2 or less, and restart at 1.4 mg/kg dose. • With bleeding: withhold dose. If planned for 1.9 mg/kg, treatment at 1.4 mg/kg may continue after thrombocytopenia has recovered to Grade 2 or less. If planned for treatment at 1.4 mg/kg, withhold belantamab mafodotin until thrombocytopenia has recovered to Grade 2 or less.
For additional occurrences, alternate dose modification of lenalidomide and belantamab mafodotin, as described above	
Neutropenia	
Interrupt all treatments until ANC returns to 1×10^9/L. • Conduct weekly hematology assessments • Consider prophylactic antimicrobials per physician discretion and institutional guidelines • Consider additional supportive treatment, per local practice (e.g., growth factors) • Immediate hospitalization for febrile neutropenia is required	
ANC fall to $<1.0\times10^9$/L	• Following recovery, resume lenalidomide at the next lower dose[1]; do not dose below 5 mg daily • Resume belantamab mafodotin at the most recent dose
Further drop in ANC to $<1.0\times10^9$/L	• Following recovery, resume lenalidomide at next lower dose daily and belantamab mafodotin at the most recent dose. Do not dose below 5 mg daily for Days 1-21 of 28-day cycle
Subsequent drop in ANC to $<1.0\times10^9$/L	Following recovery, resume lenalidomide at the most recent dose Follow guidance for belantamab mafodotin: <u>Day 1 dosing:</u> • If noted on Day 1 of any cycle, withhold belantamab mafodotin. Repeat hematology (CBC) as clinically indicated • Resume belantamab mafodotin at pre-held dose once neutropenia recovers to Grade ≤2 (ANC $\geq1.0\times10^9$/L) on Day 1 of the subsequent cycle
For additional occurrences, alternate dose modification of lenalidomide and belantamab mafodotin, as described above	

1. Next lower dose according to prescribing information.
2. For the belantamab mafodotin dose modification strategy, see Section 6.6.

6.6.4. Dose modification guidelines for belantamab mafodotin

See Table 16 for dose modification guidelines for AEs associated with belantamab mafodotin.

See Table 17 and Table 18 for dose modification guidelines for corneal events related to belantamab mafodotin. Determination of the recommended dose modification of belantamab mafodotin should be based on the most severe corneal events per the KVA scale.

Table 16 Dose modification guidelines for adverse events associated with belantamab mafodotin

Toxicity	Grade/Symptoms	Recommendations
Serum creatinine elevation which cannot be explained by concomitant sepsis, TLS, other severe infection with fever or dehydration	If absolute serum creatinine increases from baseline of >0.5 mg/dL	Repeat within 48 hours • If confirmed: withhold therapy, institute treatment and monitoring as clinically indicated, and follow for resolution. • Discuss any further dosing with the GSK Medical Monitor[1].
Serum creatinine >Grade 3	>3.0 mg/dL from baseline or 3.0-6.0 × ULN	• Provide appropriate medical treatment • Discuss any further dosing with GSK's Medical Monitor[1]
Spot urine (albumin/creatinine ratios)	>2000 mg/g (226 mg/mmol)	• Retest (at least 7 days apart) • If not confirmed, continue belantamab mafodotin at pre-held dose • If confirmed on retest and no clear evidence of disease progression – Interrupt treatment with belantamab mafodotin – Repeat testing within 4 weeks – If spot urine ≤2000 mg/g (226 mg/mmol) may restart belantamab mafodotin with 1 dose level reduction[2] – If spot urine remains >2000 mg/g (226 mg/mmol) after 4 weeks, discuss any further dosing with the GSK's Medical Monitor[1]
Urine dipstick	2+	• May continue belantamab mafodotin dosing • Confirm by quantitative assessment using albumin/creatinine (spot urine from first void) • If albumin/creatinine >2000 mg/g, at the next cycle follow guidance above for Spot Urine
	3+	• Interrupt treatment and follow-up for recovery. Implement quantification of albumin/creatinine ratio
Thrombocytopenia	See Table 15 for guidelines for overlapping toxicity	
Febrile neutropenia Graded according to NCI-CTCAE criteria	Defined as: single temp of 38.3°C, or sustained 38°C for >1 hour and ANC <1.0×10^9/L	• Withhold belantamab mafodotin and immediate hospitalization is recommended with appropriate management, per local institutional guidance • Consider additional supportive treatment per local practice (e.g., growth factors)[2]. Upon recovery, consider a dose reduction[2] of belantamab mafodotin, if neutropenia was drug-related • See also Table 15 for guidelines for overlapping toxicity

Toxicity	Grade/Symptoms	Recommendations
Neutrophil count decreased (neutropenia)	See Table 15 for guidelines for overlapping toxicity	
Infusion reaction Graded according to NCI-CTCAE criteria	Note: If symptoms resolve within 1 hour of stopping the drug infusion, the infusion may be restarted at 50% of the original infusion rate (e.g., from 100 mL/h to 50 mL/h). Otherwise dosing will be held until symptoms are resolved and the participant should be premedicated at the next scheduled dose.	
	Grade 2	• Stop infusion, provide medical treatment and continue at slower infusion rate after resolution to Grade 0-1
	Grade 3	• Further treatment with belantamab mafodotin needs to be discussed with the GSK Medical Monitor. Continuation only allowed after recovery to ≤Grade 1 and with premedication, and extension of infusion time to 2-4 hours. Any future infusion needs to be premedicated
	Grade 4	• Permanently discontinue
Pneumonitis Graded according to NCI-CTCAE criteria	Grade 2	• Withhold treatment with belantamab mafodotin • Upon recovery, restart treatment with 1 dose level reduction • If participant is already at the lowest dose level, then rechallenge with the same dose must be discussed with the GSK Medical Monitor
	Grade 3-4	• Permanently discontinue treatment with belantamab

1 GSK Medical Monitor may consult GSK's renal panel about plans to continue therapy.
2 Belantamab mafodotin dose reduction strategy; see Section 6.6.

Table 17 KVA scale for treatment-related corneal events

Grade per KVA Scale[1]		Grade 1	Grade 2	Grade 3	Grade 4
Corneal events	Corneal examination finding(s)	Mild superficial keratopathy[2]	Moderate superficial keratopathy[3]	Severe superficial keratopathy[4]	Corneal epithelial defect[5]
	Change in BCVA	Decline from baseline of 1 line on Snellen Visual Acuity	Decline from baseline of 2 or 3 lines (and Snellen Visual Acuity not worse than 20/200)	Decline from baseline by more than 3 lines (and Snellen Visual Acuity not worse than 20/200)	Snellen Visual Acuity worse than 20/200

1. Grade 0 will indicate clear cornea and that BCVA shows no change from baseline or re-baseline. The recommended dose for Grade 0 is to start or continue treatment at current dose.
2. Mild superficial keratopathy (documented worsening from baseline), **with or without** symptoms.
3. Moderate superficial keratopathy = any/or a combination of moderate superficial punctate keratopathy, patchy microcyst-like deposits, sub-epithelial haze (peripheral), or a new peripheral stromal opacity.
4. Severe superficial keratopathy = any/or a combination of severe superficial punctate keratopathy, diffuse microcyst-like deposits, sub-epithelial haze (central), or a new central stromal opacity.
5. Corneal epithelial defect such as corneal ulcers. Corneal ulcer by definition means an epithelial defect with underlying stromal infiltration.

Table 18 Dose modification for belantamab mafodotin based on the KVA scale

Grade per KVA Scale	Grade 1	Grade 2	Grade 3	Grade 4
Dose modifications[1]	Continue treatment at current dose.	Withhold belantamab mafodotin until improvement in both corneal examination findings and changes in BCVA to Grade 1 or better and resume dosing as per Section 6.6.2	Withhold belantamab mafodotin until improvement in both corneal examination findings and changes in BCVA to Grade 1 or better and resume dosing as per Section 6.6.2	Withhold belantamab mafodotin. Participant may be restarted following benefit/risk assessment and discussion between the eye care specialist, the investigator, and the sponsor. Restart cannot occur until improvement in **both** corneal examination findings and changes in BCVA to Grade 1 or better. Resume dosing as per Section 6.6.2

Note: Changes in VA due to treatment-related corneal findings:

Note: For participants who have BCVA worse than 20/20 in either eye at baseline, dose modification for that eye will be determined by the worsening of vision from baseline only (not by absolute BCVA at the visits).

Note: If a participant has a baseline BCVA of 20/200 or worse in 1 eye, then belantamab mafodotin-related changes in vision in the other eye will drive the change in BCVA grading.

Note: If a participant has baseline BCVA of 20/200 or worse in both eyes, then the decision to delay or reduce belantamab mafodotin dose will be based on PI's assessment of benefit vs. risk-based on corneal exam findings following a discussion with the qualified eye care specialist.

Note: If already on the lowest allowed dose level, the participant should resume treatment at that dose.

1 Dose modification should be based on the most severe grade. If eyes differ in severity, dose modification guideline should be applied based on the more severe eye.

6.6.5. Dose modification guidance for lenalidomide

See Table 19 for lenalidomide dose modification guidelines.

Table 19 Dose modification guidelines for hematologic and other toxicities attributed to lenalidomide[1]

Toxicity	Grade/Symptoms	Recommendations
Thrombocytopenia	See Table 15 for guidelines for overlapping toxicity	
Neutropenia ANC	See Table 15 for guidelines for overlapping toxicity	
For other toxicities judged to be related to lenalidomide	Grade 3-4	Hold lenalidomide treatment and restart at the physician's discretion at next lower dose level when toxicity has resolved to ≥Grade 2

Note: For further information, refer to local/regional prescribing information (where available) for lenalidomide.

1 Investigators should use their clinical judgment for lenalidomide dose reductions, following local/regional prescribing information. Available doses of lenalidomide are 25 mg, 20 mg, 15 mg, 10 mg, 5 mg. Participants should not be dosed below 5 mg.

6.6.6. Dose modification guidance for dexamethasone

Dexamethasone should be taken at the same time of the day and may be taken at home. For participants who are older than 75 years, underweight (BMI <18.5), have poorly controlled diabetes mellitus, or prior intolerance/AE to steroid therapy, the dexamethasone dose may be administered at a dose of 20 mg on the days indicated in Table 11.

Dose modifications for dexamethasone other than those above should be made in line with local prescribing information, including reduction below 10 mg if necessary.

6.6.7. Management of hepatitis B positive participants

- Management by local hepatology or infectious disease services is required. If no subspecialist support is available, consultation with GSK Medical Monitor is required prior to enrollment into the study for participants with positive titers to hepatitis B.
- Participants should be monitored according to SoA, Table 4.
- Participants who experience clinically significant elevations in liver chemistry should follow liver event monitoring and stopping criteria (Section 7.1.2 and Section 10.6), and careful evaluation should be immediately initiated for evaluation of etiology including HBV DNA testing.
- Participants who develop detectable HBV DNA levels during study intervention should be reviewed by local specialist(s) immediately (within 1 week) and appropriate therapy and monitoring should be instituted.
- Study treatment should be withheld, and the GSK's Medical Monitor should be contacted for any participant who develops detectable HBV DNA levels.

Table 20 Dose modification guidelines for Hepatitis B reactivation

Toxicity	Grade/Description of Toxicity	Recommendations
Hepatitis B reactivation	Detectable HBV DNA	• Immediate consult with local specialist to initiate/modify treatment • Hold study intervention • Contact GSK Medical Monitor promptly, agreement with GSK Medical Monitor must be obtained prior to further dosing of study intervention • Follow liver monitoring/stopping guidelines per protocol for elevated liver function tests

6.7. Continued access to study intervention after the end of the study

After final analysis, and should study intervention(s) not be commercially available for the concerned indication at the time of study conclusion, provisions will be made to enable access to the study intervention(s) for participants with an acceptable safety profile and who would continue to benefit from treatment at the discretion of the investigator, if eligible, except if there are study intervention(s) safety concerns or manufacturing is ceased. Participants may transition to either an extension study, PACT, a rollover study, and/or expanded access route and/or other potential mechanism of access.

A follow-up visit should be carried out within 70 days of the last dose of study intervention if the participant:

- Chooses not to, or is ineligible to, continue treatment by poststudy access route.
- Is withdrawn due to an AE considered related to study intervention(s).

Study participants who transition to the PACT phase will continue to receive study treatment(s) beyond the DCO date until the EOS as defined in Section 4.4. There is no planned study intervention following the EOS.

The investigator is responsible for ensuring that consideration has been given to the poststudy care of the participants' medical condition.

Refer to the SoA (Section 1.3) for follow-up assessments of participants who are to be followed for disease progression and survival after they permanently discontinue from study intervention(s) until the DCO date and to Section 1.3 and Section 4.1 for follow-up assessments from the DCO date to the EOS.

6.7.1. Continued access to study intervention after DCO prior to EOS

Participants receiving study intervention(s) at the time of the final analysis DCO date, may continue to receive study intervention(s) at the discretion of the investigator if they are benefiting from continued treatment, and they do not meet any protocol-defined treatment discontinuation criteria (see Section 7.1). If participants receive study intervention(s) by transitioning to the PACT phase of the study, study intervention will continue until a study intervention discontinuation criterion (see Section 7.1, as assessed by the investigator, has been met.

Participants who continue study intervention in the PACT phase will be cared for in accordance with local standard clinical practice. Additional guidance on treatment with the study intervention and participant management is provided in the Pharmacy Manual (or equivalent). Participants will continue to be monitored for all SAEs, AEs leading to study intervention discontinuation, overdoses, pre-specified ocular data, and pregnancy while receiving study intervention(s). Information relating to participant care will be recorded on participant medical records, with the exception of SAEs, AEs leading to study intervention discontinuation, overdoses, pre-specified ocular data, and pregnancy cases that must continue to be reported to GSK, will not otherwise be reported for the purposes of this study. Investigators must report all SAEs, AEs leading to study intervention discontinuation, overdoses, pre-specified ocular data, and pregnancy cases until 70 days after the participant's last dose of study intervention in accordance with Section 8.4. Post final analysis DCO date, reporting and follow-up of SAEs, overdoses, pre-specified ocular data, and pregnancy cases will be done via paper/electronic forms (see Section 8.4.5 for details). For dispensing of study intervention and maintaining study intervention accountability in the PACT phase please refer to the Pharmacy Manual. All other assessments will revert to SOC at their site.

6.8. Treatment of overdose

6.8.1. Belantamab mafodotin overdose

There is no specific information on overdose of belantamab mafodotin. GSK does not recommend specific treatment for an overdose of belantamab mafodotin.

In the event of an overdose (defined as administration of more than the protocol-specified dose) of belantamab mafodotin, the investigator should:

- Contact the GSK Medical Monitor immediately
- Evaluate the participant to determine, in consultation with the GSK Medical Monitor, if possible, whether study intervention should be interrupted or delayed.
- Institute appropriate supportive treatment immediately. Further management should be as clinically indicated.
- Closely monitor the participant for AEs, SAEs, and laboratory abnormalities until these have resolved and belantamab mafodotin concentrations are predicted to be within the anticipated range in absence of the overdose.

- Obtain an additional blood PK sample for measurement of belantamab mafodotin (ADC) and cys-mcMMAF as soon as the overdose situation is recognized, and contact the GSK Medical Monitor for further guidance with regards to additional sample collection (determined on a case-by-case basis)
- Document the quantity of the excess dose as well as the duration of the overdosing in the eCRF.

Decisions regarding dose interruptions or modifications will be made by the investigator in consultation with the GSK Medical Monitor based on the clinical evaluation of the participant.

6.8.2. Lenalidomide overdose

There is no specific experience in the management of lenalidomide overdose in participants with MM. In dose-ranging studies in healthy participants, some were exposed to up to 200 mg (administered 100 mg twice daily) and in single-dose studies, some participants were exposed to up to 400 mg. Pruritus, urticaria, rash, and elevated liver transaminases were the primary reported AEs. In clinical studies, the dose-limiting toxicities were neutropenia and thrombocytopenia.

6.8.3. Daratumumab overdose

There is no known specific antidote for daratumumab overdose. In the event of an overdose, the participant should be monitored for any signs or symptoms of adverse effects and appropriate symptomatic treatment should be instituted immediately (refer to the local/regional prescribing information [where available] for daratumumab for further information).

6.8.4. Dexamethasone overdose

An overdose or prolonged use of dexamethasone may exaggerate glucocorticoid adverse effects. Treatment should be symptomatic and supportive with the dosage of dexamethasone being reduced or slowly withdrawn where possible.

The participants' dosing diary should accurately reflect what dose has been taken.

6.9. Concomitant therapy

Participants will be instructed to inform the investigator prior to starting any new medications from the time of first dose of study intervention until the EOT visit. Concomitant medications administered after the EOT should be recorded for SAEs/AESIs as defined in Section 8.4. Any concomitant medication (including over-the-counter or prescription medicines, vitamins, and/or herbal supplements) that the participant is receiving at the time of enrollment or receives during the study must be recorded along with:

- Reason for use

- Dates of administration including start and end dates
- Dosage information including dose and frequency

The GSK Medical Monitor should be contacted if there are any questions regarding concomitant therapy.

6.9.1. Permitted concomitant therapies

Participants should receive full supportive care during the study, including transfusions of blood products, growth factors, and treatment with antibiotics, antiemetics, antidiarrheals, and analgesics, as appropriate. Concomitant therapy with bisphosphonates is allowed and recommended. See Section 6.1.2.1 for recommended concomitant therapies to be used with daratumumab.

Participants may receive local irradiation for pain or stability control.

Use of approved monoclonal antibody treatments for serious conditions unrelated to MM, such as pre-exposure prophylaxis of COVID-19, may be permitted, but needs to be discussed with the GSK Medical Monitor.

While on study, a participant who is diagnosed with an unrelated malignancy that can be addressed by local therapy can remain on study, study intervention may be resumed as per investigator discretion after discussion with the GSK Medical Monitor. The participant should continue to be followed for disease progression of MM as per the SoA (Section 1.3).

6.9.2. Prohibited and cautionary concomitant therapies

Chronic treatment with oral steroids other than treatment specified in the SoA are prohibited, with the following exceptions:

- Low dose prednisolone (≤10 mg/day) as substitution in participants with adrenal insufficiency
- Short course (up to 7 days) of steroids to manage an AE
- Steroids to treat IRRs
- Inhaled, intranasal, topical, and topical ophthalmic steroids

Other prohibited therapies include:

- Plasmapheresis: prohibited from 7 days prior to first dose through the EOS
- Any other approved or investigational anti-myeloma therapy not specified in this protocol (including but not limited to immunomodulatory and antineoplastic drugs or proteasome inhibitors). This is inclusive of all medications with activity against MM and medications used for other indications that have anti-myeloma properties (e.g., cyclophosphamide)

- Investigational agents other than belantamab mafodotin

Administration of live or live-attenuated vaccines are contraindicated 30 days prior to the first dose of study intervention and while receiving study intervention. Use of live or live-attenuated vaccines is further contraindicated until 70 days following the last dose of belantamab mafodotin. Killed or inactivated vaccines may be administered; however, the response to such vaccines cannot be predicted.

For participants receiving belantamab mafodotin:

- Elimination pathways for belantamab mafodotin and cys-mcMMAF have not been characterized in humans. Cys-mcMMAF was not an inhibitor, an inducer, or a good substrate of CYP enzymes in vitro. Cys-mcMMAF was shown to be a substrate of P-gp, OATP1B1, and OATP1B3 transporters in vitro
- Caution should be exercised when belantamab mafodotin is combined with strong inhibitors of P-gp; and strong inhibitors of OATP1B1 and OATP1B3 should be avoided unless considered medically necessary
- For participants receiving anti-HIV and antimicrobials: the anti-HIV and antimicrobials that are OATP inhibitors are prohibited unless considered medically necessary. Preferably alternative antimicrobials and anti-HIV drugs would need to be prescribed to these participants
- Refer to [FDA, 2024] for lists of clinical inhibitors of P-gp, OATP1B1, and OATP1B3

7. DISCONTINUATION OF STUDY INTERVENTION AND PARTICIPANT DISCONTINUATION/WITHDRAWAL

7.1. Discontinuation of study intervention

Study intervention must be permanently discontinued in the case of:

- Confirmed PD, as defined by IMWG criteria. Confirmation of PD requires 2 consecutive central laboratory assessments meeting criteria for PD, or an imaging assessment confirming new bone lesions, definite progression of existing, or development of new, extramedullary plasmacytomas (as per IMWG criteria [Kumar, 2016]). NOTE: See below for advice on unacceptable toxicity for individual agents
- Meeting any of the protocol-defined safety stopping criteria
- Pregnancy

Before discontinuing study intervention strictly due to disease progression, sites must contact the GSK Medical Monitor to confirm progression meets IMWG criteria [Kumar, 2016]. The GSK Medical Monitor will review the central laboratory results as well as imaging results (if applicable) to confirm PD as per IMWG criteria. If PD criteria are not met, the investigator retains the decision to determine whether the treatment should be discontinued. However, the reason for treatment discontinuation should only be documented as disease progression if IMWG criteria for confirmed disease progression are met.

NOTE: As per IMWG criteria, all response categories require 2 consecutive assessments made at any time before the institution of any new therapy. All categories also require no known evidence of progressive or new bone lesions or extramedullary plasmacytomas if radiographic studies were performed.

In addition, study intervention may be permanently discontinued for any of the following reasons:

- Deviation(s) from the protocol consisting of serial noncompliance with the protocol that could potentially impact participant safety
- Request of the participant or proxy (withdrawal of consent by participant or proxy)
- Investigator's discretion
- Participant is lost to follow-up
- All or part of the study is closed or terminated

If the participant voluntarily discontinues from treatment due to toxicity, the AE will be recorded as the primary reason for permanent discontinuation on the eCRF.

Participants who have treatment-related ocular AEs at the EOT should be followed to resolution, as described in the SoA, before discontinuing from the study.

Once a participant has permanently discontinued from a study intervention, the participant will not be allowed to restart study intervention.

All participants who permanently discontinue study intervention will have safety assessments at the EOT follow-up as specified in the SoA.

Participants who permanently discontinue study intervention for reasons other than confirmed disease progression will remain in the study and will be followed for PFS until:

- Disease progression according to IMWG criteria [Kumar, 2016]
- Death
- Withdrawal of consent for study procedures, loss of follow-up or EOS (whichever occurs first)

See the SoA (Section 1.3) for details on data to be collected at the time of discontinuation of study intervention (i.e., the EOT visit).

Participants with documented disease progression while on study intervention, or during PFS follow-up, will be followed up for PFS2 and OS until death, withdrawal of consent for OS and PFS2 follow-up, loss to follow-up, or EOS, whichever occurs first.

7.1.1. Discontinuation of individual components of combination regimens

At the discretion of the investigator, participants meeting the stopping criteria for lenalidomide and dexamethasone may continue belantamab mafodotin (Arm A) or daratumumab (Arm B) monotherapy until PD, unacceptable toxicity, death, or EOS. Similarly, participants meeting the stopping criteria for belantamab mafodotin (Arm A) or daratumumab (Arm B) may continue lenalidomide and dexamethasone until PD, unacceptable toxicity, or death (whichever occurs first).

The primary reason for discontinuation of any of the individual components of the combination regimen study intervention must be documented independently in the medical record and eCRF.

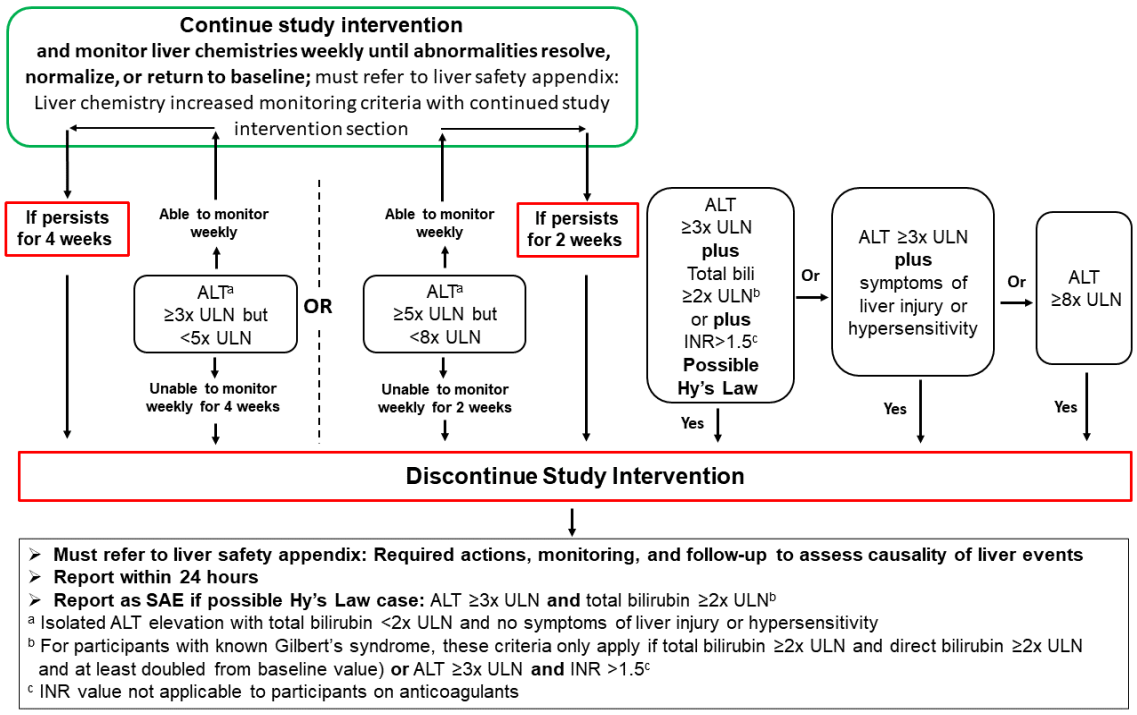
The primary reason for premature discontinuation of the study intervention will be documented in the eCRF based on the list below:

Reasons
AE
Lost to follow-up
Participant reached protocol-defined stopping criteria
Physician decision
Pregnancy
Protocol deviation
PD
Site terminated by sponsor
Study terminated by sponsor
Sponsor terminated study intervention
Withdrawal by participant
Completed analysis, but continuing with study intervention/treatment
Death

7.1.2. Liver event stopping criteria

Liver chemistry stopping and increased monitoring criteria have been designed to assure participant safety and evaluate liver event etiology. Discontinuation of study intervention for abnormal liver tests is required by the investigator when a participant meets 1 of the conditions outlined in Figure 2 or a participant has abnormal liver chemistries not meeting protocol-specified stopping rules if the investigator believes that it is in best interest of the participant.

Figure 2 Liver event study intervention stopping criteria and liver event increased monitoring criteria with continued study intervention



Refer to Section 10.6.1 for required liver safety actions, monitoring, and follow-up to assess the causality of liver events.

Participants who do not meet protocol-specified liver event stopping criteria but meet protocol-defined increased monitoring criteria may continue to receive study intervention with increased (i.e., weekly) liver chemistry monitoring. Refer to Section 10.6.2 for details regarding required increased monitoring of liver chemistry with continued study intervention.

7.1.2.1. Study intervention restart or rechallenge after liver stopping criteria are met

Study intervention restart after liver chemistry stopping criteria are met is allowed in this study. If the participant meets liver chemistry stopping criteria, do not restart participant with study intervention unless:

- GSK Medical Governance approval is granted (as described below),
- IRB/IEC approval, if required, is obtained and,
- A separate ICF consent for intervention restart is signed by the participant and participant is informed of any associated risks.

If GSK Medical Governance approval to restart the participant with study intervention is not granted, the participant must permanently discontinue study intervention and should continue in the study for protocol-specified follow-up assessments.

Refer to Section 10.6 for full guidance.

If GSK approval to restart and/or rechallenge the participant with study intervention is not granted, then the participant must permanently discontinue study intervention and may continue in the study for protocol-specified follow-up assessments.

7.1.3. Corneal event stopping criteria

All belantamab mafodotin dose modifications and stopping criteria for corneal events are based on the KVA scale.

Participants who develop Grade 4 corneal events according to the KVA scale must be discussed in detail between the treating eye care specialist (Section 10.10) and the GSK Medical Monitor to determine whether the participant can be allowed to continue treatment with belantamab mafodotin or permanently discontinue treatment (Section 7.1). The decision will be documented in the study files, together with an individual assessment of risk-benefit.

7.1.4. Allergic reactions and anaphylactic reactions stopping criteria

All participants will be monitored carefully for evidence of an allergic response to treatment. A participant who exhibits signs or symptoms of severe hypersensitivity or anaphylaxis will receive appropriate medical treatment and will permanently discontinue study intervention.

7.1.5. Pregnancy

Participants who become pregnant should immediately stop study intervention and follow the pregnancy reporting requirements in Section 10.3.5.7, as well as those outlined in country-specific lenalidomide Pregnancy Prevention Program.

7.2. Participant discontinuation/withdrawal from the clinical study

A participant may withdraw from the study at any time at their own request for any reason (or without providing any reason) or may be withdrawn at any time at the discretion of the investigator for safety, behavioral, compliance, or administrative reasons. This is expected to be uncommon.

Withdrawn participants will not be replaced.

The primary reason for participant withdrawal from the study will be documented in the eCRF based on the list in Section 7.1.

Participants who are withdrawn from the study because of AEs/SAEs must be clearly distinguished from participants who are withdrawn for other reasons. The investigator will follow participants who are withdrawn from the study due to an AE/SAE until the event is resolved (see Section 10.3.5.5).

At the time of withdrawal, the site must ascertain whether:

- The participant wishes to discontinue study intervention but can continue with follow-up visits, even with a reduced schedule. At the time of discontinuation, if the participant is still on-treatment, an EOT visit should be conducted as shown in the SoA (Section 1.3). A modified follow-up allows for reduced collection of data such as survival status (as a minimum), disease assessments, subsequent anti-myeloma therapies and progression (PFS2). A participant who agrees to modified follow-up is not considered to have withdrawn consent or withdrawn from the study.
- The participant wishes to discontinue also from the follow-up visits but consents to remote collection of data where countries allow (e.g., hospital records, telephone calls, death registers, etc.). To maintain the data integrity, the site should continue collecting data to determine subsequent anti-myeloma therapies and progression following the PFS2, and survival status. Site personnel or an independent third party will attempt to collect publicly available survival status information of the participant, within legal and ethical boundaries, for all participants

randomized/enrolled, including those who did not receive study intervention. Public sources may be searched for survival status information. If the survival status of the participant is determined to be deceased or known alive, this will be documented along with other relevant study information. Sponsor personnel will not be involved in any attempts to collect survival status information.

- The participant wishes full consent withdrawal for future data collection. If this occurs every effort should be made to discuss the implications on data integrity and the importance of obtaining survival status. If the participant withdraws consent for disclosure of future information, the sponsor may retain and continue to use any data collected before the withdrawal of consent. All data and samples collected up to and including the data of withdrawal of/last contact with the participant will be available for the study analyses. If, however, the participant requests destruction of any samples taken and not tested, the investigator must document this in the site study records.

The agreed follow-up approach should be recorded in medical records.

7.3. Lost to follow-up

A participant will be considered lost to follow-up if they repeatedly fail to return for scheduled visits and is unable to be contacted by the study site. The following actions must be taken if a participant fails to return to the clinic for a required study visit:

- The site must attempt to contact the participant and reschedule the missed visit as soon as possible, counsel the participant on the importance of maintaining the assigned visit schedule and ascertain whether the participant wishes to and/or should continue in the study providing options for modified follow-up when applicable (Section 7.2).
- Before a participant is deemed lost to follow-up, the investigator or designee must make every effort to regain contact with the participant (where possible, 3 telephone calls and, if necessary, a certified letter to the participant's last known mailing address or local equivalent methods). These contact attempts should be documented in the participant's medical record.
- Should the participant continue to be unreachable, they will be considered to have withdrawn from the study.
- Site personnel, or an independent third party, will attempt to collect the vital status of the participant within legal and ethical boundaries for all participants randomized, including those who did not get study intervention. Public sources may be searched for vital status information. If vital status is determined as deceased, this will be documented, and the participant will not be considered lost to follow-up. Sponsor personnel will not be involved in any attempts to collect vital status information.

8. STUDY ASSESSMENTS AND PROCEDURES

- Study procedures and their timing are summarized in the SoA (Section 1.3). Protocol waivers or exemptions are not allowed.
- Adherence to the study design requirements, including those specified in the SoA (Section 1.3), is essential and required for study conduct.
- All screening evaluations must be completed and reviewed to confirm that potential participants meet all eligibility criteria. Participants who have signed informed consent but are not eligible to proceed should be recorded in the eCRF with a status of "screen failure".
- Procedures conducted as part of the participant's routine clinical management and obtained before signing of the ICF may be utilized for screening or baseline purposes provided the procedures met the protocol-specified criteria and were performed within the timeframe defined in the SoA (Section 1.3).
- In the event of a significant study-continuity issue (e.g., caused by a pandemic), alternate strategies for participant visits, assessments, study intervention distribution, and monitoring may be implemented by the sponsor or the investigator, per local health authority/ethics requirements.
- If allowed by country regulation/ethics considerations, study assessments may be conducted remotely (by an HHS professional) and/or virtually (TM, secure video conferences, telephone calls, or a web portal and/or mobile application); however, on-site visits are required per the SoA (Section 1.3).

8.1. Administrative and general/baseline procedures

8.1.1. Collection of demographic data

Record demographic data such as year of birth, sex, race, and ethnicity in the participant's eCRF.

Collection of sex, race and ethnicity data is necessary to assess and monitor the diversity of the study participants, and to determine if the study participants are truly representative of the impacted population.

8.1.2. Medical/vaccination history

Obtain the participant's vaccination history by interviewing the participant. Obtain the participant's medical history by interviewing the participant and review of the participant's medical records. Record any pre-existing conditions, signs, and/or symptoms present prior to the first dose of study intervention in the eCRF.

8.2. Efficacy assessments

Planned time points for all efficacy assessments are provided in the SoA (Section 1.3).

8.2.1. Minimal residual disease

MRD status will be assessed using a NGS in vitro diagnostic assay and tested at a central laboratory. After the initial BMA sample is collected during screening, additional BMA samples will be collected at the time of achieving suspected CR or better, then repeated at each of the following applicable timepoints after CR: post-C1D1 +6, +12, +18, +24, +30, +36 months (±1 month), thereafter annually (±1 month) until PD. When a timepoint is less than 3 months from the suspected CR or better MRD assessment, a repeat BMA sample for MRD is not required. When the BMA samples from Screening Visit cannot be calibrated or missing, archived diagnostic BM sample may be used for calibration. If it is determined that no screening BM samples can be successfully calibrated, future BMA sample collection for MRD assessment will be stopped. Response evaluation will be performed according to the IMWG Uniform Response Criteria for MM [Kumar, 2016].

MRD by NGS will not be used for modification of participant treatment.

8.2.2. Response evaluation

The dual primary endpoint of this study is PFS and MRD. Response will be assessed according to the IMWG criteria [Kumar, 2016] by the investigator as well as an IRC. Response assessments will be performed Q4W for 104 weeks, then Q8W thereafter. These are fixed time point visits regardless of any delays or modification to **treatment cycles**.

Standard disease assessments for NDMM will include the following assessments performed at central laboratory:

- UPEP, urine immunofixation, 24-hour collection for urine M-protein
- SPEP, serum M-protein, serum immunofixation
- Calcium corrected for albumin
- IgG, IgM, IgA
- IgD, IgE (only in participants with IgD or IgE myeloma)
- Serum kappa, lambda FLC ratio
- BM at Screening and to confirm CR/sCR

Additional testing required where:

- CR+ is achieved: BM aspirate to assess MRD status, and biopsy to confirm CR and for IHC to confirm sCR
- Imaging of EMD (in participants with EMD)

- Physical examination (as indicated for palpable/superficial lesions)
- Skeletal surveys at Screening and as clinically indicated

For evaluation purposes the date of initial test is considered as the date of response as per IMWG.

Baseline serum/urine disease assessment will be completed during the Screening Period (within 28 days prior to the first dose of study intervention) and baseline imaging within 30 days prior to the first dose of study intervention. On study serum and urine-based assessments (M-protein, FLC, immunofixation) will be performed Q4W for 104 weeks, and Q8W thereafter. Details for the preparation and shipment of samples for central laboratory assessments will be provided in the Laboratory Manual.

In participants with extramedullary myeloma, the disease assessments must include imaging (e.g., CT, MRI, or PET-CT scans: the same method should be used throughout the study) and physical examination (as indicated for palpable/superficial lesions).

For participants who are followed by imaging for EMD the imaging should be performed as described in the SoA (Section 1.3). Imaging will be collected and stored centrally and may be reviewed by an independent central reviewer.

All assessments on study must be performed on a calendar schedule and must not be affected by dose interruptions/delays.

For participants who are discontinuing IP due to PD, the confirmation of laboratory parameters must be performed from a different blood collection preferably within 14 days of the original date of suspected disease progression, and before initiation of any new anti-myeloma therapy. For participants with PD due to radiographic progression, confirmatory scans are not required. The laboratory parameters do not need to be repeated if radiologic assessment is the determinant of progression.

8.2.3. Assessment of CR where daratumumab interference is suspected

Daratumumab is an IgG kappa monoclonal antibody that can be detected in serum electrophoresis and immunofixation assays. This can have an impact on the detection of CR and/or disease progression in participants who achieve CR/sCR with IgG kappa myeloma. All SPEP samples from participants in Arm B with IgG kappa myeloma are required to undergo central testing with a daratumumab-specific interference assay (HYDRASHIFT 2/4, FDA-approved).

8.2.4. Overall survival collection

OS is defined as the time from randomization to the date of death due to any cause. To avoid adverse effects on OS assessment, the time between randomization and date of first dose of study intervention must not exceed 3 days. Survival status will be collected during the Follow-up Period per the SoA (Section 1.3) and during survival sweeps. Modified follow-up visits may be implemented in cases where a participant is considering withdrawing from the study. These modified follow-up visits allow for continued collection of OS and other endpoint data with minimal contact with the participant (see Section 7.2). Participants who are alive at the point of analysis will be censored at the date of last known alive.

8.3. Safety assessments

Planned time points for all required safety assessments are provided in the SoA (Section 1.3). Key assessments are summarized below. This list details study-related procedures that are required by GSK. Other clinical procedures (e.g., vital signs and physical examination) should be performed as per standard local clinical practice.

8.3.1. ECOG performance status

The participant's PS will be assessed using the ECOG scale (Section 10.11) as specified in the SoA (Section 1.3).

8.3.2. IMWG Frailty Score

Participants frailty score at screening will be assessed using the IMWG Frailty Score [Palumbo, 2015]. See Section 10.7 for further details.

8.3.3. New York Heart Association functional scale

Criteria for rating participants for heart failure status according to the NYHA criteria are provided in Section 10.8.

8.3.4. Electrocardiograms

A 12-lead ECG is obtained at screening/baseline and then as clinically indicated as specified in the SoA (Section 1.3). The ECG machine must automatically calculate the heart rate and measure PR, QRS, QT, and QTcF. No further ECGs are required, but if obtained as part of medical care a 12-lead ECG should be performed by qualified personnel at the site after the participant has at least a 5-minute rest.

Triplicate ECG should be performed if during treatment QTcF is >530 msec.

Where triplicate ECGs are required, 3 individual ECG tracings should be obtained as closely as possible in succession, but no more than 2 minutes apart. The full set of triplicate ECGs should be completed in 4 minutes or less.

8.3.5. Clinical safety laboratory assessments

All protocol-required laboratory assessments, as defined in Section 10.2, must be conducted in accordance with the Laboratory Manual and the SoA (Section 1.3).

Prior to administration of the first dose of study intervention, results of laboratory assessments should be reviewed and documented by the investigator. Any laboratory test with a value outside the normal range may be repeated (prior to the first dose) at the discretion of the investigator. The laboratory results must be retained with source documents.

If additional non-protocol-specified laboratory assessments are performed at the institution's local laboratory and result in a change in participant management or are considered clinically significant by the investigator (e.g., SAE or AE or dose modification) the results must be recorded. Clinically significant abnormal laboratory findings are those which are not associated with the underlying disease, unless judged by the investigator to be more severe than expected for the participant's condition.

In the absence of a diagnosis, abnormal laboratory findings assessments, or other abnormal results considered clinically significant by the investigator, will be recorded as an AE or SAE, if they meet the definition of an AE or SAE (see Section 10.3.1 and Section 10.3.2).

All laboratory tests with values considered clinically significantly abnormal during participation in the study or until 70 days after the last dose of study intervention should be repeated until the values return to normal, or baseline are no longer considered significantly abnormal by the investigator or GSK Medical Monitor. If such values do not return to normal/baseline within a period of time judged reasonable by the investigator, the etiology should be identified, and the sponsor notified.

Asymptomatic elevations of LDH and CK and AST have been observed in Study BMA117159. Participants with significant elevations (≥3×ULN) should, where possible, have a sample sent for central testing of CK and LDH isoenzyme levels.

8.3.6. Ophthalmic examinations

Study sites are required to establish a close collaboration with a qualified eye care specialist who will be responsible for assessing participants while they are on-study and managing participants who develop a corneal event in close communication with the GSK Medical Monitor.

Participants will be assessed by a qualified eye care specialist (see Section 10.10 for qualifications and requirements of the eye care specialist) at screening/baseline and then on Day 1 prior to belantamab mafodotin dosing (assessment window of up to 5 days prior to dosing, but all effort should be made to schedule as close to belantamab mafodotin dosing as possible).

If a participant subsequently develops vision changes or other ocular symptoms, the participant should be promptly evaluated by a qualified eye care specialist.

Participants who have treatment-related ocular AEs or KVA (Grade 2 and above for both corneal exam findings and VA) at the EOT visit will be followed Q12W (±7 days) until full resolution of ophthalmic changes (Grade 1 or baseline for both corneal exam findings and VA) by a qualified eye care specialist.

For participants who have treatment-related ocular AEs or KVA (Grade 2 and above for both corneal exam findings and VA) at EOS (Section 4.4) GSK recommends that they are followed until full resolution of ophthalmic changes (Grade 1 or baseline for both corneal exam findings and VA) by a qualified eye care specialist.

If steroid eye drops are deemed medically necessary and prescribed for >7 days, intraocular pressure must be monitored.

8.3.6.1. Ocular examinations and procedures

A full screening/baseline ophthalmic examination for all participants must include for both eyes (OU):

1. BCVA.
2. Documentation of manifest refraction and the method used to obtain BCVA.
3. Current glasses prescription (if applicable).
4. Selected anterior segment (slit lamp) examination with focus on the cornea and lens, including fluorescein staining of the cornea.
5. Intraocular pressure measurement.
6. Dilated funduscopic exam.

The on-treatment and follow-up ophthalmic exams should be performed as described below and in the SoA (Section 1.3).

This should include:

1. BCVA.
2. Documentation of manifest refraction and the method used to obtain BCVA.
3. Anterior segment (slit lamp) examination with focus on the cornea and lens, including fluorescein staining of the cornea.
4. Intraocular pressure measurement.
5. Dilated funduscopic exam (if clinically indicated).

The EOT, and **last follow-up** ophthalmic exam, if required, should match the **screening/baseline** exam.

Additional examinations should be performed at the discretion of the treating eye care specialist.

8.4. Adverse events, serious adverse events, and other safety reporting

For definitions relating to safety information, see Section 10.3.

The investigator and any qualified designees are responsible for detecting, documenting, and recording events that meet the definition of an AE or SAE and other safety information and remain responsible for following up AEs that are serious, considered related to the study intervention or study procedures, or that caused the participant to discontinue the study intervention (see Section 7). This includes events reported by the participant.

The method of recording, evaluating, and assessing causality of AEs and SAEs and the procedures for completing and transmitting SAE reports are provided in Section 10.3.

8.4.1. Time period and frequency for collecting AE, SAE, and other safety information

- All SAEs will be collected from the start of study intervention for at least 70 days after discontinuing all study interventions, regardless of initiation of a new cancer therapy or transfer to hospice, at the time points specified in the SoA (Section 1.3). However, any SAEs assessed as related to study participation (e.g., study intervention, protocol-mandated procedures, invasive tests, or change in existing therapy) or related to a GSK product will be recorded from the time a participant consents to participate in the study up to and including any follow-up.
  Note: In China *only*, all SAEs should be collected from the signing of the ICF.
- All AEs will be collected from the start of treatment until 70 days following discontinuation of study intervention regardless of initiation of a new cancer therapy or transfer to hospice at the time points specified in the SoA (Section 1.3). AEs assessed as related to study participation (e.g., study intervention, protocol-mandated procedures, invasive tests, or change in existing therapy) will be collected from the time a participant consents to participate in the study.
- Medical occurrences that begin before the start of study intervention but after obtaining informed consent will be recorded on the Medical History/Current Medical Conditions section of the eCRF, not the AE section.
- All SAEs will be recorded and reported to the sponsor or designee immediately and under no circumstance should this exceed 24 hours, as indicated in Section 10.3. The investigator will submit any updated SAE data to the sponsor within 24 hours of it being available.
- A poststudy AE/SAE is defined as any event that occurs outside of the AE/SAE reporting period defined in Table 21.

- Investigators are not obligated to actively seek information on AEs or SAEs after conclusion of the study participation. However, if the investigator learns of any SAE, including a death, after a participant has been discharged from the study, the investigator must record it in the medical records per the local country requirements. If the investigator considers the event to be reasonably related to the study intervention or study participation, the investigator must promptly notify the sponsor. Refer to Table 22 for contact information.
- For participants in the PACT phase of the study, the sponsor will continue to collect safety information including SAEs, AEs leading to study intervention discontinuation, overdose, pre-specified ocular data, and pregnancy cases via paper/electronic forms which will be reported directly to the sponsor (see Section 8.4.5 for details). SAEs, AEs leading to study intervention discontinuation, overdose, pre-specified ocular data, and pregnancy cases will be reported during the PACT study intervention period and for up to 70 days after the last dose. Additionally, any SAEs that are ongoing at the time of the final DCO must be followed up to resolution unless the event is considered by the investigator unlikely to resolve, or the participant is lost to follow-up. Updates to these events will also occur via paper/electronic forms directly to the sponsor (see Section 8.4.5 for details). Pre-specified ocular data will continue to be collected for participants receiving study intervention after the start of PACT as follows:
  - Participants will be required to have an ocular assessment at time of dosing and/or as clinically indicated, until the EOT.
  - It is advised that participants with treatment-related ocular examination findings after treatment discontinuation continue to have ocular assessments as SOC until resolution. Note that participants who continue to receive belantamab mafodotin via commercial source or a drug supply program should follow ocular examination guidance provided by the source/program from which they are receiving belantamab mafodotin.
- The sponsor retains the right to request additional information for any participant with ongoing AEs/SAEs at EOS, if judged necessary.

8.4.2. Method of detecting AEs and SAEs

The method of recording, evaluating, and assessing causality of AEs and SAEs and the procedures for completing and transmitting SAE reports are provided in Section 10.3.

Care will be taken not to introduce bias when detecting AEs and SAEs. Open-ended and nonleading verbal questioning of the participant is the preferred method to inquire about AE occurrences.

8.4.3. Follow-up of AEs and SAEs

After the initial AE/SAE report, the investigator is required to proactively follow each participant at subsequent visits/contacts. All SAEs and nonserious AESIs as defined in Section 10.3 will be followed until resolution, stabilization, the event is otherwise explained, or the participant is lost to follow-up (as defined in Section 7.3). Further information on follow-up procedures is provided in Section 10.3.5.5.

8.4.4. AESIs

AESIs for belantamab mafodotin are corneal events and thrombocytopenia. Severity of all AESIs will be graded using NCI-CTCAE, v5.0. Guidelines for dose modifications and interruptions for management of common toxicities associated with the study intervention(s) are provided in Section 6.6. Dose modifications for belantamab mafodotin corneal events (Table 18) will be based on grading of corneal events according to the guidelines of KVA scale (Table 17).

8.4.5. Regulatory reporting requirements for SAEs/AESIs

Prompt notification by the investigator to the sponsor of an SAE is essential so that legal obligations and ethical responsibilities toward the safety of participants and the safety of a study intervention under clinical investigation are met. See Section 8.4.1 for reporting timeframes.

For SAEs, the investigator must always provide an assessment of causality at the time of the initial report, as defined in the Section 10.3.5.

An investigator who receives an investigator safety report describing an SAE or other specific safety information (e.g., summary or listing of SAEs) from the sponsor will review and then file it along with the IB and will notify the IRB/IEC, if appropriate according to local requirements.

Investigators have to report to the sponsor pregnancies, medication errors, abuse, and misuse even in an absence of an AE/SAE, as these may be subjected to local regulatory reporting requirements for the sponsor.

Investigator safety reports must be prepared for SUSARs according to local regulatory requirements and sponsor policy and forwarded to investigators as necessary.

The timelines for reporting SUSARs by the sponsor are:

- In the case of fatal or life-threatening SUSARs, within 7 calendar days after the sponsor became aware of the reaction;
- In the case of non-fatal or non-life-threatening SUSARs, within 15 calendar days after the sponsor became aware of the reaction;

- In the case of a SUSAR which was initially considered to be non-fatal or non-life threatening but which turns out to be fatal or life-threatening, within 7 calendar days after the sponsor became aware of the reaction being fatal or life-threatening.

Table 21 Timeframes for submitting SAE, pregnancy, and other events reports to GSK

Type of Event	Initial Reports		Follow-up of Relevant Information on a Previous Report	
	Timeframe	Documents	Timeframe	Documents
SAEs (including serious AESIs)	24 hours[1,2]	paper/electronic AEs Report	24 hours[1]	paper/electronic AEs Report
Pregnancies	24 hours[1]	paper pregnancy notification report/electronic pregnancy report	24 hours[1]	paper pregnancy follow-up report/electronic pregnancy report

1 Timeframe allowed after receipt or awareness of the information by the investigator/site staff.
2 Paper AEs Report will be dated and signed by the investigator (or designee). For each SAE, the investigator(s) must document in the medical notes that they have reviewed the SAE and have provided an assessment of causality.

8.4.6. Pregnancy

- Details of all pregnancies in female participants will be collected after the start of study intervention and for 4 weeks following the last dose of lenalidomide, or 4 months for belantamab mafodotin and daratumumab, whichever is longest.
- Details of pregnancies from female partners of male participants will be collected after the start of study intervention and for 4 months following the last dose of belantamab mafodotin or 4 weeks for lenalidomide, whichever is longest.
- If a pregnancy is reported, the investigator will record pregnancy information on the appropriate form and submit it to the sponsor within 24 hours of learning of the pregnancy (female participant and/or female partner of male participant after obtaining the necessary signed ICF from the female partner) and should follow the procedures outlined in Section 10.3 as well as those outlined in Participant the Lenalidomide Global Pregnancy Prevention Plan or other country-specific program.
- Any pregnancy complication or elective termination of a pregnancy for medical reasons will be reported as an AE or SAE.
- Abnormal pregnancy outcomes (e.g., spontaneous abortion, fetal death, stillbirth, congenital anomalies, ectopic pregnancy) are considered SAEs and will be reported as such.
- The participant/pregnant female partner (when contraception is required for male participant) will be followed to determine the outcome of the pregnancy. The investigator will collect follow-up information on the participant/pregnant female partner and the neonate and the information will be forwarded to the sponsor. See Table 21 for reporting timeframes.

- Any poststudy pregnancy-related SAE considered reasonably related to the study intervention by the investigator will be reported to the sponsor as described in Section 8.4.4. While the investigator is not obligated to actively seek this information in former study participants/pregnant female partner when contraception is required for male participant), they may learn of an SAE through spontaneous reporting.
- Any female participant who becomes pregnant while participating in the study will discontinue study intervention.
- Any male participant whose partner becomes pregnant while participating in the study will discontinue study intervention or be withdrawn from the study OR may request continuation of study intervention.

Prior to continuation of study intervention following pregnancy in a study participant, the following must occur:

- The sponsor and the relevant IRB/IEC give written approval.
- The participant gives signed informed consent.
- The investigator agrees to monitor the outcome of the pregnancy and the status of the participant and her offspring.

8.4.7. Cardiovascular and death events

The Death eCRF is provided immediately after the occurrence or outcome of death is reported. Initial and follow-up reports regarding death must be completed within 1 week of when the death is reported.

For any CV events detailed in Section 10.8 (whether or not they are considered SAEs) and all deaths, specific CV, and Death sections of the eCRF will be required to be completed. These sections include questions regarding CV (including sudden cardiac death) and non-CV death.

The CV eCRFs are presented as queries in response to reporting of certain CV MedDRA terms. The CV information should be recorded in the specific CV section of the eCRF within 1 week of receipt of a CV event data query prompting its completion.

Death due to the disease under study is to be recorded on the Death eCRF.

For death due unequivocally to disease under study/SAEs/event that occurred during the Treatment Period, or poststudy intervention not related to the study intervention, record death details in the Death eCRF.

8.4.8. Contact information for reporting SAEs, AESIs, pregnancies, and study holding rules

Table 22 Contact information for reporting SAEs, AESIs, pregnancies, and study holding rules

Study contact for questions regarding SAEs, AESIs, pregnancies, and SAEs linked to device deficiencies Contact GSK's local and/or medical contacts	**Study contact for reporting of study holding/stopping rules** If a holding/stopping rule is met, the investigator must immediately inform the GSKs Local and/or Medical contacts.
Contacts for reporting SAEs, AESIs, pregnancies, and SAEs linked to device deficiencies Available 24/24 hours and 7/7 days For medicines: uk.gsk-rd-gcsp-ctsm-admin@gsk.com	

8.5. Pharmacokinetics

8.5.1. Belantamab mafodotin blood sample collection for pharmacokinetics

In treatment Arm A, blood samples for PK analysis of belantamab mafodotin (ADC) and cys-mcMMAF will be collected at the time points indicated in the SoA (Table 23). Each PK sample must be collected as close as possible to the planned time relative to the dose administered to the participant on PK sampling days. The actual data and time of each blood sample collection will be recorded.

Details on PK blood sample collection including blood volumes, processing, storage, and shipping procedures are provided in the Laboratory Manual.

All PK, sBCMA, and ADA samples once collected (regardless of dosing), can be analyzed if the sample date and time have been recorded. PK sample collection may be terminated when sufficient data have been collected.

Table 23 Schedule for PK, ADA, and sBCMA sample collection

Dose/Day[1]	Sample Timepoint	PK Sampling (Arm A only)	ADA (Arm A only)	sBCMA Sampling (Arm A and Arm B)	Notes
Dose1 D1	Predose	X	X	X[2]	Within 2 hours prior to SOI
	EOI	X			Within 30 min **after** EOI
Dose1 D2	24 hours after SOI	X			Within ±2 hours of SOI+24 hours
Dose1 D29	Day 29	X	X		Collect within ±7 days of Day 29
Dose1 D57[3]	Day 57	X			Collect within ±7 days of Day 57
Dose2/3/4 D1	Predose	X	X (Dose 2 and 3 only)		Within 2 hours prior to SOI
	EOI	X			Within 0-30 min **after** EOI
Dose6/9/12 D1	Predose	X (Dose 6 only)	X		Within 2 hours prior to SOI
Dose12+: Every 6th dose (Dose18 D1, etc.)	Predose	--	X		Within 2 hours prior to SOI
EOT	Anytime	--	X	X[4]	

Note: PK/ADA/sBCMA sample collection may be terminated when sufficient data have been collected at the discretion of the sponsor.

1. PK sample is tied to the dose of belantamab mafodotin. This may not correspond to the belantamab mafodotin cycle number.
2. sBCMA samples for Arm B should be collected on C1D1 predose.
3. This sample will only be collected if Dose 2 is delayed.
4. Sample should be collected at EOT or PD whichever comes first.

8.5.2. Belantamab mafodotin pharmacokinetic sample analysis

Plasma analysis will be performed under the control of GSK, the details of which will be included in the Laboratory Manual. Concentration of belantamab mafodotin (ADC) and cys-mcMMAF will be determined in plasma samples using the currently approved bioanalytical methodologies. Raw data will be archived at the bioanalytical sites (detailed in the Laboratory Manual)

Once the plasma has been analyzed for belantamab mafodotin (ADC) and cys-mcMMAF, any remaining plasma may be analyzed for other compound-related metabolites and the results may be reported under a separate GSK protocol.

8.5.3. Daratumumab and lenalidomide

No PK sampling or analysis is required for daratumumab, lenalidomide, or dexamethasone in arms A and B. For daratumumab, no anti-daratumumab antibody sampling or analysis is required.

8.6. Pharmacodynamics

The pharmacodynamic parameters evaluated in this study pertain to the biomarkers described in Section 8.8.

8.7. Genetics

A 5-mL blood sample for DNA isolation may be collected from participants who have consented to participate in the genetics analysis component of the study. Participation is optional. Participants who do not wish to participate in the genetic research may still participate in the study.

In the event of DNA extraction failure, a replacement genetic blood sample may be requested from the participant. Signed informed consent will be required to obtain a replacement sample unless it was included in the original consent.

See Section 10.5 for information regarding genetic research. Details on processes for collection and shipment and destruction of these samples can be found in the Laboratory Manual.

8.8. Biomarkers

Biomarker research is part of this study and will involve the use of peripheral blood (whole blood, serum, PBMCs, or plasma) and BM. Samples collected during this study may be used to measure novel biomarkers (and may involve RNA, DNA, or protein analysis) to explore the mechanistic and biological relationship with belantamab mafodotin and SOC treatments.

Additional novel biomarkers may also be incorporated, as data warrants, and data generated in this study may be compared to other studies involving belantamab mafodotin and/or belantamab.

Biomarker analysis may be adjusted based on emerging data and so not all biomarker investigations can be prospectively detailed but will however, focus primarily on, but not necessarily limited to:

1. Investigating the impact of therapeutic intervention, in both the control and experimental arms, on BCMA expression at the start and EOT, via sBCMA measurements.
2. Providing mechanistic understanding of the effect of belantamab mafodotin compared to daratumumab, via immune cell phenotyping.

All analysis will be exploratory in nature.

- Samples are mandatory and will be collected according to the schedule described in the SoA (Section 1.3) and as detailed in the Laboratory Manual, provided separately to sites. Details regarding study vendors are documented in the TMF. The sample collection strategy may be adjusted based on emerging data.

- Any samples collected following obtaining informed consent may be used for research to develop methods, assays, prognostics, and/or diagnostics.
- Any leftover samples, including but not limited to, those associated with disease assessment analysis can be used for biomarker research.
- Successful biomarker analyses are dependent on a multitude of factors. These include, but are not limited to, the availability of appropriate biomarker assays and sample quality/quantity. Should it become clear at any time during the study or at the end of the study that the factors needed for successful biomarker analysis are missing or there is no scientific value in undertaking the analysis, the sponsor may defer or not undertake the analysis.
- Should the sponsor terminate the study early, the justification and intended utility of the data will determine whether biomarker assessments are undertaken.
- GSK may store samples for up to 20 years after the end of the study to achieve study objectives. Additionally, with participants' consent, samples may be used for further research by GSK or others such as universities or other companies to contribute to the understanding of MM or other diseases, the development of related or new treatments, diagnostics, or research methods.

8.8.1. Screening

Participants must sign a screening ICF consenting to collection of blood samples for testing using appropriate technologies.

8.8.2. Sample management

Samples will be analyzed at GSK or a GSK designated central laboratory.

Should samples be taken from participants who fail screening, but samples are either available or have already been analyzed, sample/data use can still be used at the discretion of the sponsor provided the ICF has been signed and consent is not withdrawn. The sponsor will only include use of the sample/data provided there is scientific value in using the sample/data.

Samples will be collected and tested in compliance with local policy/restrictions.

Details of sample preparation, storage, and analysis will be provided in the Laboratory Manual. Raw data will be archived at the bioanalytical site.

8.8.3. Blood biomarkers

Blood specimen collections are outlined in the SoA (Section 1.3), and sample requirements are outlined in the Laboratory Manual.

8.8.3.1. sBCMA sample analysis

Cleavage of the tumor expressed BCMA receptor leads to release of the BCMA extracellular domain into the peripheral circulation as soluble BCMA (sBCMA, [Laurent, 2015]). sBCMA can be used as a surrogate for tumor expressed BCMA.

8.8.3.2. Immune cell phenotyping

The multimodal mechanism of action of belantamab mafodotin and daratumumab are postulated to include, among other MoAs, induction of antibody-dependent cellular cytotoxicity, CDC, and phagocytosis (ADCC/CDC/ADCP). To elucidate the interactions of the therapeutic agents and their potential diverging impact on immune cell populations, immune cell populations will undergo phenotyping. Furthermore, immune cell activation markers and immune cell suppression markers will also be analyzed. Analysis will be carried out on PBMCs using appropriate technologies (including, but not limited to, flow cytometry). Several different immune cell populations, including, but not limited to, the major lymphocytic cell populations (B-, T-, natural killer cells) and their associated phenotypic markers will be monitored, in both the experimental and control arms.

8.8.4. Additional assessments

As stated above not all biomarker analysis can be prospectively named due to emerging scientific data that may necessitate implementation of biomarkers that are not currently discussed. Alternate omics research may involve the use of samples to investigate:

- Transcriptomic expression patterns, using appropriate technologies, to analyze the pathogenesis of MM in NDMM, the absorption, distribution, metabolism, or excretion of belantamab mafodotin or response dynamics of participants exposed to either the experimental or control arm.
- Genomic analysis, using appropriate technologies, to probe potential mechanisms of resistance in both the experimental and control arms.
- Proteomic analysis to elucidate for example pathophysiology of NDMM in both the experimental and control arms.

8.9. Immunogenicity assessments

Immunogenicity (ADA) samples in Arm A will be collected at the time points specified in the SoA (Section 1.3).

ADA sample collection may be terminated when sufficient data have been collected. These samples will be tested by the sponsor or sponsor's designee.

Anti-belantamab mafodotin antibody samples will be tested for anti-belantamab mafodotin antibodies using a tiered-testing scheme consisting of validated screening, confirmation, and titration assays. Briefly, all samples will be tested in the screening assay. Samples that screen positive are considered potentially positive and will be tested

for specificity in a confirmation assay. Finally, titer values will be obtained for confirmed positive samples using a titration assay. The sample results (e.g., positive or negative) and titer values (positive samples only) will be reported. Samples that test positive for anti-belantamab mafodotin antibodies may be further characterized in a validated neutralizing antibody assay to determine the neutralizing activity of the antibodies.

Anti-belantamab mafodotin antibody samples will be disposed 3 months after final approved results are provided to the Clinical Study Team or its designee or upon documented study termination.

8.10. Health-related quality of life

Five HRQoL assessments will be performed in this study. The questionnaires will be administered to participants in different regions based on the availability of translated versions.

All participants will complete the self-administered version of the PRO questionnaires unless their vision prevents them from being able to complete the questionnaire on their own. Participants who are not able to complete the questionnaire on their own and require assistance should have the PROs administered to them in an interview format. The questionnaires should be read to the participants verbatim, and participant responses should be recorded directly without any interpretation. For any additional assessments conducted via telephone (either during participation in the Treatment Period or during Follow-up), the interviewer-administered format should be used. The questionnaires will be administered to participants in different regions based on the availability of translated versions.

The PROs in this study (Table 24) aim to measure changes in general cancer and disease-specific symptoms and symptomatic side effects and tolerability (PRO-CTCAE and FACT-GP5). Cancer-related QoL will be measured with the EORTC QLQ-C30, and the participants' global impression of disease severity and change therein will be measured via the PGIS and PGIC, respectively. Planned time points for all PRO assessments are listed in the SoA (Section 1.3).

Table 24 Patient-reported outcomes

Instrument	Role	Rationale and Overview
EORTC QLQ-C30 v3.0	General Cancer HRQoL	The EORTC modules are widely used disease-specific instruments for assessing cancer HRQoL in clinical studies.
EORTC QLQ-MY20	Disease-specific HRQoL	MM specific PRO that assesses symptoms and impacts relevant to the disease.
FACT-GP5 v4	Symptomatic AEs (Global)	Single question from the well-established FACT-G (V4), to assess overall tolerability from patient's perspective.
OSDI	Ocular symptoms, vision-related functioning, environmental triggers	12-item assessment of the impact of potential ocular toxicity on function and HRQoL.
PGIS v1.0 PGIC v1.0	Global impression of symptom severity and change	Each is a single item that serves as an anchor to establish thresholds of clinically meaningful change for the questionnaires in the study.
PRO-CTCAE v1.0	Symptomatic AEs (Specific)	A standard and flexible approach to describe and evaluate symptomatic AEs as they are experienced and reported by patients themselves.

Completion of PRO questionnaires

The PRO questionnaires are to be administered at the beginning of a visit in the order presented in the electronic device. PRO assessments will be administered based on the availability of appropriate translated versions. PROs will be completed at the start of every applicable visit, either before or after laboratory sample collection, and prior to any other procedures or provision of results. PROs may be completed while results are pending. Adequate time must be allowed to complete all items on the questionnaires, and if necessary, the participant must be encouraged to complete any missing items. Where possible, participants should complete the PRO independently in a quiet place. Assistance to complete the questionnaires can be provided if absolutely necessary, but care should be taken to ensure the participant's answers are faithfully recorded. If unavoidable, safety laboratory assessments may be completed prior to PRO questionnaire completion. In this case, to avoid biasing responses, the participants should not be told the results of diagnostic tests prior to completing the questionnaires.

8.10.1. European Organization for Research and Treatment of Cancer Quality of Life questionnaire (EORTC QLQ-C30)

The EORTC QLQ-C30 is a 30-item questionnaire containing both single- and multi-item measures [Aaronson, 1993]. These include 5 functional scales (Physical, Role, Cognitive, Emotional, and Social Functioning), 3 symptom scales (Fatigue, Pain, and Nausea/Vomiting), a Global Health Status/QoL scale, and 6 single items (Constipation, Diarrhea, Insomnia, Dyspnea, Appetite Loss, and Financial Difficulties). Scores for each scale and single item measure are averaged and transformed linearly to a score ranging from 0–100. A high score for functional scales and for Global Health Status/QoL represent better functioning ability or HRQoL, whereas a high score for symptom scales and single items represents significant symptomatology [Proskorovsky, 2014].

8.10.2. EORTC questionnaire 20-item MM module (EORTC QLQ-MY20)

The EORTC QLQ-MY20 is a supplement to the QLQ-C30 instrument used in patients with MM [Aaronson, 1993; Cocks, 2007]. Only the Disease Symptoms domain of the QLQ-MY20 will be administered, which includes bone aches or pain, back pain, hip pain, arm or shoulder pain, chest pain, and pain increasing with activity. The disease symptom domain of the EORTC QLQ-MY20 will be referred to as the EORTC IL52. As with the QLQ-C30, QLQ-MY20 domain scores are averaged and transformed linearly to a score ranging from 0–100. A high score for Disease Symptoms represents a high level of symptomatology or problems [Proskorovsky, 2014].

8.10.3. FACT-GP5

The FACT-GP5 is a 27-item compilation of general questions divided into 4 primary QoL domains: Physical Well-Being, Social/Family Well-Being, Emotional Well-Being, and Functional Well-Being [Cella, 1993]. It is considered appropriate for use with patients with any form of cancer and has also been used and validated in other chronic illness conditions (e.g., HIV/AIDS and multiple sclerosis) and in the general population (using a slightly modified version).

The FACT-GP5 item is a single item from the FACT-G that assesses how bothersome the side effects of treatment are for cancer patients. The recall period is the past 7 days, and the item has a 5-category response scale ranging from "0=Not at all" to "4=Very much." This item is being included to assess the overall tolerability of treatment from the participant's perspective.

8.10.4. PRO-CTCAE

The PRO-CTCAE is a PRO measure developed to evaluate symptomatic toxicity in patients in clinical studies of cancer [Basch, 2014]. The PRO-CTCAE was designed to be used as a companion to the CTCAE, the standard lexicon for AE reporting in cancer studies. The PRO-CTCAE includes a library of 124 items representing 78 symptomatic toxicities drawn from the CTCAE. PRO-CTCAE provides a systematic yet flexible tool for descriptive reporting of symptomatic treatment side effects in clinical studies of cancer. In the present study, a subset of items selected from the PRO-CTCAE Version

1.0 Item Library will be administered [Cella, 1993; Cocks, 2007; Kluetz, 2016]. The PRO-CTCAE will be completed by the participants at the times shown in the SoA (Section 1.3).

8.10.5. OSDI

The impact of potential ocular toxicity on function and HRQoL will be assessed with the use of the OSDI. The OSDI is a 12-item questionnaire designed to assess both the frequency of dry eye symptoms and their impact on vision-related functioning [Dougherty, 2011; Schiffman, 2000]. The OSDI has demonstrated good reliability, validity, sensitivity, and specificity, and can be used as a complement to other clinical and subjective measures of dry eye disease by providing a quantifiable assessment of dry eye symptom frequency and the impact of these symptoms on vision-related functioning. The OSDI will be completed by the participants at the times shown in the SoA (Section 1.3).

8.10.6. Patient global impression items

The PGIS assesses the global impression of symptom severity at baseline and subsequent time points. The PGIC serves to rate the global change in symptoms at subsequent time points. In addition to evaluating symptom severity and change, these questions serve as anchors to establish thresholds of clinically meaningful change for the questionnaires in the study.

8.10.7. EuroQol questionnaire (EQ-5D-3L)

The EQ-5D-3L is a standardized instrument for use as a measure of health utility. It is designed for self-completion or interview administration and is cognitively simple, taking only a few minutes to complete.

The EQ-5D-3L self-assessment questionnaire has 2 parts. The first part consists of 5 items covering 5 dimensions (mobility, self-care, usual activities, pain/discomfort, and anxiety/depression). Each dimension is measured by a 3-point Likert scale (no problems, some or moderate problems, and unable or extreme problems). Respondents are asked to choose 1 level that reflects their "own health state today" for each of the 5 dimensions. Respondents can be then classified into 1 of 243 distinct health states. The second part is a 20-cm visual analogue scale (EQ-VAS) that has endpoints labeled "best imaginable health state" and "worst imaginable health state" anchored at 100 and 0, respectively. Respondents are asked to indicate how they rate their own health by drawing a line from an anchor box to that point on the EQ-VAS which best represents their own health on that day. EQ-5D-3L health states are converted to a single summary index by applying a formula that essentially attaches weights to each of the levels in each dimension. The formula is based on the valuation of EQ-5D health states from general population samples.

8.11. Nonprotocol-specified Health Care Resource Utilization

Nonprotocol-specified Healthcare Resource Utilization will be assessed in this study and reported by participants to study staff at each visit. Parameters to be measured include:

- Number of office/outpatient/hospital clinic visits by specialty.
- Number of emergency room/urgent care facility visits.
- Number and duration of inpatient hospitalizations (total nights, including duration by wards [intensive care unit vs. general ward]).

The data collected may be used to conduct exploratory economic analyses.

8.12. General guidance for treatment continuity when participants are unable to come into the clinic

Prior to utilization of any of the measures outlined in this section, discussion, and approval must be obtained from the sponsor.

It is expected that sites participating in clinical studies will make every effort to ensure proper monitoring and well-being of enrolled participants by adhering to safety monitoring as outlined in the SoA (Section 1.3). The use of local laboratories and local radiology centers to reduce the need for a participant to come into the clinic is supported if deemed necessary for the well-being of the participant. These local facilities should be added to regulatory documents, as required.

Any restrictions in place at the site that will impact monitoring and/or participant access to the site and care providers should be communicated to the sponsor.

A global TM platform that allows for continued monitoring of AEs, concomitant medications, protocol deviations, etc., may be engaged. Discussions around utilization of this technology should be held on a per-site basis, and appropriate documentation of utilization should be captured.

In general, for participants with limited possibility to travel or decreased capacity to come to the clinic, it is acceptable to replace non-dosing in-person visits with telephone contact, home healthcare, or use of an alternative location for assessment (e.g., local laboratories and imaging centers, at-home collection of PROs).

9. STATISTICAL CONSIDERATIONS

The SAP will include a detailed description of the statistical analyses described in this section.

This section is a summary of the planned statistical analyses of the primary and secondary endpoints.

9.1. Statistical hypotheses

The dual primary objectives of the study are to demonstrate if either:

- BRd prolongs PFS compared with DRd,

  H_0: $S_c(t) \geq S_t(t)$ vs. H_1: $S_c(t) < S_t(t)$ for all timepoints t where $S_t(t)$ and $S_c(t)$ is the BRd and DRd PFS, respectively.

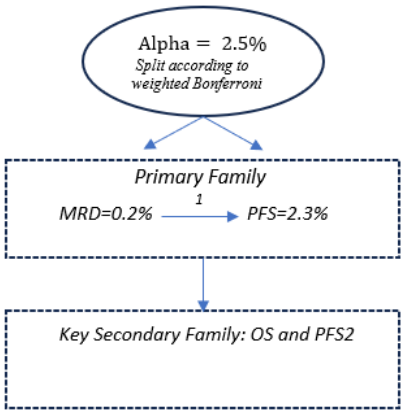
- BRd improves the MRD negative rate compared with DRd,

  H_0: $MRD_t - MRD_c \leq 0$ vs. H_1: $MRD_t - MRD_c > 0$, where MRD_t is the MRD negative rate of BRd and MRD_c is the MRD negative rate of DRd.

9.1.1. Multiplicity adjustment

The multiplicity strategy for the study is presented graphically in Figure 3 [Bretz, 2009]. The study is strongly controlled at the 2.5% 1sided alpha level with 0.2% allocated to MRD and 2.3% allocated to PFS. The alpha can be recycled between the primary endpoints of MRD and PFS so that if MRD is statistically significant, PFS can then be tested at the 2.5% level. Once the primary endpoints have been tested, and assuming PFS is statistically significant, alpha can be passed on to the key secondary family. The hierarchy of the second family of hypotheses (OS and PFS2) will be specified in the SAP.

Figure 3 Multiplicity strategy



PFS will be evaluated in 3 interim analyses and at the PFS primary analysis. The Lan DeMets O'Brien and Fleming [Lan, 1983; O'Brien, 1979] method will be used for defining the futility (non-binding) and early efficacy boundaries. The boundaries will be adjusted based on the actual number of PFS events observed at the time of each analysis.

MRD will be evaluated only once (any MRD subsequent analyses will be descriptive), and the PFS IA1 will coincide with it.

If the PFS null hypothesis is rejected, the key secondary endpoints will be sequentially tested each at the available alpha, either at the IA2 or IA3 (if PFS is positive) or otherwise at the PFS primary analysis. The PFS primary analysis will be skipped if PFS is positive either at IA2 or IA3.

OS will be tested across 5 planned analyses. The Lan DeMets O'Brien and Fleming method will be used to control alpha [Lan, 1983; O'Brien, 1979] while a fixed HR=1.3 will be used to provide guidance for OS futility (non-binding). The boundaries will be adjusted based on the actual number of OS events observed at the time of analysis and the alpha allocated.

9.2. Analysis sets

Table 25 Analysis sets

Analysis Set	Definition/Criteria	Main Analysis in Scope
Screened	• All participants who were screened for eligibility and signed the ICF. This population will be used for the screen failure summary.	• Study population
Enrolled	• The Enrolled population is defined as all participants who have entered the study (e.g., participants who are identified on the Screen Failure form as non-screen failures).	• Study population
Randomized (ITT)	• All participants who were randomly assigned to study intervention in the clinical study. • Data will be reported according to the assigned randomized study intervention.	• Study population, Demography, Efficacy
Safety	• All participants in the Randomized analysis set, who received at least 1 dose of study intervention. • Participants will be assigned to the actual study intervention group if they received that intervention for more than 50% of the cumulative dose.	• Safety, Exposure
PK	• All participants in the Safety analysis set who had at least 1 non-missing belantamab mafodotin PK assessment (nonquantifiable values will be considered as non-missing values). • Data will be reported according to the actual study intervention.	• PK
ADA	• All participants in the Safety analysis set with at least 1 immunogenicity sample with result.	• Immunogenicity

Additional analysis sets may be defined in the SAP, if required.

9.3. Statistical analyses

9.3.1. General considerations/definitions

Analyses will be conducted in the analysis sets as per Table 25. All confidence intervals will be 2-sided at the 95% confidence level unless otherwise specified. Disease assessments will be based on the IMWG criteria [Kumar, 2016] and will be determined by an IRC unless otherwise stated.

9.3.1.1. Extent of exposure

The extent of exposure, dose intensity, and relative dose intensity as well as number of cycles, cumulative dose, and dose modifications will be summarized by treatment component and by treatment arm.

9.3.1.2. Adverse events

AEs will be graded by the investigator according to the NCI-CTCAE v5.0.

Events will be summarized by frequency and proportion of total participants, by System Organ Class and Preferred Term in decreasing frequency. Separate summaries will be given for all AEs, TEAEs, SAEs, AEs leading to discontinuation of study intervention, and fatal AEs. AEs, if listed in the NCI-CTCAE v5.0, will be summarized by the maximum grade. Otherwise, AEs will be summarized by maximum intensity.

The characteristics (e.g., number of occurrences, action taken, grade, etc.) of the AESIs (as per Section 8.4.4) will be summarized separately.

The incidence and the primary cause of death will be summarized.

9.3.1.3. Clinical laboratory evaluations

Selected hematology and clinical chemistry data will be summarized using frequencies and proportions according to NCI-CTCAE v5.0. Laboratory test results, outside the reference ranges, that do not have an associated NCI-CTCAE criteria, will be summarized using proportions. Further details will be provided in the SAP.

9.3.1.4. Other safety measures

Data for ophthalmic examination findings will be summarized. For continuous variables, these summaries will include sample size (n), mean, median, standard deviation, minimum, and maximum. For categorical variables, the summaries will include frequencies and corresponding percentages. Details will be provided in the SAP.

9.3.1.5. Biomarker analyses

The results of biomarker investigations may be reported separately from the main CSR. All endpoints of interest from all comparisons will be summarized descriptively and/or graphically as appropriate to the data.

Additional exploratory analyses may be performed to further characterize novel biomarkers.

Additional exploratory biomarker analyses (including, but not limited to, blood samples, tumor tissue, and imaging) to identify factors important for MM may also be pursued to identify the following: responsiveness and resistance mechanism to belantamab mafodotin. Blood samples (including plasma and serum) and tumor tissue may undergo genomic, proteomic, metabolomic, and/or transcriptional analyses. Images may be used for analysis by radiomics, tumor growth kinetics, etc.

Further details will be provided in the SAP and/or a separate biomarker analysis plan.

9.3.1.6. Pharmacokinetic analyses

Concentration-Time Data: Linear and semi-logarithmic mean and median concentration-time profiles (when appropriate) will be plotted for belantamab mafodotin. Concentrations of belantamab mafodotin (ADC) and cys-mcMMAF will be listed for each participant and summarized (when appropriate) by planned time point.

Derived Pharmacokinetic Parameters: Plasma belantamab mafodotin (ADC) and cys-mcMMAF concentration-time data may be combined with data from other studies and analyzed using a population PK approach. The initial analysis may use the current population PK model at the time of the analysis to generate post hoc PK parameter estimates for the individual participants in Arm A. Results may be reported separately.

9.3.1.7. Pharmacokinetic/pharmacodynamic analyses

If deemed appropriate and if data permit, exposure-response relationships between belantamab mafodotin exposure (e.g., concentration, Cmax, or average concentration) and clinical activity and/or toxicity (e.g., response, corneal event) may be explored using population methods. Data from this study may be combined with data from other studies for this purpose. If data permit, the effects of covariates may be explored. Results may be reported separately.

9.3.1.8. Pharmacogenetic analyses

Further details regarding pharmacogenetics analyses will be addressed in Section 10.5 and the pharmacogenetics SAP.

9.3.2. Primary endpoints/estimands analysis

9.3.2.1. Definition of endpoints and estimands

- PFS is defined as the time from randomization until the earliest date of confirmed PD, determined by an IRC, according to the IMWG criteria [Kumar, 2016], or death due to any cause. Censoring rules are provided in Section 10.11.

- MRD negative status is defined at the 10^{-5} sensitivity threshold assessed by NGS at least once during the time of confirmed CR or better response per IMWG criteria by IRC. Participants with a confirmed CR or better response who do not achieve MRD negative status (including missing/inconclusive assessment(s)) and participants without a confirmed CR or better response will be considered as having non-negative MRD.

9.3.2.2. Main analytical approach

Section 9.1 details the statistical hypotheses tested. Analysis of additional estimands will be included in the SAP.

- For the analysis of PFS, the median, 25th, and 75th percentiles will be estimated along with the corresponding 95% CI using the Brookmeyer-Crowley method [Brookmeyer, 1982].

  The distribution of PFS will be compared, between the treatment arms, using log-rank test stratified by the randomization factors used for randomization, at the nominal alpha in accordance with the multiplicity strategy. The HR and corresponding 95% CI will be estimated from a Cox proportional hazard model stratified by the randomization factors with treatment arm as the sole explanatory variable.

  The proportional hazard assumption will be checked through the Kaplan-Meier plot, log(log[survival]) against log(survival time) plot, Schoenfeld residuals, and evaluation of time dependency of HR by adding an interaction term of time by treatment in the Cox proportional hazard model. If the proportional hazard

assumption does not hold, RMST or piecewise HR may be provided additionally, as appropriate, as additional estimands.

- The number, percentage of participants who are MRD negative and the difference in proportions between treatment arms will be presented along with the corresponding exact 95% CI. The p-value will be calculated using the Cochran Mantel Haenszel test stratified by the randomization factors.

9.3.2.3. Sensitivity analyses

PFS analyses exploring differing censoring/event rules to the primary ones, using the same methodology for PFS as described in Section 9.3.2.2, will be conducted. The alternative censoring rules for the sensitivity analyses will be provided in the SAP.

9.3.2.4. Supplementary/supportive analyses

Supplementary analyses for both PFS and MRD will be conducted using investigator-assessed response.

MRD analyses (as per Section 9.3.2.2) will be repeated at the sensitivity threshold of 10^{-6}.

For assessment of internal consistency and investigation of homogeneity of the treatment effect across subgroups, analyses of the primary (and selected secondary efficacy endpoints) will be conducted based on pre-specified subgroups explored, e.g., by age (<75 years, ≥75 years), sex (Female, Male), ISS (I, II, III), region (North America, Rest-of-World), High Risk Cytogenetics (High Risk, Standard Risk). Details will be provided in the SAP.

9.3.3. Secondary endpoint/estimand analyses

9.3.3.1. Definition of endpoints and estimands

- OS is defined as time from the date of randomization to the date of death due to any cause.
- PFS2 is defined as the time from the date of randomization to the date of disease progression (investigator-assessed) following the first subsequent anti-myeloma therapy, or death from any cause, whichever is earlier. If progression after starting the new anti-myeloma therapy cannot be measured, a PFS2 event is defined as the date of discontinuation of first new anti-myeloma therapy, or death from any cause, whichever is earlier.
- CR+, is defined as confirmed CR or better (per IMWG criteria by IRC) and will be based on BOR from treatment start date until confirmed disease progression or initiation of new anti-myeloma therapy (whichever is first).
- VGPR+ is defined as confirmed VGPR, CR or better (per IMWG criteria by IRC), also based on BOR.

- sMRD rate is defined as the percentage of participants who achieve MRD negative status at 10^{-5} sensitivity threshold assessed by NGS at least twice, a minimum of 12 months apart and with no MRD positive (or indeterminate [insufficient information to determine MRD status]) result in between, during the time of confirmed CR+ response per IMWG criteria [Kumar, 2016] by IRC. Participants who do not achieve MRD negative status or participants without a confirmed CR+ response will be considered as not achieving sMRD negativity.
- DoR is defined as the time from first documented evidence of PR until PD or death due to PD, among participants who achieve confirmed PR (responses per IMWG criteria by IRC).
- TSST is defined as the time from randomization until the date of start of second subsequent line of anti-myeloma therapy (irrespective of PD) or death due to any cause, whichever is earlier.

9.3.3.2. Main analytical approach

PFS2 and OS analyses will be conducted using stratified log-rank test and Cox regression as for the main analysis of PFS. Determination of dates of PFS2 events and dates for censoring will be provided in the SAP.

For OS, participants who are alive at the time of analysis will be censored at last date at which they were known to be alive. Attempts to obtain survival status may occur following DCO and prior to data extract. If participants are confirmed to be alive or if the death date is after the DCO, then the participant will be censored at the date of DCO.

The number and percentage of participants with CR+, VGPR+ and sMRD will be summarized by treatment arm. The difference between treatment arms along with the corresponding 95% exact CIs will be provided.

The distribution of DoR will be summarized using the Kaplan-Meier method by treatment arm for those participants with confirmed PR or better. TSST will also be summarized using the Kaplan-Meier method with median, 25th, and 75th percentiles estimated along with the corresponding 95% CI.

Safety analyses are described in Section 9.3.1.2.

Patient-reported outcome questionnaires will be analyzed descriptively either by counts and proportions or mean, standard deviation, median, min, and max of the actual value and the change from baseline if applicable.

9.3.4. Safety analyses

The Safety analysis set will be used for the analysis of safety data. All serially collected safety endpoints (e.g., laboratory tests, ECGs) will be summarized according to the scheduled, nominal visit at which they were collected and across all on-treatment time points using a "worst-case" analysis. Complete details of the safety analyses will be provided in the SAP.

9.4. Interim analyses

The study will include 3 PFS interim analyses: a) the first IA is at approximately 69 events and will coincide with the MRD analysis, b) 2 subsequent IA will be at approximately 138 and 175 PFS events.

The Lan DeMets O'Brien and Fleming [Lan, 1983; O'Brien, 1979] method will determine the early efficacy and the non-binding futility boundaries, which are summarized in Table 26. Boundaries will be adjusted according to the exact IF at the analysis time. The statistical analysis plan will describe the planned interim analyses in greater detail.

Table 26 Boundaries for PFS interim analyses

PFS Analysis	IF	Events	Cum. Alpha Spent	Cum. Beta Spent	Futility Boundary (Non-binding) P-value Scale	Efficacy Boundary P-value Scale
PFS a=2.3%						
IA1	35%	69	0.0001	0.002	0.794	0.0001
IA2	70%	138	0.007	0.002	Not applicable	0.006
IA3	89%	175	0.016	0.002	Not applicable	0.014
Primary PFS	100%	197	0.023	0.062	0.018	0.018
PFS a=2.5% *(assumes MRD is statistically significant)*						
IA1	35%	69	0.0002	0.001	0.809	0.0002
IA2	70%	138	0.007	0.001	Not applicable	0.007
IA3	89%	175	0.017	0.001	Not applicable	0.015
Primary PFS	100%	197	0.025	0.058	0.019	0.019

9.4.1. Sequence of interim and other planned analyses

Table 27 Summary of planned interim analyses

Analysis	Purpose	Endpoint	Timing
IA1	MRD efficacy and PFS futility (nominal alpha allocated as well)	MRD, PFS, OS, key safety and exposure	At MRD data maturity for the first 371 participants (expected PFS IF ~35% in the Randomized (ITT) population, i.e., ~69 PFS events)
IA2	PFS efficacy	PFS (key and other supportive secondary endpoints if PFS successful, MRD descriptive).	~138 PFS events (~70% IF)
IA3	PFS efficacy	PFS (key and other supportive secondary endpoints if PFS successful, MRD descriptive)	~175 PFS events (~89% IF)
Primary PFS	Primary PFS Analysis	All endpoints	~197 PFS events (100% IF)
OS at ~80% IF	OS efficacy	OS, PFS (descriptive), key safety	~203 OS events (~80% IF)
Final OS	Final OS analysis	OS, PFS (descriptive), key safety	~254 OS events (100% IF)

Notes:

- IA1 is driven by MRD data maturity for the first 371 participants and may not be conducted prior to accrual completion for the entire study sample size. Subsequent PFS interim analyses will be PFS event driven. All PFS interims will use all randomized study participants.
- All available data, specific to the endpoints analyzed, at the time of data cut will be used, unless otherwise specified.
- Analysis of secondary endpoints will occur only once when PFS is statistically significant. Upon successful rejection of the PFS hypothesis and regardless of the timing of rejection, the full alpha allocated to testing the hypothesis can be propagated. The PFS and OS alpha will be adjusted to account for multiple testing across timepoints. The boundaries will be adjusted based on the observed number of events at the time of analysis. MRD may be descriptively summarized at the time of PFS statistical significance.

9.4.2. Statistical considerations associated with the interim analyses

The interim analyses will be conducted such that the ongoing study integrity is maintained. Any interim analyses of the primary endpoints will be performed by an independent statistical support group, who is responsible for providing the IA results to the IDMC (Section 10.1.6.2).

The analysis methods for the primary efficacy endpoints, as described in Section 9.3.2.2 will be used for the interim analyses.

Only the independent statistical support group will be unblinded to aggregated data by treatment group. The IA results will not be shared with investigators, participants, or the study team who are involved in the conduct of the study before the final database lock.

9.4.3. CPMS early access

GSK CPMS analysts or delegate(s) not involved in the study conduct will have access to a blinded population PK dataset (including, but not limited to, concentration, actual dosing information, demographics, and laboratory information, but excluding AE and efficacy information) at several time points throughout the study for population PK model development/refinement. Additionally, designated representatives not involved with study conduct may be unblinded for performing population PK and PKPD dataset preparation in support of planned analyses and PK display review. All other personnel will remain blinded to aggregate data by treatment group until database lock.

9.5. Sample size determination

The 2.5% 1-sided alpha will be prospectively split between PFS and MRD dual primary endpoints as 2.3% and 0.2% respectively. If the MRD null hypothesis is rejected, the alpha will be passed on to PFS. Calculations were conducted using East 6.5.4.

PFS

Based on data from the MAIA study, the median PFS in DRd arm is expected to be approximately 62 months [Facon, 2019; Moreau, 2022]. It is envisaged that belantamab mafodotin, in combination with lenalidomide and dexamethasone, will lead to at least 40% reduction in the risk of progression or death, i.e., an expected HR=0.60. This corresponds to an increase in median PFS from 62 to 103 months under the exponential assumption.

With an overall alpha of 2.3%, the PFS primary analysis will be conducted after observing approximately 197 PFS events in approximately 520 participants. The study will have 93.7% power to detect a HR of 0.60. If the MRD null hypothesis is rejected, with a 1-sided alpha of 2.5%, the study will have slightly increased power (94.1%) for the same HR of 0.60.

Assuming a uniform enrollment rate of approximately 30 participants per month, and randomization ratio 1:1, enrollment is expected to continue for approximately 17 months. With an annual dropout rate of 5% it is estimated that the targeted PFS events will be observed at approximately 72 months from the time when the first participant is randomized.

MRD

The study can achieve 90% power to detect a difference in the MRD negative rate between BRd and DRd arms of 20% or more with a subset of the first 371 participants. This is assuming a DRd MRD negative rate of 24% [Facon, 2019], with an accrual rate of approximately 30 participants/month, alpha 0.2% (1-sided) and lead time for the MRD assessment of approximately 12 months.

The MRD read-out is expected, under these assumptions, at approximately 24 months following first participant randomization. MRD will be statistically tested only once with any subsequent analyses being descriptive in nature.

OS

Long-term survival follow-up will continue until 254 deaths have been observed or 5 years after the last participant is randomized. With 1-sided alpha 2.5%, the study can achieve 80% power (79.2% if alpha 2.3%) to detect a 30% risk reduction of death (HR=0.70). The calculation allows for 5 OS analyses using Lan DeMets O'Brien and Fleming method for alpha spending [Lan, 1983; O'Brien, 1979] and a fixed non-binding futility boundary of HR >1.30. The IF may shift depending on the actual observed OS events at the timing of each analysis and therefore, the boundaries will be adjusted accordingly. Further details are provided in the SAP.

10. SUPPORTING DOCUMENTATION AND OPERATIONAL CONSIDERATIONS

10.1. Appendix 1: Regulatory, ethical, and study oversight considerations

10.1.1. Regulatory and ethical considerations

- This study will be conducted in accordance with the protocol and with the following:
  - Consensus ethical principles derived from international guidelines including the Declaration of Helsinki and Council for International Organizations of Medical Sciences international ethical guidelines.
  - Applicable ICH GCP guidelines.
  - Applicable laws and regulations.
- The protocol, protocol amendments, ICF, IB, and other relevant documents (e.g., advertisements) must be submitted to an IRB/IEC by the investigator and reviewed and approved by the IRB/IEC before the study is initiated.
- Any amendments to the protocol will require IRB/IEC approval before implementation of changes made to the study design, except for changes necessary to eliminate an immediate hazard to study participants.
- Protocols and any substantial amendments to the protocol will require health authority approval prior to initiation except for changes necessary to eliminate an immediate hazard to study participants.
- The investigator will be responsible for the following, as applicable:
  - Providing written summaries of the status of the study to the IRB/IEC annually or more frequently in accordance with the requirements, policies, and procedures established by the IRB/IEC.
  - Notifying the IRB/IEC of SAEs or other significant safety findings as required by IRB/IEC procedures.
  - Providing oversight of the conduct of the study at the site and adherence to requirements of 21 CFR, ICH guidelines, the IRB/IEC, European regulation 536/2014 for clinical studies (if applicable), European Medical Device Regulation 2017/745 for clinical device research, and all other applicable local regulations.

10.1.2. Financial disclosure

Investigators and sub-investigators will provide the sponsor with sufficient, accurate financial information as requested to allow the sponsor to submit complete and accurate financial certification or disclosure statements to the appropriate regulatory authorities. Investigators are responsible for providing information on financial interests during the course of the study and for 1 year after completion of the study.

10.1.3. Informed consent process

- The investigator or the investigator's representative will explain the nature of the study, including the risks and benefits, to the participants and answer all questions regarding the study.
- Potential participants must be informed that their participation is voluntary. They will be required to sign a statement of informed consent that meets the requirements of 21 CFR 50, local regulations, ICH guidelines, privacy and data protection (e.g., HIPAA and GDPR requirements), where applicable, and the IRB/IEC or study center.
- Sample testing will be done in accordance with the recorded consent of the individual participant.
- By default, collected samples for the study will be stored for a maximum of 20 years. This storage period begins when the last participant performs the last study visit. This timeline can be adapted based on local laws, regulations, or guidelines requiring different timeframes or procedures. In all cases, the storage period should be aligned with participant's consent. These additional requirements must be formally communicated to, discussed, and agreed with GSK.
- The medical record must include a statement that written informed consent was obtained before the participant was enrolled in the study and the date the written consent was obtained. The authorized person obtaining the informed consent must also sign the ICF.
- Participants must be reconsented to the most current version of the ICF(s) during their participation in the study.
- A physical copy of the ICF(s) must be provided to the participant.
- A participant who is rescreened is not required to sign another ICF if the rescreening occurs within 30 days from the previous ICF signature date.
- In case of unexpected pregnancy, participant must be informed that personal information such as date of birth, sex of the baby will be collected as part of safety follow-up. Consent for collection of information about the baby may be obtained from the participant and/or their partner per local regulations.
- If partners of male participants become pregnant during the study, consent will need to be obtained or notification given per local regulation to the partner before collecting their personal information such as last menstrual period, year of birth or date of birth, sex of their baby as part of safety follow-up.

10.1.4. Recruitment strategy

Recruitment will be performed by the investigators at the participating sites. The sponsor will NOT participate in recruitment of participants to the study. Prior to selecting a site for inclusion in the study, data will be gathered to understand the numbers of participants that they may be able to enroll from their own patients and networks. The sponsor and a third party vendor will prepare informational material such as infographics, pamphlets, poster, SoA table and a patient letter to be distributed to the sites to support the investigators in the recruitment process. In addition, the study will be published in public databases such as ClinicalTrials.gov.

Recruitment will be monitored throughout the study and mitigation plans put in place if needed.

10.1.5. Data protection

- Participants will be assigned a unique identifier by the investigator. Any participant records or datasets that are transferred to the sponsor will contain the identifier only; participant names or any information which would make the participant identifiable will not be transferred.
- GSK will ensure protection of the personal data of the investigator and site staff which is collected within the framework of and for the purpose of the study.
- The participant must be informed that his/her personal study-related data will be used by the sponsor in accordance with local data protection law. The level of disclosure must also be explained to the participant, that his/her data will be used as described in the informed consent.
- The participant must be informed that his/her medical records may be examined by Clinical Quality Assurance auditors or other authorized personnel appointed by the sponsor, by appropriate IRB/IEC members, and by inspectors from regulatory authorities.
- The contract between sponsor and study sites specifies responsibilities of the parties related data protection, including handling of data security breaches and respective communication and cooperation of the parties.
- Information technology systems used to collect, process, and store study-related data are secured by technical and organizational security measures designed to protect such data against accidental or unlawful loss, alteration, or unauthorized disclosure, or access. GSK and/or trusted third parties working on behalf of GSK and/or institutions working with GSK for the purposes of this study are contractually bound to protect participant coded data. GSK will protect participant coded data and will only share it as described in the ICF.
- GSK has a global, internal policy that requires all GSK staff and complementary workers to report data incidents or breaches immediately, using dedicated tools. Clear procedures are defined for assessing and investigating data breaches to identify and to take appropriate remediation steps, to contain and to mitigate any risks for individuals resulting from a breach, in compliance with applicable laws.

10.1.6. Committees structure

An SRT is in place for each GSK product. It comprises a global cross-functional team responsible for the ongoing assessment of benefit-risk for a product. The SRT contribute to the continual assessment of incoming new efficacy and safety information. Participant safety will be continuously monitored by the GSK Medical Monitor, designated safety lead (or delegate) and sponsor's internal committee (if in place), throughout the study. Pertinent findings and conclusions are shared with the product's SRT for review of the overall benefit-risk profile of the product.

10.1.6.1. Independent Review Committee

This study will utilize an IRC to evaluate response and disease progression per IMWG criteria.

The IRC is a blinded, external committee composed of specialists who are experts in the diagnosis, assessment, and treatment for TI-NDMM, as well as the medical aspects of clinical studies. The members of the IRC cannot be employees of GSK, and no member of the IRC may be an investigator for the study or involved in the conduct of the study in any capacity. Details concerning the IRC process are documented in the IRC Charter (e.g., the clinical data required, how events will be classified, the IRC adjudication process, consensus algorithm). The IRC will remain blinded to all treatment arm assignments, as no treatment information will be provided to the IRC members. Data required for IRC assessment (e.g., laboratory, imaging and BM data) will be reviewed and adjudicated. Further details on the IRC process will be defined in the IRC Charter. Results from the IRC will not be provided to the investigator sites.

10.1.6.2. Independent Data Monitoring Committee

An IDMC consisting of at least 2 physicians and 1 statistician as defined in the IDMC Charter will review data from the IA and periodic safety reviews. Details of the interim analyses are included in Section 9.4. Details of safety reviews will be included in the IDMC Charter.

Ad hoc meetings may be convened at the discretion of the IDMC or if requested by the sponsor. Additional details will be provided in the IDMC Charter.

10.1.7. Dissemination of clinical study data

- The key design elements of this protocol and results summaries will be posted on www.ClinicalTrials.gov and/or GSK Study Register in compliance with applicable regulations/GSK policy. GSK will aim to register protocols summaries prior to study start and target results summaries submission within 6 months of primary/study completion date (pediatric population) and within 12 months of primary/study completion date (adult population). Where external regulations require earlier disclosure, GSK will follow those timelines.

- Where required by regulation, summaries will also be posted on applicable national or regional clinical study registers.
- Where required by applicable regulatory requirements, an investigator signatory will be identified for the approval of the study report, and provided reasonable access to statistical tables, figures, and relevant reports. GSK will also provide the investigator with the full summary of the study results, including a summary of clinical study results understandable to laypersons. The investigator is encouraged to share the layperson summary of results with the study participants, as appropriate. The full study report will be made available upon request, after decision on marketing authorization by regulatory authorities.
- Where required by regulation, the names of the sponsor signatory and investigator signatory will be made public.
- GSK will provide the investigator with the randomization codes and participant-level line listings for their site only after completion of the full statistical analysis.
- GSK intends to make anonymized participant-level data from this study available to external researchers for scientific analyses or to conduct further research that can help advance medical science or improve patient care. This helps ensure the data provided by study participants are used to maximum effect in the creation of knowledge and understanding. Data will be shared with researchers in a non-identifying way, and appropriate measures will be taken to protect personal information; these measures will comply with data protection and privacy laws that apply.

10.1.8. Data quality assurance

- All participant data relating to the study will be recorded on printed or electronic CRFs unless transmitted to the sponsor or designee electronically (e.g., laboratory data). The investigator is responsible for verifying that data entries are accurate and correct by physically or electronically signing the CRF.
- Guidance on completion of eCRFs will be provided in eCRF completion guidelines.
- The investigator must permit study-related monitoring, audits, IRB/IEC review, and regulatory agency inspections, and provide direct access to source documents.
- QTLs will be predefined in the QTL plan to identify systematic issues that can impact participant right, safety, and/or reliability of study results. These predefined parameters will be monitored during the study, and important deviations from the QTLs and remedial actions taken will be summarized in the CSR.
- Monitoring details describing strategy, including definition of study critical data items and processes (e.g., risk-based initiatives in operations and quality such as risk management and mitigation strategies and analytical risk-based monitoring, involvement of central reading mechanism) methods, responsibilities, and requirements, including handling of noncompliance issues and monitoring techniques (central, remote, or on-site monitoring) are provided in the monitoring plan.

- The sponsor or designee is responsible for the data management of this study, including quality checking of the data.
- The sponsor assumes accountability for actions delegated to other individuals (e.g., contract research organizations).
- Records and documents, including signed ICF, pertaining to the conduct of this study must be retained by the investigator for a minimum period of 15 years from the issue of the final CSR/equivalent summary, or in accordance with applicable law, whichever is longer. No records may be destroyed during the retention period without the written approval of the sponsor. No records may be transferred to another location or party without written notification to the sponsor. In the event of a conflict between this protocol and the fully executed clinical study agreement, the protocol shall prevail with respect to records retention.
- When source data are sent for external assessment or adjudication (e.g., endpoint adjudication committee; expert reader), source data are stored by the external body for agreed timeframe according to the charter or contract.

10.1.9. Source documents

- For this study there will not be source data recorded directly into the eCRF (i.e., no prior written or electronic record of data is available).
- Source documents provide evidence for the existence of the participant and substantiate the integrity of the data collected. Source documents are filed at the investigator's site.
- Data reported on the eCRF or entered in the eCRF that are transcribed from source documents must be consistent with the source documents or the discrepancies must be explained. The investigator may need to request previous medical records or transfer records, depending on the study. Also, current medical records must be available.
- Definition of what constitutes source data and its origin can be found in Investigator Source Agreement.
- The investigator must maintain accurate documentation (source data) that supports the information entered in the eCRF.
- The sponsor or designee will perform monitoring to confirm that data entered into the eCRF by authorized site personnel are accurate, complete, and verifiable from source documents; that the safety and rights of participants are being protected; and that the study is being conducted in accordance with the currently approved protocol and any other study agreements, ICH GCP, and all applicable regulatory requirements.

- Source data are shared with third parties contracted by GSK for external assessment or adjudication (e.g., endpoint adjudication committee; expert reader). The non-exhaustive list of source data shared may include discharge summaries, imaging reports, scans, videos, pathology reports, biological specimens, ECG reports, etc. Participant names or any information that would make the participant identifiable or is not essential for the external assessment or adjudication will be redacted by the investigator sites prior to transfer. Details of the participant information redaction strategy are provided in the relevant third party manuals and/or study plans. These source data will be used by the third party solely for the purpose indicated within this protocol.

10.1.10. Study and site start and closure

Start of study and first act of recruitment

The start of study and the first act of recruitment are defined as FSFV (first ICF signature date) at a country-level.

Study/Site termination

GSK or designee reserves the right to close the study site or terminate the study at any time for any reason at the sole discretion of GSK. Study sites will be closed upon study completion. A study site is considered closed when all required documents and study supplies have been collected and a study site closure visit has been performed.

The investigator may initiate study site closure at any time, provided there is reasonable cause and sufficient notice is given in advance of the intended termination.

Reasons for the early closure of a study site by the sponsor or investigator may include but are not limited to:

For study termination:

- Discontinuation of further study intervention development.

For site termination:

- Failure of the investigator to comply with the protocol, the requirements of the IRB/IEC or local health authorities, the sponsor's procedures, or GCP guidelines.
- Inadequate or no recruitment (evaluated after a reasonable amount of time) of participants by the investigator.
- Total number of participants included earlier than expected.

If the study is prematurely terminated or temporarily suspended, the sponsor shall promptly inform the investigators, the IECs/IRBs, the regulatory authorities, and any contract research organization(s) used in the study of the reason for termination or temporary suspension, as specified by the applicable regulatory requirements. The investigator shall promptly inform the participant and should assure appropriate participant therapy and/or follow-up.

10.1.11. Publication policy

GSK seeks to publish medically or scientifically significant results in searchable peer-reviewed scientific literature within 18 months from LSLV. GSK follows International Committee of Medical Journal Editors standards for authorship and uses good publications practices to guide publications.

10.2. Appendix 2: Clinical laboratory tests

- The tests detailed in Table 28 will be performed by either the local laboratory or the central laboratory as indicated.
- Protocol-specific requirements for inclusion or exclusion of participants are detailed in Section 5 of the protocol.
- Additional tests may be performed at any time during the study as determined necessary by the investigator or required by local regulations.
- Pregnancy Testing: Refer to Section 5.1 Inclusion Criteria for screening pregnancy criteria and the SoA (Section 1.3) for pregnancy testing schedule.
- Additional serum or urine pregnancy tests may be performed, as determined necessary by the investigator, or required by local regulation, to establish the absence of pregnancy at any time during the participant's participation in the study.
- The addresses of the clinical laboratories in charge of the testing can be found in the "List of Clinical Laboratories and Key Vendors."

Table 28 Protocol-required laboratory assessments

Hematology[1]			
Platelet count	**RBC indices:** MCV MCH MCHC	**Automated WBC differential:** Neutrophils Lymphocytes Monocytes Eosinophils Basophils	
RBC count			
WBC count (absolute)			
Reticulocyte count			
Hemoglobin			
Hematocrit			
Clinical chemistry[1]			
Creatinine	Potassium	AST	Total and direct bilirubin
Glucose	Chloride	ALT	Uric acid
Sodium	Total bicarbonate	GGT	Albumin
Magnesium	Calcium	ALP	Total protein
eGFR	Phosphorus	CK[7]	LDH[7]
Urinalysis[1]			
Optional urine dipstick for protein [8]			
Other laboratory tests			
Spot urine (from first void) for albumin/creatinine ratio[1, 3]; Coagulation screen ± INR if on coumarin (warfarin)[1]; Pregnancy Test (urine or blood)[1]; FSH[1] and estradiol[1] (as needed in WONCBP only); HBsAg[1]; HBcAb[1], Hepatitis C Antibody, Hepatitis C RNA test[1,3] (optional), eGFR[1] Note: Hepatitis C RNA testing will be done to determine participant's eligibility if Hepatitis C antibody positive. HBV DNA testing will be done to determine participant's eligibility if HBcAb positive or HBsAg positive.			
PK and ADA[2]			
Belantamab mafodotin (ADC) and cys-mcMMAF PK ADAs to belantamab mafodotin			

Disease evaluation laboratory tests			
UPEP[2]	Serum immunofixation[2]	Ca corrected for albumin (serum)[2]	Serum M-protein calculation[2]
SPEP[2]	Urine immunofixation[2]	IgG[2], IgM[2], IgA[2], IgD/IgE[2,4]	
Serum kappa/lambda FLC ratio[2]	24-hour urine collection for M-protein[2]	Beta-2 microglobulin[2]	
BM aspiration/biopsy			
BM (aspirate and biopsy) for disease assessment[1], BM aspirate for risk assessment (FISH + tp53 mutation) analysis[1, 3, 5]; BM biopsy to confirm sCR (by IHC) [1, 3]; BM aspirate for MRD testing[2]			
Biomarker measurements[2]			
sBCMA (serum); Immunophenotyping			
Optional testing[2]			
Pharmacogenomics sample[6]			

1. To be performed at local laboratory.
2. To be performed at central laboratory.
3. If not available locally it can be performed centrally.
4. Only for participants with IgD/E myeloma.
5. FISH and tp53 mutation testing results are not required prior to randomization or dosing. Central confirmation of FISH and tp53 mutation testing results will be performed when local risk assessment testing is performed for screening.
6. Informed consent for optional sub-studies (e.g., genetic research) must be obtained before collecting a sample.
7. Participants with significant lab elevations (≥3×ULN) should, where possible, have a sample sent for central testing of CK and LDH isoenzyme levels.
8. Urine dipstick for protein may be used to assess for presence of urine protein. Albumin/creatinine ratio needs to be done in any participant with urine dipstick result of ≥1+ (at Screening Visit), or ≥2+ (during study intervention), or with positive protein if urine dipstick protein quantification is not available.

10.3. Appendix 3: AEs and SAEs: Definitions and procedures for recording, evaluating, follow-up, and reporting

10.3.1. Definition of AE

AE definition
• An AE is any untoward medical occurrence in a clinical study participant, temporally associated with the use of a study intervention, whether or not considered related to the study intervention. • Note: An AE can therefore be any unfavorable and unintended sign (including an abnormal laboratory finding), symptom, or disease (new or exacerbated) temporally associated with the use of a study intervention.
Events meeting the AE definition
• Any abnormal laboratory test results (hematology, clinical chemistry, or urinalysis) or other safety assessments (e.g., ECG, radiological scans), including those that worsen from baseline, considered clinically significant in the medical and scientific judgment of the investigator (i.e., not related to progression of underlying disease). • Exacerbation of a chronic or intermittent pre-existing condition including either an increase in frequency and/or intensity of the condition. • New condition detected or diagnosed after study intervention administration even though it may have been present before the start of the study. • Signs, symptoms, or the clinical sequelae of a suspected intervention- intervention interaction. • Signs, symptoms, or the clinical sequelae of a suspected overdose of either study intervention or a concomitant medication. Overdose per se will not be reported as an AE/SAE unless it is an intentional overdose taken with possible suicidal/self-harming intent. Such overdoses should be reported regardless of sequelae. • Significant failure of an expected pharmacologic or biological action. • Events that occur as a result of protocol-mandated procedures (i.e., invasive procedures, modification of participant's previous therapeutic regimen). • "Lack of efficacy" or "failure of expected pharmacological action" per se will not be reported as an AE or SAE. Such instances will be captured in the efficacy assessments. However, the signs, symptoms, and/or clinical sequelae resulting from lack of efficacy will be reported as AE or SAE if they fulfill the definition of an AE or SAE.
Events <u>NOT</u> meeting the AE definition
• Any clinically significant abnormal laboratory findings or other abnormal safety assessments that are associated with the underlying disease, unless judged by the investigator to be more severe than expected for the participant's condition.

- The disease/disorder being studied or expected progression, signs, or symptoms of the disease/disorder being studied, unless more severe than expected for the participant's condition.
- Medical or surgical procedure (e.g., endoscopy, appendectomy): the condition that leads to the procedure is the AE.
- Situations in which an untoward medical occurrence did not occur (social and/or convenience admission to a hospital, admission for routine examination.).
- Anticipated day-to-day fluctuations of pre-existing disease(s) or condition(s) present or detected at the start of the study that do not worsen. Pre-existing diseases will be recorded in the medical history section of the eCRF.
- Hospitalization for elective treatment of a pre-existing condition (known or diagnosed before signing the informed consent) that did not worsen from baseline.
- The event is, in the investigator's opinion, of greater intensity, frequency, or duration than expected for the individual participant, or
- The investigator considers that there is a reasonable possibility that the event was related to the administration of the study intervention.

10.3.2. Definition of SAE

An SAE is defined as any untoward medical occurrence that, at any dose, meets 1 or more of the criteria listed:

a. Results in death

b. Is life-threatening

The term "life-threatening" in the definition of "serious" refers to an event in which the participant was at risk of death at the time of the event. It does not refer to an event that hypothetically might have caused death if it were more severe.

c. Requires inpatient hospitalization or prolongation of existing hospitalization

- In general, hospitalization signifies that the participant has been admitted (usually involving at least an overnight stay) at the hospital or emergency ward for observation and/or treatment that would not have been appropriate in the physician's office or outpatient setting. Complications that occur during hospitalization are AE. If a complication prolongs hospitalization or fulfills any other serious criteria, the event is serious. When in doubt as to whether "hospitalization" occurred or was necessary, the AE should be considered serious.
- Hospitalization for elective treatment of a pre-existing condition that did not worsen from baseline is not considered an AE.

d. Results in persistent or significant disability/incapacity

- The term disability means a substantial disruption of a person's ability to conduct normal life functions.

•	This definition is not intended to include experiences of relatively minor medical significance such as uncomplicated headache, nausea, vomiting, diarrhea, influenza, and accidental trauma (e.g., sprained ankle) that may interfere with or prevent everyday life functions but do not constitute a substantial disruption.
e.	**Is a congenital anomaly/birth defect**
f.	**Abnormal pregnancy outcomes (e.g., spontaneous abortion, fetal death, stillbirth, congenital anomalies, ectopic pregnancy)**
g.	**Other situations:** • Possible Hy's Law case: ALT ≥3×ULN AND total bilirubin ≥2×ULN (for participants with known Gilbert's syndrome these criteria only apply if total bilirubin ≥2×ULN, and direct bilirubin ≥2×ULN and at least doubled from baseline value) or INR >1.5 must be reported as SAE • Medical or scientific judgment should be exercised by the investigator in deciding whether SAE reporting is appropriate in other situations such as significant medical events that may jeopardize the participant or may require medical or surgical intervention to prevent 1 of the other outcomes listed in the above definition. These events should usually be considered serious. – Examples of such events include invasive or malignant cancers, intensive treatment for allergic bronchospasm, blood dyscrasias, convulsions, or development of intervention dependency or intervention abuse.

10.3.3. Definition of cardiovascular events

CV definition
Investigators will be required to fill out the specific CV event page of the eCRF for the following AEs and SAEs: • MI/unstable angina • Congestive heart failure • Arrhythmias • Valvulopathy • Pulmonary hypertension • Cerebrovascular events/stroke and transient ischemic attack • Peripheral arterial thromboembolism • Deep venous thrombosis/pulmonary embolism • Revascularization

10.3.4. Definition of TEAE

TEAE Definition
• A TEAE is an event that emerges during treatment, having been absent pre-treatment or worsens relative to the pre-treatment state.

10.3.5. Recording, assessment, and follow-up of AE, SAE, AESIs, and pregnancies

10.3.5.1. AE and SAE recording

- When an AE/SAE occurs, it is the responsibility of the investigator to review all documentation (e.g., hospital progress notes, laboratory reports, and diagnostics reports) related to the event.
- The investigator will then record all relevant AE/SAE information in eCRF.
- It is not acceptable for the investigator to send photocopies of the participant's medical records to GSK in lieu of completion of the GSK AE/SAE CRF page.
- There may be instances when copies of medical records for certain cases are requested by GSK. In this case, all participant identifiers, with the exception of the participant identifier, will be redacted on the copies of the medical records before submission to GSK.
- The investigator will attempt to establish a diagnosis of the event based on signs, symptoms, and/or other clinical information. Whenever possible, the diagnosis (not the individual signs/symptoms) will be documented as the AE/SAE.

10.3.5.2. Assessment of intensity

Investigators should assess the severity of AEs according to the NCI-CTCAE v5.0 [US Department of Health and Human Services, 2017]. In general, NCI-CTCAE severity grades are as follows:

- Grade 1: Mild; asymptomatic or mild symptoms; clinical or diagnostic observations only; intervention not indicated; easily tolerated.
- Grade 2: Moderate; minimal, local, or noninvasive intervention indicated; limiting age-appropriate instrumental ADLs (instrumental ADLs refer to preparing meals, shopping for groceries or clothes, using the telephone, or managing money).
- Grade 3: Severe or medically significant but not immediately life-threatening; hospitalization or prolongation of hospitalization indicated; disabling; limiting self-care ADL (self-care ADLs refer to bathing, dressing and undressing, feeding self, using the toilet, taking medications, and being not bedridden).
- Grade 4: Life-threatening consequences; urgent intervention indicated.
- Grade 5: Death related to AE.

A distinction should be made between serious and severe AEs. Severity is a measure of intensity, whereas seriousness is defined by the criteria presented in Section 10.3.2. For example, a mild degree of GI bleeding requiring an overnight hospitalization for monitoring purposes may be considered an SAE but is not necessarily severe. Similarly, an AE that is severe in intensity is not necessarily an SAE. For example, alopecia may be assessed as severe in intensity but may not be considered an SAE.

10.3.5.3. Assessment of causality

The investigator is obligated to assess the relationship between study intervention and each occurrence of each AE/SAE. The investigator will use clinical judgment to determine the relationship.

- A *reasonable possibility* of a relationship conveys that there are facts, evidence, and/or arguments to suggest a causal relationship, rather than a relationship cannot be ruled out.
- Alternative causes, such as underlying disease(s), concomitant therapy, and other risk factors, as well as the temporal relationship of the event to study intervention administration, will be considered and investigated.
- For causality assessment, the investigator will also consult the IB and/or product information, for marketed products.
- The investigator must review and provide an assessment of causality for each AE/SAE and document this in the medical notes. There may be situations in which an SAE has occurred and the investigator has minimal information to include in the initial report to GSK. However, it is very important that the investigator always make an assessment of causality for every event before the initial transmission of the SAE data to GSK.
- Where multiple interventions are administered in the same visit, the investigator should specify, when possible, if the AE/SAE could be causally related to a specific study intervention. When a causal relationship to a specific study intervention cannot be determined, the investigator should indicate the AE/SAE to be related to all interventions.
- The investigator may change their opinion of causality in light of follow-up information and send a SAE follow-up report with the updated causality assessment.
- The causality assessment is 1 of the criteria used when determining regulatory reporting requirements.

10.3.5.4. Assessment of outcomes

The investigator will assess the outcome of all serious and nonserious AEs recorded during the study as:

- Recovered/resolved
- Recovering/resolving
- Not recovered/not resolved
- Recovered with sequelae/resolved with sequelae
- Fatal (SAEs only)

10.3.5.5. Follow-up of AEs, SAEs, AESIs, pregnancies, or any other events of interest

- The investigator is obligated to perform or arrange for the conduct of supplemental measurements and/or evaluations as medically indicated or as requested by GSK to elucidate the nature and/or causality of the AE or SAE as fully as possible. This may include additional laboratory tests or investigations, histopathological examinations, or consultation with other health care professionals.
- New or updated information will be recorded in the originally submitted documents (CRF).
- The investigator will submit any updated SAE data to GSK within 24 hours of receipt of the information.
- After the initial AE/SAE/AESI/pregnancy or any other event of interest, the investigator is required to proactively follow each participant at subsequent visits/contacts. All SAEs, and nonserious AESIs (as defined in the Section 8.4.4), will be followed until the event is resolved, stabilized, otherwise explained, or the participant is lost to follow-up.
- Other nonserious AEs must be followed until the event is resolved, stabilized, otherwise explained, or until the participant is lost to follow-up.

Follow-up during the study

- AEs/AESIs corneal events and thrombocytopenia documented at a previous visit/contact and defined as not recovered/not resolved or recovering/resolving will be reviewed at subsequent visits/contacts until the participant withdraws consent for study participation, or until the participant starts subsequent anti-myeloma therapy, whichever occurs first.
- If a participant dies during their participation in the study or during a recognized Follow-up Period, GSK will be provided with any available postmortem findings, including histopathology.

Follow-up of pregnancies

- Pregnant participants/pregnant partners of male participants will be followed to determine the outcome of the pregnancy. At the end of the pregnancy, whether full-term or premature, information on the status of the mother and child will be forwarded to GSK using the paper pregnancy follow-up report/electronic pregnancy report and the AE Report if applicable. Generally, the Follow-up Period does not need to be longer than 6 to 8 weeks after the estimated date of delivery.
- Regardless of the reporting period for SAEs in this study, if the pregnancy outcome is an SAE, it should always be reported as such.
- Furthermore, the investigator must report any SAE occurring as a result of a poststudy pregnancy that is considered by the investigator to be reasonably related to the study intervention, to GSK as described in Section 10.3.5.7.

10.3.5.6. Updating of SAE, AESI, and pregnancy information after removal of write access to the participant's eCRF

When additional SAE, AESI, or pregnancy information is received after write access to the participant's eCRF is removed, new or updated information should be recorded on the appropriate paper report, with all changes signed and dated by the investigator. The updated report should be sent to the study contact for reporting SAEs (refer to Section 8.4.3).

10.3.5.7. Reporting of SAEs, AESIs, and pregnancies

SAE reporting to GSK via an electronic data collection tool

- The primary mechanism for reporting an SAE to GSK will be the electronic data collection tool.
- If the electronic system is unavailable, then the site will use the paper SAE data collection tool (see next section) to report the event within 24 hours.
- The site will enter the SAE data into the electronic system as soon as it becomes available.
- After the study is completed at a given site, the electronic data collection tool will be taken offline to prevent the entry of new data or changes to existing data.
- If a site receives a report of a new SAE from a study participant or receives updated data on a previously reported SAE after the electronic data collection tool has been taken offline, then the site can report this information on a paper SAE form (see next section).
- If the site during the course of the study or poststudy becomes aware of any serious, nonserious AEs, pregnancy exposure, related to any GSK product that is not part of the study design, they will report these events to GSK or to the concerned Competent Authority via the national spontaneous reporting system. These will be classified as spontaneous individual case safety reports.

Contacts for SAE reporting can be found in Section 8.4.8.

SAE reporting to GSK via paper data collection tool

- Facsimile transmission of the SAE paper data collection tool is the preferred method to transmit this information to the SAE coordinator.
- In rare circumstances and in the absence of facsimile equipment, notification by telephone is acceptable with a copy of the SAE data collection tool sent by overnight mail or courier service.
- Initial notification via telephone does not replace the need for the investigator to complete and sign the SAE data collection tool (CRF) within the designated reporting timeframes.

Contacts for SAE reporting can be found in Section 8.4.8.

10.4. Appendix 4: Contraceptive and barrier guidance

Contraceptive use by men and women should be consistent with local regulations regarding the methods of contraception for those participating in clinical studies.

10.4.1. Definitions

10.4.1.1. Woman of childbearing potential

Women in the following categories are considered WOCBP (fertile):

- Adolescents of childbearing potential: Tanner stage ≥2 (post-thelarche) irrespective of the occurrence of menarche or following menarche.
- From the time of menarche until becoming postmenopausal unless permanently sterile (see below).

  NOTE: Menarche is the first onset of menses in a young female. Menarche is normally preceded by several changes associated with puberty including breast development and pubic hair growth.

If fertility is unclear (e.g., amenorrhea in adolescents or athletes) and a menstrual cycle cannot be confirmed before first dose of study intervention, additional evaluation should be considered.

10.4.1.2. Woman of non-childbearing potential

Women in the following categories are considered WONCBP:

- Premenarchal: Tanner stage 1 (prepubertal).
- Permanently sterile due to 1 of the following procedures:
  - Documented hysterectomy
  - Documented bilateral salpingectomy
  - Documented bilateral oophorectomy

    For permanently sterile individuals due to an alternate medical cause other than the above, (e.g., Mullerian agenesis, androgen insensitivity, gonadal dysgenesis), investigator discretion should be applied to determining study entry. If reproductive status is questionable, additional evaluation should be considered.

    NOTE: Documentation can come from the site personnel's review of the participant's medical records, medical examination, or medical history interview.
- Postmenopausal female
- A postmenopausal state is defined as no menses for at least 24 consecutive months without an alternative medical cause.

- A high FSH level in the postmenopausal range may be used to confirm a postmenopausal state in women not using hormonal contraception or HRT. However, in the absence of 24 months of amenorrhea, confirmation with more than 1 FSH measurement (>40 IU/L) is required. In questionable cases a blood sample with simultaneous FSH >40 MIU/mL and estradiol <40 pg/mL (<147 pmol/L) is confirmatory.
- Females on HRT and whose menopausal status is in doubt must discontinue HRT to allow confirmation of postmenopausal status before study enrollment.

10.4.2. Contraception guidance

Because of the increased risk of venous thromboembolism in participants with MM taking lenalidomide and dexamethasone, combined oral contraceptive pills are not recommended. If a participant is currently using combined oral contraception the participant should switch to another 1 of the highly effective methods listed below in Table 29. The risk of venous thromboembolism continues for 4 to 6 weeks after discontinuing combined oral contraception.

Table 29 Contraceptives allowed during the study

Contraceptives[1] Allowed During The Study Include:
Highly Effective Methods[2] That Have Low User Dependency Failure rate of <1% per year when used consistently and correctly.
• Implantable progestogen-only hormone contraception associated with inhibition of ovulation[3]
• Intrauterine device
• Intrauterine hormone-releasing system[3]
• Bilateral tubal occlusion
• Vasectomized partner *Article I. Note: Vasectomized partner is a highly effective contraceptive method provided that the partner is the sole sexual partner of the WOCBP and the absence of sperm has been confirmed. If not, an additional highly effective method of contraception should be used. Spermatogenesis cycle is approximately 90 days.*
Highly Effective Methods[2] That Are User Dependent Failure rate of <1% per year when used consistently and correctly.
• Combined (estrogen- and progestogen-containing) hormonal contraception associated with inhibition of ovulation[3] – oral – intravaginal – transdermal – injectable
• Progestogen-only hormone contraception associated with inhibition of ovulation[3] – oral – injectable
• Sexual abstinence *Article II. Note: Sexual abstinence is considered a highly effective method only if defined as refraining from heterosexual intercourse during the entire period of risk associated with the study intervention. The reliability of sexual abstinence needs to be evaluated in relation to the duration of the study and the preferred and usual lifestyle of the participant*

Note: Periodic abstinence (calendar, sympto-thermal, post-ovulation methods), withdrawal (coitus interruptus), spermicides only, and lactational amenorrhea method are not acceptable methods of contraception. Male condom and female condom should not be used together (due to risk of failure with friction).

1 Contraceptive use by men or women should be consistent with local regulations regarding the use of contraceptive methods for those participating in clinical studies.

2 Failure rate of <1% per year when used consistently and correctly. Typical use failure rates differ from those when used consistently and correctly.

3 Male condoms must be used in addition to hormonal contraception. If locally required, in accordance with Clinical Trial Facilitation Group guidelines, acceptable contraceptive methods are limited to those which inhibit ovulation as the primary mode of action.

10.5. Appendix 5: Genetics

10.5.1. Use/analysis of DNA

Use/Analysis of DNA

- Genetic variation may impact a participant's response to study intervention, susceptibility to, and severity and progression of disease. Variable response to study intervention may be due to genetic determinants that impact drug absorption, distribution, metabolism, and excretion; mechanism of action of the drug; disease etiology; and/or molecular subtype of the disease being treated. Therefore, where local regulations and IRB/IEC allow, a blood sample will be collected for DNA analysis from consenting participants.
- DNA samples will be used for research related to study intervention or MM and related diseases. They may also be used to develop tests/assays, including diagnostic tests related to belantamab mafodotin alone or in combination with lenalidomide and dexamethasone, daratumumab, lenalidomide, and dexamethasone or concomitant medications and/or interventions of these drug classes and MM. Genetic research may consist of the analysis of 1 or more candidate genes or the analysis of genetic markers throughout the genome, as appropriate.
- Additional analyses of DNA samples may be conducted if it is hypothesized that this may help further understand the clinical data.
- The samples may be analyzed as part of a multi-study assessment of genetic factors involved in the response to study intervention (belantamab mafodotin in combination with lenalidomide and dexamethasone, daratumumab, lenalidomide, and dexamethasone or concomitant medications) or study interventions of these classes to understand the disease under study or related conditions.
- The results of genetic analyses may be reported in the CSR or in a separate study summary.
- The sponsor will store the DNA samples in a secure storage space with adequate measures to protect confidentiality.
- The samples will be retained while research on study intervention in this study or study interventions of these classes or for MM continues but no longer than 20 years or other period per local requirements.

10.6. Appendix 6: Liver safety: Requirements and guidelines

10.6.1. Liver safety: Required actions, monitoring, and follow-up to assess causality of liver events

Liver event stopping criteria with increased monitoring and required follow-up assessments have been designed to assure participant safety and evaluate liver event etiology.

Liver Event Stopping Criteria	
ALT absolute	**Both ALT ≥8×ULN and ≥2×baseline value**
ALT increase	**Unable to monitor weekly for 4 weeks: Both ALT ≥3×ULN and ≥1.5×baseline value** **Able to monitor weekly: Both ALT ≥3×ULN and ≥1.5×baseline value that persists for 4 weeks** **Note: if values reduce to <3×ULN and <1.5×baseline or return to within baseline or normal limits for 2 consecutive weekly assessments, weekly monitoring may return to regular per protocol schedule**
Bilirubin[1,2]	**ALT ≥3×ULN and total bilirubin ≥2×ULN (for participants with known Gilbert's syndrome, these criteria only apply if total bilirubin ≥2×ULN and direct bilirubin ≥2×ULN and at least doubled from baseline value)**
INR[2]	**ALT ≥3×ULN and INR >1.5**
Symptomatic[3]	**ALT ≥3×ULN and ≥1.5×baseline value associated with symptoms (new or worsening) believed to be related to liver injury or hypersensitivity**

Required Actions, Monitoring, and Follow-up to Assess Causality of Liver Event	
Actions and Monitoring	**Follow-up to Assess Causality of Liver Event**
• **Immediately discontinue study intervention** • **Report the event to GSK within 24 hours** • **Complete the liver event form and complete SAE data collection tool if the event also meets the criteria for an SAE[2]** • **Perform liver event follow-up to assess causality of liver event.** • **Monitor the participant liver chemistries (see MONITORING)** For the purpose of these guidelines, "baseline" refers to laboratory assessments performed closest and prior to first dose of study intervention.	• **Viral serology[4]** • **Anti-nuclear antibody, anti-smooth muscle antibody, Type 1 anti-liver kidney microsomal antibodies, and quantitative total IgG or gamma globulins** • **Blood sample for PK analysis of belantamab mafodotin (ADC), obtained within 70 days after last dose of study intervention[5]** • **Serum CPK and LDH, GGT, GLDH, and serum albumin** • **Fractionate bilirubin, if total bilirubin ≥2×ULN** • **Obtain complete blood count with differential to assess eosinophilia**

MONITORING:
If ALT ≥3×ULN AND total bilirubin ≥2×ULN or INR >1.5:
• **Repeat liver chemistries (include ALT, AST, alkaline phosphatase, total bilirubin and INR) and perform liver event follow-up to assess liver event causality within 24 hours**
• **Monitor participants twice weekly until liver chemistries reduce to <3×ULN for ALT, <2×ULN for total bilirubin or ≤1.5 for INR or return to or remain within baseline or normal limits**
• **A specialist or hepatology consultation is recommended**
For all other stopping criteria (bilirubin <2×ULN and INR ≤1.5):
• **Repeat liver chemistries (include ALT, AST, alkaline phosphatase, total bilirubin, and INR) and perform liver event follow-up to assess liver event causality within 24-72 hours**
• **Monitor participants weekly until liver chemistries resolve, reduce to <3×ULN for ALT or return to or remain within baseline or normal limits**
 	
RESTART and/or RECHALLENGE
• **Restart and/or rechallenge is allowed per protocol but do not resume study intervention unless GSK approval is granted**; If restart and/or rechallenge is not granted, permanently discontinue study intervention and continue participant in the study for any protocol-specified follow-up assessments. Refer to restart and/or rechallenge guidelines in Section 10.6.3.
	• **Record the appearance or worsening of clinical symptoms of liver injury, or hypersensitivity, on the liver event form**
	• **Record use of concomitant medications on the concomitant medications report form including acetaminophen, herbal remedies, recreational drugs and other over-the-counter medications**
	• **Record alcohol use on the liver event alcohol intake form**
	If ALT ≥3×ULN AND total bilirubin ≥2×ULN or INR >1.5 obtain the following in addition to the assessments listed above:
	• **Serum acetaminophen adduct assay should be conducted (where available) to assess potential acetaminophen contribution to liver injury**
	• **Liver imaging (ultrasound, magnetic resonance, or computed tomography) to evaluate liver disease, complete liver imaging forms**
	• **Liver biopsy may be considered and discussed with local specialist if available:**
	  ○ **In participants when serology raises the possibility of autoimmune hepatitis**
	  ○ **In participants when suspected DILI progresses or fails to resolve on withdrawal of study intervention**
	  ○ **In participants with acute or chronic atypical presentation**
	• **If liver biopsy conducted complete liver biopsy form**

1. Serum bilirubin fractionation should be performed if testing is available. Additionally, if serum bilirubin fractionation testing is unavailable, record presence of detectable urinary bilirubin on dipstick, indicating direct bilirubin elevations and suggesting liver injury.
2. All events of ALT ≥3×ULN and total bilirubin ≥2×ULN (for participants with known Gilbert's syndrome, these criteria only apply if total bilirubin ≥2×ULN and direct bilirubin ≥2×ULN and ≥2×baseline value) or ALT ≥3×ULN and INR >1.5, which may indicate severe liver injury (possible 'Hy's Law) must be reported as an SAE (excluding studies of hepatic impairment or cirrhosis); the INR threshold value stated will not apply to participants receiving anticoagulants.
3. New or worsening symptoms believed to be related to liver injury (such as fatigue, nausea, vomiting, right upper quadrant pain or tenderness, or jaundice) or believed to be related to hypersensitivity (such as fever, rash or eosinophilia).
4. Includes: Hepatitis A IgM antibody; HBsAg and HBcAb (IgM); HCV RNA; cytomegalovirus IgM antibody; Epstein-Barr viral capsid antigen IgM antibody (or if unavailable, obtain heterophile antibody or monospot testing); Hepatitis E IgM antibody, and RNA PCR test. HBV DNA quantification, and HDV antibody should be measured if participant is known to be HBsAg and/or HBcAb-positive prior to onset of the liver event or is subsequently found to be HBsAg-positive on investigation following the liver event. If hepatitis delta antibody assay cannot be performed, it can be replaced with a PCR of HDV RNA virus (where needed and if this is feasible).
5. Record in the eCRF the date/time of the PK blood sample collection and the date/time of the last dose of study intervention prior to blood sample collection. If the date or time of the last dose is unclear, provide the best approximation. If the date/time of the last dose cannot be approximated or a PK sample cannot be collected in [specify time period], do not obtain a PK sample. Instructions for sample handling and shipping are in the Laboratory Manual.

10.6.2. Liver safety: Liver chemistry increased monitoring criteria with continued study intervention

Liver Event Increased Monitoring Criteria and Actions with Continued Study Intervention	
Criteria	**Actions**
ALT ≥5×ULN and <8×ULN and total bilirubin <2×ULN or INR ≤1.5 without symptoms believed to be related to liver injury or hypersensitivity, and who can be monitored weekly for 2 weeks. **OR** **ALT ≥3×ULN and <5×ULN and total bilirubin <2×ULN or INR ≤1.5 without symptoms believed to be related to liver injury or hypersensitivity, and who can be monitored weekly for 4 weeks.**	• **Notify the GSK Medical Monitor within 24 hours of learning of the abnormality to discuss participant safety** • **Participant can continue study intervention** • **Participant must return weekly for repeat liver chemistries (ALT, AST, alkaline phosphatase, total bilirubin, and INR) until they stabilize (i.e., ALT or AST <3×ULN and no increases in total bilirubin and INR) or return to or remain within baseline or normal limits** For the purpose of these guidelines, "baseline" refers to laboratory assessments performed closest and prior to first dose of study intervention. • **If at any time participant meets the liver chemistry stopping criteria, proceed as described above** • **If ALT decreases from ALT ≥5×ULN and <8×ULN to ≥3×ULN but <5×ULN (total bilirubin <2×ULN and INR ≤1.5), continue to monitor liver chemistries weekly** • **If, after 4 weeks of monitoring, stopping criteria have not been met but any of the monitored liver chemistry (ALT, AST, alkaline phosphatase, total bilirubin, and INR) remains abnormal/above baseline, monitor participants twice monthly until they stabilize or return to within baseline or normal limits. Alternatively, the monitoring can return to standard per protocol when the investigator and GSK Medical Monitor agree that values are stable or no longer significantly abnormal (this may require local investigation of potential causes for liver chemistry abnormality)**

10.6.3. Liver safety: Study intervention restart and/or rechallenge guidelines

Restart refers to resuming study intervention administration following liver stopping events for which there was a clear underlying cause (other than DILI) of the liver event (e.g., biliary obstruction, pancreatic events, hypotension, acute viral hepatitis). Furthermore, restart is not permitted following a liver stopping event when the underlying cause was alcohol-related hepatitis. Typically, alcohol-related hepatitis has AST>ALT with neither higher than 10×ULN and occurs in the context of reported or suspected heavy alcohol intake.

Rechallenge refers to resuming study intervention administration following study-intervention induced liver injury (DILI). Following DILI, drug rechallenge is associated with a 13% mortality across all drugs in prospective studies [Andrade, 2009]. Clinical outcomes vary by drug, with nearly 50% fatality with halothane re-administered within 1 month of initial injury. However, some drugs seldom result in recurrent liver injury or fatality. Because of the risks associated with rechallenge after DILI, this should only be considered for a participant for whom there is compelling evidence of benefit from a critical or life-saving medicine, there is no alternative approved medicine available, and a benefit-risk assessment of rechallenge is considered to be favorable.

Approval by GSK for a study intervention restart and/or rechallenge after a liver event meeting protocol-specified stopping criteria can be considered only:

- If the principal investigator requests consideration for study intervention restart and/or rechallenge.
- In the case of study intervention rechallenge, there is an absence of the risk factors for a severe or fatal drug rechallenge outcome, including:
  - hypersensitivity with initial liver injury (e.g., fever, rash, eosinophilia) [Andrade, 2009],
  - jaundice or bilirubin ≥2×ULN with initial liver injury,
  - SAE or fatality has earlier been observed with drug rechallenges [Hunt, 2010; Papay, 2009],
  - evidence of drug-related nonclinical liability (e.g., reactive metabolites; mitochondrial impairment) [Hunt, 2010].
- In the case of study intervention restart and/or rechallenge, there is an absence of the following:
  - ongoing severe liver injury defined by: ALT ≥3×ULN, bilirubin ≥2×ULN, or INR >1.5,
  - evidence that the event is an alcohol induced hepatitis.
- For participants with improving liver chemistry to normal, or both <1.5×baseline and ALT <3×ULN.
- The participant does not have known genetic marker associated with liver injury from study intervention).
- IRB/IEC approval of study intervention restart and/or rechallenge must be obtained, if required.
- If restart and/or rechallenge of study intervention is approved by GSK in writing:
  - The participant must be provided with a clear description of the possible benefits and risks of study intervention administration, including the possibility of recurrent, more severe liver injury or death.
  - The participant must also provide signed informed consent specifically for the study intervention restart and/or rechallenge. Documentation of informed consent must be recorded in the study file.
  - The study intervention must be administered at the dose specified by GSK.
- Participants approved by GSK for restart and/or rechallenge of study intervention must return to the clinic once per week (restart) or twice per week (rechallenge) for liver chemistry tests until stable liver chemistries have been demonstrated; then laboratory monitoring may resume per protocol.
- If the participant meets protocol-defined liver event stopping criteria after the study intervention restart and/or rechallenge, the study intervention should be permanently discontinued.

- The GSK Medical Monitor, and the IRB/IEC (if required) must be informed of the participant's outcome following study intervention restart or rechallenge.
- GSK must be notified of any AEs.

10.7. Appendix 7: IMWG Frailty Score

	Score
Age, years	
≤75	0
76-80	1
>80	2
ADL	
>4	0
≤4	1
IADL	
>5	0
≤5	1
CCI	
≤1	0
≥2	1

Total Score	Participant status
0	**Fit**
1	**Intermediate-fitness**
≥2	**Frail**

[Palumbo, 2015]

The following link will auto-calculate the frailty score
http://www.myelomafrailtyscorecalculator.net/

10.8. Appendix 8: New York Heart Association classification

The NYHA Functional Classification: Class I, II, III or IV Heart Failure [NYHA, 1994] provides a simple way of classifying the extent of heart failure. It places participants in 1 of 4 categories based on the level of limitation experienced during physical activity:

Class	Symptoms
I (Mild)	No limitation of physical activity. Ordinary physical activity does not cause undue fatigue, palpitation, or dyspnea (shortness of breath).
II (Mild)	Slight limitation of physical activity. Comfortable at rest, but ordinary physical activity results in fatigue, palpitation, or dyspnea.
III (Moderate)	Marked limitation of physical activity. Comfortable at rest, but less than ordinary physical activity results in fatigue, palpitation, or dyspnea.
IV (Severe)	Unable to carry out any physical activity without discomfort. Symptoms of cardiac insufficiency at rest. If any physical activity is undertaken, discomfort is increased.

10.9. Appendix 9: ECOG functional status

Performance Grade	Description
0	Normal activity. Fully active, able to carry on all pre-disease performance without restriction.
1	Symptoms, but ambulatory. Restricted in physically strenuous activity, but ambulatory and able to carry out work of a light or sedentary nature (e.g., light housework).
2	In bed <50% of the time. Ambulatory and capable of all self-care, but unable to carry out any work activities. Up and about more than 50% of waking hours.
3	In bed >50% of the time. Capable of only limited self-care, confined to bed or chair more than 50% of waking hours.
4	100% bedridden. Completely disabled. Cannot carry on any self-care. Totally confined to bed or chair.
5	Dead.

Source: [Oken, 1982]

10.10. Appendix 10: Eye care specialist's qualifications and requirements

For examiners with a degree in ophthalmology or optometry, those involved in eye evaluations in the protocol will, at a minimum be able to provide comprehensive eye care to participants, which includes routine check-ups, treatment, and ongoing management of visual disease.

Specifically, qualified eye care specialists must be able to:

- Perform comprehensive eye exams.
- Perform VA with manual refraction tests and analyze results.
- Perform slit lamp tests and analyze results.
- Perform intraocular pressure examination.
- Dilated fundoscopic examination.
- Diagnose and treat ocular issues and diseases such as keratopathy or glaucoma.

10.11. Appendix 11: PFS event and censoring rules

Situation	Date of Progression or Censoring	Outcome
No adequate baseline or post-baseline tumor assessments[1] and the participant is alive	Randomization	Censored
No confirmed PD and no death	Date of last adequate[3] assessment of response	Censored
Confirmed progression or death documented at or between scheduled visits (without extended loss of follow up[4])	Date of progression or death (whichever occurs first)	Event
Start of subsequent anti-myeloma treatment prior to confirmed PD/death	(S1) Date of last adequate[3] assessment of response on or prior to starting anti-myeloma therapy[2]	(S1) Censored
	(S2) Date of starting new anti-myeloma therapy	(S2) Event
Confirmed progression or death after an extended loss to follow-up time[4]	Date of last adequate[3] assessment of response prior to the consecutive missed assessments	Censored
	(S4) Date of death or confirmed progression (ignoring extended loss of follow-up time)	(S4) Event
Discontinuation of study intervention due to unconfirmed/clinical PD	(S3) Date of Treatment Discontinuation	(S3) Event

1 Adequate baseline assessment defined as: the patient has at least 1 of the following measurements: i) Serum M-protein ≥0.5 g/dL (≥5 g/L); OR ii) Urine M-protein ≥200 mg/24 hour; OR iii) Serum FLC assay: Involved FLC level ≥10 mg/dL (≥100 mg/L) AND an abnormal serum free light chain ratio (<0.26 or >1.65).

2 If PD or death and new anti-myeloma therapy occur on the same day assume the outcome is progression or death, and the date is the date of the assessment of progression or the date of death. If anti-myeloma therapy is started prior to any adequate assessments, censoring date should be the date of randomization.

3 An adequate assessment is defined as an assessment where the confirmed response is sCR, CR, VGPR, PR, MR, or SD. If the adequate assessment occurred on the same date as new anti-myeloma therapy, it is assumed that the assessment occurred first.

4 Extended loss to follow-up time defined as 2 or more missed consecutive disease assessments (details provided in the SAP).

10.12. Appendix 12: Considerations for participants with HIV infection

10.12.1. Considerations regarding switching combination antiretroviral therapy

A switch of combination antiretroviral therapy, defined as a change of a single drug or multiple drugs simultaneously, is allowed in the 3 months prior to screening only due to tolerability/safety concerns, access to medications, and/or convenience/simplification; the change must NOT have been made for suspected or established treatment failure.

The following switches, if they are the only switches, are not considered a change in regimen:

- A switch from a protease inhibitor boosted with ritonavir to the same protease inhibitor boosted with cobicistat (and vice versa).
- A switch from lamivudine to emtricitabine (and vice versa).
- A switch from tenofovir disoproxil fumarate to tenofovir alafenamide (and vice versa).

10.12.2. Additional considerations regarding inclusion of HIV-seropositive individuals

- Prophylactic antimicrobials may be excluded if PK and/or pharmacodynamic DDIs or possible overlapping toxicities with study intervention are of clinically relevant concern, keeping in mind that these may have implications for inclusion of those individuals with AIDS who might otherwise be eligible. If study-specific DDIs are not specified as exclusion criteria, these can be assessed/managed on a case-by-case basis.
- Study personnel should consider exclusion of participants receiving certain antiretroviral agents based on PK and pharmacodynamic interaction potential and possible overlapping toxicity profiles that may overlap with study intervention (e.g., zidovudine [neutropenia], stavudine [neuropathy], didanosine [neuropathy], atazanavir [QT prolongation], ritonavir-boosted lopinavir [QT prolongation], and saquinavir [QT prolongation]). Such potential DDIs can be assessed on a case-by-case basis if not otherwise identified as study-specific exclusion criteria.
- Although uncommon to be clinically relevant, study intervention agents with immuno-modulatory effects may result in PK DDIs mediated through cytokine changes and their effects on metabolizing enzymes. Such potential DDIs are best assessed on a case-by-case basis.
- Other classes of agents with known potential DDIs should also be taken into consideration, including some antifungal agents and immunosuppressants.

10.12.3. Management of participants with HIV/AIDS

HIV infection is to be managed as part of SOC by the study participant's HIV specialist/HIV care provider and is not dictated by the protocol. The responsibility for treatment and monitoring of HIV infection is to be delineated in the ICF.

Components of antiretroviral therapy should be considered concomitant medications, and any antiretroviral agents contraindicated due to study intervention-specific DDIs and/or overlapping toxicities are to be avoided throughout treatment with study intervention, as applicable.

During induction of cancer therapy, the participant's HIV specialist/HIV care provider should be encouraged to monitor and communicate clinical status and laboratory assessment results at least Q12W, or sooner if indicated, when there may be a possibility that study intervention could impact immune status and/or HIV virologic control. Any impact on HIV stability is to be captured via the standard AE eCRF, and any applicable unscheduled laboratory assessments obtained from the local laboratory and recorded in the eCRF.

When applicable, laboratory assessments should confirm an increased viral load ≥50 c/mL and/or reduction of CD4 cell counts below standard thresholds. For example, if a participant begins the study with a CD4 cell count >350 cells/mm^3, any reduction to ≤350 cells/mm^3 is to be confirmed; and if a participant begins the study with a CD4 cell count >200 and ≤350 cells/mm^3, any reduction to ≤200 cells/mm^3 is to be confirmed.

When a viral load increase and/or CD4 cell count decrease is confirmed, factors other than study intervention should be explored (e.g., lack of adherence to antiretroviral therapy). If a causal relationship to study intervention cannot be ruled out, unstable HIV infection is a possibility and will require delay of study intervention administration or withdrawal from study intervention, as applicable.

10.13. Appendix 13: Country-specific requirements

Inclusion Criteria:

- In the **Republic of Korea**, a participant must be over 19 years of age inclusive, at the time of signing the informed consent.
- In **France**, a participant will be eligible for inclusion in this study only if either affiliated to or a beneficiary of a social security category.
- In **Japan**, Hepatitis B participants who are HBsAg- (e.g., HBsAb+/HBsAg-, HBcAb+/HBsAg-, HBcAb+/HBsAb+/HBsAg-) are eligible for inclusion in this study if HBV DNA is undetectable. A patient with HBsAg+ are eligible if HBV DNA is undetectable after assessing HBsAg, HBsAb, and HBV DNA, and if a hepatologist agrees with the participation into this study.

Adverse Events:

- For all studies conducted in China, SAEs will be collected from signing of the ICF and NOT from the start of treatment.
- For studies conducted in the EU, the sponsor will provide written summaries of the status of the study, information on SAEs or other significant safety findings to the agency in accordance with regulatory requirements.

Other Assessments:

- In **China**, the following assessment will not be performed as a follow-up to bilirubin or INR meeting Liver Chemistry stopping criteria: Serum acetaminophen adduct assay.
- In **China**, immunophenotyping assessments will be waived and samples will not be collected.
- In **China**, sample collection and testing for MRD assessment will not be performed until approval from HGRAC is obtained. See Table 3 for details on collection and timepoints. Bone marrow samples will be collected and then transported to a central laboratory in China (Q Squared Solutions [Beijing] Co. Ltd.) until exportation to a third-party laboratory (Adaptive Biotechnologies Corporation, USA) for MRD testing. Adaptive will be the only MRD testing laboratory for this study globally. After testing, the remaining samples will be stored for a period of time approved by HGRAC after the end of the study and will be destroyed in the third-party laboratory. Additional details can be found in the Protocol Addendum and Laboratory Manual. Full list of laboratory names and addresses are provided in the "List of Clinical Laboratories and Key Vendors".

Contraception Guidance:

Of the contraceptive methods defined in Section 10.4.2 Contraceptives Allowed During the Study, the following are NOT approved in **Japan** as contraceptive method:

Highly Effective Methods That Have Low User Dependency
• Implantable progestogen-only hormone contraception associated with inhibition of ovulation
Highly Effective Methods That Are User Dependent
• Combined (estrogen- and progestogen-containing) hormonal contraception associated with inhibition of ovulation Article III. intravaginal Article IV. transdermal Article V. injectable
• Progestogen-only hormone contraception associated with inhibition of ovulation Article VI. oral Article VII. injectable

Additional requirements:

For participants from China, any description in connection with collection and using of Human Genetic Resources in this protocol shall be subject to the filing and/or approval of HGRAC.

10.14. Appendix 14: De-centralized options for healthcare

De-centralized or Home Healthcare (General Visit)

Where applicable country and local regulations and infrastructure allow, de-centralized or home healthcare may be permitted. De-centralized visits are defined as visits that are conducted outside of the clinical trial site. Home healthcare is defined as a remote visit(s) that is/are performed at the participant's home by qualified personnel (e.g., nurse).

Activities that may be done as part of a de-centralized or home healthcare visit must follow the schedule provided in the SoA (Section 1.3) and include:

- Collection of blood and urine samples including:
  - Safety assessments which may include routine blood and urine sampling
  - PK and ADA specimen collection
  - Efficacy assessments to be sent to central lab
  - Biomarker, immunogenicity, and genetic assessments
  - Pregnancy tests
  - 12-Lead ECG

  NOTE: Biological samples should not be collected if they cannot be processed in a timely manner or appropriately stored until their intended use. Please refer to the Laboratory Manual for sample collection and storage requirements.
- Weight
- Administration of study drug (oral only, and subcutaneous daratumumab if performed by a qualified healthcare professional)
- Administration of pre/post-medication
- Identification and reporting of concomitant medications
- Dosing diary review for medication compliance
- Identification and reporting of AEs/SAEs

  NOTE: The investigator or other qualified medical professional must ensure reporting of AEs and SAEs is completed in accordance with the protocol and applicable regulations and that the appropriate medical intervention, therapeutic intervention, and/or support measures are instituted, as necessary.

The participant should be informed of any potential risks associated with de-centralized or home healthcare and sign a revised ICF if required. IRB/Ethics committee should be informed and/or approve of this change in approach and the process documented in study files.

De-centralized or Home Healthcare (Ophthalmologic Exam)

Where applicable country and local regulations and infrastructure allow, protocol-required eye exams may be done in the participant's home or specified alternative eye care specialist clinic. Activities that may be done as part of in-home eye exams must follow the schedule provided in the SoA (Section 1.3) and include:

- VA by near-chart VA or pinhole
- Slit lamp exam
- Tonometry
- Ophthalmoscopy

The participant should be informed of any potential risks associated with de-centralized or home healthcare ophthalmologic exams and sign a revised ICF if required. IRB/Ethics committee should be informed and/or approve of this change in approach and the process documented in study files.

Telemedicine

Where applicable country and local regulations and infrastructure allow, telemedicine visits may be permitted. Telemedicine visits are defined as online (virtual) visits which will use secure video conference, telephone calls, a web portal and/or mobile application as a way of communicating with and monitoring the participant's progress. Telemedicine visits are conducted by an investigator or other qualified medical professional and may be done in combination with visits from Home Healthcare personnel (see above).

Activities that may be done as part of a telemedicine visit include:

- Medical evaluation of the participant
- Identification and reporting of concomitant medications
- Dosing diary review for medication compliance
- Identification, management, and reporting of AEs and SAEs

  NOTE: The investigator or other qualified medical professional must ensure reporting of AEs and SAEs is completed in accordance with the protocol and applicable regulations and that the appropriate medical intervention, therapeutic intervention, and/or support measures are instituted, as necessary. Participants utilizing telemedicine can report AEs at any time via an app, telephone call, or videoconference with site staff.

The participant should be informed of any potential risks associated with the virtual medium and sign a revised ICF if required. IRB/Ethics committee should be informed and/or approve of this change in approach and the process documented in study files.

Direct Shipment of Study Drug to Participants (Oral Only)

Where applicable country and local regulations and infrastructure allow, direct shipment of study drug to participants may be permitted for lenalidomide and dexamethasone. However, due to storage condition and stability requirements of lenalidomide and dexamethasone, direct-to-participant shipment of these oral drugs may only be possible in exceptional circumstances.

Remote PRO Administration

Where applicable country and local regulations and infrastructure allow, remote PRO administration may be permitted. Remote PRO administration is defined administration of protocol PROs by a qualified third party over the telephone. The remote PRO Administrator will use the versions of the PROs designed for verbal administration. The remote PRO Administrator will have access to the electronic PRO portal for the study and input participant responses as the interview is being conducted.

The participant should be informed of any potential risks associated with the remote PRO administration and sign a revised ICF if required. IRB/Ethics committee should be informed and/or approve of this change in approach and the process documented in study files.

10.15. Appendix 15: Clinical study tokenization

Connecting clinical study data to data about a patient collected outside of the study ("real-world data" or "RWD") can be beneficial for better understanding a clinical study population, efficiently conducting long-term follow-up, reducing burden on clinical operations, and more. Where applicable country and local regulations and infrastructure allow, GSK is seeking to connect clinical study data and RWD to further investigate belantamab mafodotin administered in combination with lenalidomide and dexamethasone vs. daratumumab, lenalidomide, and dexamethasone in participants with newly diagnosed MM who are ineligible for autologous stem cell transplantation (TI-NDMM). However, participant IDs and other randomization numbers in Electronic Data Capture systems are unique to each study, and do not provide a way to connect study data to RWD in a privacy preserving manner.

To link conventional study data to RWD, a common, universal ID (herein named a "token") that is unique to each study participant, persistent, and also available in real-world datasets is necessary. A token is a universal, deidentified key that can be used to reference patients across datasets. It is created based on elements of personally identifiable information using an industry-leading privacy preserving technology, powered by Datavant.

The key features below make tokens a secure and consistent mechanism for linking disparate healthcare data sources at the patient level, without unblinding the study or

- Tokens are irreversible and blinded: Hashing by Datavant's technology destroys the input PII, which makes tokens de-identified. The input PII cannot be backward-engineered from the output token.
- Tokens are unique: Each token is unique to the input PII, avoiding false matches. In addition, multiple tokens can be created from different PII elements, improving the precision and granularity of matching.
- Tokens are consistent: The same set of input PII will always create the same tokens. Tokens created in any dataset are compatible and can be matched to one another.
- Tokens are secure: Multiple tokens derived from different PII elements are created for each patient to enable accurate matching. Datavant's tokenization software is used by hundreds of RWD sources and has been used to tokenize more than 100 billion records, for over 200 million patients.

Process and Methodology

As part of their participation in the trial, participants will be offered the consent for tokenization and linking of their clinical trial data. Outside of tokenization and linking of their clinical trial data, no additional study procedures beyond what is outlined in this clinical study protocol, would be completed as result of their participation. Participation in tokenization is optional and participants who decline to participate in the tokenization process will still be able to participate in the clinical study. The study participant's tokenized data will be available for linking to other future data indefinitely or until they withdraw their consent for future data linking, whichever comes first. The participant will be instructed of the withdrawal consent process with information provided in the ICF.

10.16. Appendix 16: Vaccination status questionnaire

What vaccines has patient received?

	Yes	No	UNK	If yes, when?
Covid*	☐	☐	☐	MM/YYYY or UNK/YYYY
Seasonal flu*	☐	☐	☐	MM/YYYY or UNK/YYYY
RSV*	☐	☐	☐	MM/YYYY or UNK/YYYY
Pneumococcal	☐	☐	☐	MM/YYYY or UNK/YYYY or UNK/UNK
Herpes Zoster/shingles	☐	☐	☐	MM/YYYY or UNK/YYYY or UNK/UNK

**Please record only the last vaccination received for the season.*

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